UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2023

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from _____ to _____.

Commission file number: **001-37580**

Alphabet Inc.

(Exact name of registrant as specified in its charter)

Delaware	**61-1767919**
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)

1600 Amphitheatre Parkway
Mountain View, CA 94043
(Address of principal executive offices, including zip code)
(650) 253-0000
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Class A Common Stock, $0.001 par value	**GOOGL**	**Nasdaq Stock Market LLC (Nasdaq Global Select Market)**
Class C Capital Stock, $0.001 par value	**GOOG**	**Nasdaq Stock Market LLC (Nasdaq Global Select Market)**

Securities registered pursuant to Section 12(g) of the Act:

Title of each class
None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of “large accelerated filer,” “accelerated filer,” “smaller reporting company,” and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
Emerging growth company	☐		

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C.7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of June 30, 2023, the aggregate market value of shares held by non-affiliates of the registrant (based upon the closing sale prices of such shares on the Nasdaq Global Select Market on June 30, 2023) was approximately $1,331.2 billion. For purposes of calculating the aggregate market value of shares held by non-affiliates, we have assumed that all outstanding shares are held by non-affiliates, except for shares held by each of our executive officers, directors, and 5% or greater stockholders. In the case of 5% or greater stockholders, we have not deemed such stockholders to be affiliates unless there are facts and circumstances which would indicate that such stockholders exercise any control over our company, or unless they hold 10% or more of our outstanding common stock. These assumptions should not be deemed to constitute an admission that all executive officers, directors, and 5% or greater stockholders are, in fact, affiliates of our company, or that there are not other persons who may be deemed to be affiliates of our company. Further information concerning shareholdings of our officers, directors, and principal stockholders is included or incorporated by reference in Part III, Item 12 of this Annual Report on Form 10-K.

As of January 23, 2024, there were 5,893 million shares of Alphabet's Class A stock outstanding, 869 million shares of Alphabet's Class B stock outstanding, and 5,671 million shares of the Alphabet's Class C stock outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant's Proxy Statement for the 2024 Annual Meeting of Stockholders are incorporated herein by reference in Part III of this Annual Report on Form 10-K to the extent stated herein. Such proxy statement will be filed with the Securities and Exchange Commission within 120 days of the registrant's fiscal year ended December 31, 2023.

Alphabet Inc.
Form 10-K
For the Fiscal Year Ended December 31, 2023

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Note About Forward-Looking Statements

This Annual Report on Form 10-K contains forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. These include, among other things, statements regarding:

- the growth of our business and revenues and our expectations about the factors that influence our success and trends in our business;
- fluctuations in our revenues and margins and various factors contributing to such fluctuations;
- our expectation that the continuing shift from an offline to online world will continue to benefit our business;
- our expectation that the portion of our revenues that we derive beyond advertising will continue to increase and may affect our margins;
- our expectation that our traffic acquisition costs (TAC) and the associated TAC rate will fluctuate, which could affect our overall margins;
- our expectation that our monetization trends will fluctuate, which could affect our revenues and margins;
- fluctuations in paid clicks and cost-per-click as well as impressions and cost-per-impression, and various factors contributing to such fluctuations;
- our expectation that we will continue to periodically review, refine, and update our methodologies for monitoring, gathering, and counting the number of paid clicks and impressions;
- our expectation that our results will be affected by our performance in international markets as users in developing economies increasingly come online;
- our expectation that our foreign exchange risk management program will not fully offset our net exposure to fluctuations in foreign currency exchange rates;
- the expected variability of gains and losses related to hedging activities under our foreign exchange risk management program;
- the amount and timing of revenue recognition from customer contracts with commitments for performance obligations, including our estimate of the remaining amount of commitments and when we expect to recognize revenue;
- our expectation that our capital expenditures will increase, including the expected increase in our technical infrastructure investment to support the growth of our business and our long-term initiatives, in particular in support of artificial intelligence (AI) products and services;
- our plans to continue to invest in new businesses, products, services and technologies, and systems, as well as to continue to invest in acquisitions and strategic investments;
- our pace of hiring and our plans to provide competitive compensation programs;
- our expectation that our cost of revenues, research and development (R&D) expenses, sales and marketing expenses, and general and administrative expenses may increase in amount and/or may increase as a percentage of revenues and may be affected by a number of factors;
- estimates of our future compensation expenses;
- our expectation that our other income (expense), net (OI&E), will fluctuate in the future, as it is largely driven by market dynamics;
- our expectation that our effective tax rate and cash tax payments could increase in future years;

- seasonal fluctuations in internet usage and advertiser expenditures, underlying business trends such as traditional retail seasonality, which are likely to cause fluctuations in our quarterly results;
- the sufficiency of our sources of funding;
- our potential exposure in connection with new and pending investigations, proceedings, and other contingencies, including the possibility that certain legal proceedings to which we are a party could harm our business, financial condition, and operating results;
- our expectation that we will continue to face heightened regulatory scrutiny and changes in regulatory conditions, laws, and public policies, which could affect our business practices and financial results;

- the expected timing, amount, and effect of Alphabet Inc.'s share repurchases;
- our long-term sustainability and diversity goals;

as well as other statements regarding our future operations, financial condition and prospects, and business strategies. Forward-looking statements may appear throughout this report and other documents we file with the Securities and Exchange Commission (SEC), including without limitation, the following sections: Part I, Item 1 "Business;" Part I, Item 1A "Risk Factors;" and Part II, Item 7 "Management's Discussion and Analysis of Financial Condition and Results of Operations." Forward-looking statements generally can be identified by words such as "anticipates," "believes," "could," "estimates," "expects," "intends," "may," "plans," "predicts," "projects," "will be," "will continue," "will likely result," and similar expressions. These forward-looking statements are based on current expectations and assumptions that are subject to risks and uncertainties, which could cause our actual results to differ materially from those reflected in the forward-looking statements. Factors that could cause or contribute to such differences include, but are not limited to, those discussed in this Annual Report on Form 10-K, including the risks discussed in Part I, Item 1A "Risk Factors" and the trends discussed in Part II, Item 7 "Management's Discussion and Analysis of Financial Condition and Results of Operations," and those discussed in other documents we file with the SEC. We undertake no obligation to revise or publicly release the results of any revision to these forward-looking statements, except as required by law. Given these risks and uncertainties, readers are cautioned not to place undue reliance on such forward-looking statements.

As used herein, "Alphabet," "the company," "we," "us," "our," and similar terms include Alphabet Inc. and its subsidiaries, unless the context indicates otherwise.

"Alphabet," "Google," and other trademarks of ours appearing in this report are our property. We do not intend our use or display of other companies' trade names or trademarks to imply an endorsement or sponsorship of us by such companies, or any relationship with any of these companies.

PART I

ITEM 1. BUSINESS

Overview

As our founders Larry and Sergey wrote in the original founders' letter, "Google is not a conventional company. We do not intend to become one." That unconventional spirit has been a driving force throughout our history, inspiring us to tackle big problems and invest in moonshots. It led us to be a pioneer in the development of AI and, since 2016, an AI-first company. We continue this work under the leadership of Alphabet and Google CEO, Sundar Pichai.

Alphabet is a collection of businesses — the largest of which is Google. We report Google in two segments, Google Services and Google Cloud, and all non-Google businesses collectively as Other Bets. Alphabet's structure is about helping each of our businesses prosper through strong leaders and independence.

Access and Technology for Everyone

The Internet is one of the world's most powerful equalizers; it propels ideas, people, and businesses large and small. Our mission to organize the world's information and make it universally accessible and useful is as relevant today as it was when we were founded in 1998. Since then, we have evolved from a company that helps people find answers to a company that also helps people get things done.

We are focused on building an even more helpful Google for everyone, and we aspire to give everyone the tools they need to increase their knowledge, health, happiness, and success. Google Search helps people find information and make sense of the world in more natural and intuitive ways, with trillions of searches on Google every year. YouTube provides people with entertainment, information, and opportunities to learn something new. Google Assistant offers the best way to get things done seamlessly

across different devices, providing intelligent help throughout a person's day, no matter where they are. Google Cloud helps customers solve today's business challenges, improve productivity, reduce costs, and unlock new growth engines. We are continually innovating and building new products and features that will help our users, partners, customers, and communities and have invested more than $150 billion in research and development in the last five years in support of these efforts.

Making AI Helpful for Everyone

AI is a transformational technology that can bring meaningful and positive change to people and societies across the world, and for our business. At Google, we have been bringing AI into our products and services for more than a decade and making them available to our users. Our journey began in 2001, when machine learning was first incorporated into Google Search to suggest better spellings to users searching the web. Today, AI in our products is

used by billions of people globally through features like autocomplete suggestions in Google Search; translation across 133 languages in Google Translate; and organization, searching, and editing in Google Photos.

Large language models (LLMs) are an exciting aspect of our work in AI based on deep learning architectures, such as the Transformer, a neural network architecture that we introduced in 2017 that helped with language understanding. This led to the Bidirectional Encoder Representations from Transformers, or BERT, in 2019 that helped Search understand the intent of user search queries better than ever before.

Google was a company built in the cloud, and we continue to invest in our Google Cloud offerings, including Google Cloud Platform and Google Workspace, to help organizations stay at the forefront of AI innovation with our AI-optimized infrastructure, mature AI platform and world-class models, and assistive agents.

We believe AI can solve some of the hardest societal, scientific and engineering challenges of our time. For example, in 2020, Google DeepMind's AlphaFold system solved a 50-year-old protein folding challenge. Since then, we have open-sourced to the scientific community 200 million of AlphaFold's protein structures which are used to work on everything from accelerating new malaria vaccines to advancing cancer drug discovery and developing plastic-eating enzymes. As another example, AI can also have a transformative effect on climate progress by providing helpful information, predicting climate-related events, and optimizing climate action. Using advanced AI and geospatial analysis, Google Research has developed flood forecasting models that can provide early warning and real-time flooding information to communities and individuals.

As AI continues to improve rapidly, we are focused on giving helpful features to our users and customers as we deliver on our mission to organize the world's information and make it universally accessible and useful. With a bold and responsible approach, we continue to take the next steps to make this technology even more helpful for everyone.

Deliver the Most Advanced, Safe, and Responsible AI

We aim to build the most advanced, safe, and responsible AI with models that are developed, trained, and rigorously tested at scale powered by our continued investment in AI technical infrastructure. In December 2023, we launched Gemini, our most capable and general model. It was built from the ground up to be multimodal, which means it can generalize and seamlessly understand, operate across, and combine different types of information, including text, code, audio, images, and video. Our teams across Alphabet will leverage Gemini, as well as other AI models we have previously developed and announced, across our business to deliver the best product and service experiences for our users, advertisers, partners, customers, and developers.

We believe our approach to AI must be both bold and responsible. That means developing AI in a way that maximizes the positive benefits to society while addressing the challenges, guided by our AI Principles. We published these in 2018, as one of the first companies to articulate principles that put beneficial use, users, safety, and avoidance of harms above business considerations. While there is natural tension between being bold and being responsible, we believe it is possible — and in fact critical — to embrace that tension productively.

Enable Organizations and Developers to Innovate on Google Cloud

AI is not only a powerful enabler, it is also a major platform shift. Globally, businesses from startups to large enterprises, and the public sector are thinking about how to drive transformation. That is why we are focused on making it easy and scalable for others to innovate, and grow, with AI. That means providing the most advanced computing infrastructure and expanding access to Google's latest AI models that have been rigorously tested in our own products. Our Vertex AI platform gives developers the ability to

train, tune, augment, and deploy applications using generative AI models and services such as Enterprise Search and Conversations. Duet AI for Google Cloud provides pre-packaged AI agents that assist developers to write, test, document, and operate software.

Improve Knowledge, Learning, Creativity, and Productivity

Things that we now consider routine – like spell check, mobile check deposit, or Google Search, Google Translate, and Google Maps – all use AI. As AI continues to improve rapidly, we are focused on giving helpful features to our users as we continue to deliver on our mission to organize the world's information and make it universally accessible and useful.

While we have been integrating AI into our products for years, we are now embedding the power of generative AI to continue helping our users express themselves and get things done. For example, Duet AI in Google Workspace helps users write, organize, visualize, accelerate workflows, and have richer meetings. Bard allows users to collaborate with experimental AI with new features that include image capabilities, coding support, and app integration. Dream Screen, a new experimental feature in YouTube, allows for the creation of AI-generated video or image backgrounds to Shorts by typing an idea into a prompt.

We also know businesses of all sizes around the world rely on Google Ads to find customers and grow their businesses — and we make that even easier with AI. With Performance Max, advertisers simply tell us their campaign goals and share their creative assets, and AI will automatically produce and run a highly effective ad campaign across all of Google's properties, to meet their budget. Product Studio brings the benefits of AI to businesses of all sizes, helping them easily create uniquely-tailored imagery featuring their products — for free. Additionally, we are experimenting with Search and Shopping ads that are directly integrated into the AI-powered snapshot and conversational mode in Search Generative Experience.

Build the Most Helpful Personal Computing Platforms and Devices

Over the years, our Pixel phones have incorporated AI compute directly into the device and built experiences on top of it. Our latest Pixel devices were built around AI, bringing the best AI-assistive experiences to our users, such as Best Take, Magic Editor, and Audio Magic Eraser. As we look ahead, we are designing our Android and Chrome operating systems with new AI-forward user experiences.

Moonshots

Many companies get comfortable doing what they have always done, making only incremental changes. This incrementalism leads to irrelevance over time, especially in technology, where change tends to be revolutionary, not evolutionary. People thought we were crazy when we acquired YouTube and Android and when we launched Chrome, but those efforts have matured into major platforms for digital video and mobile devices and a safer, popular browser. Our early investments in AI started out as moonshots but are now incorporated into our core products and central to future developments. We continue to look toward the future and to invest for the long term, most notably for the application of AI to our products and services, as well as other frontier technologies such as quantum computing. As we said in the original founders' letter, we will not shy away from high-risk, high-reward projects that we believe in, as they are the key to our long-term success.

Privacy and Security

We make it a priority to protect the privacy and security of our products, users, and customers, even if there are near-term financial consequences. We do this by continuously investing in building products that are secure by default; strictly upholding responsible data practices that emphasize privacy by design; and building easy-to-use settings that put people in control. We are continually enhancing these efforts over time, whether by enabling users to auto-delete their data, giving them tools, such as My Ad Center, to control their ad experience, or advancing anti-malware, anti-phishing, and password security features.

Google

For reporting purposes Google comprises two segments: Google Services and Google Cloud.

Google Services

Serving Our Users

We have always been committed to building helpful products that can improve the lives of millions of people worldwide. Our product innovations are what make our services widely used, and our brand one of the most recognized in the world. Google Services' core products and platforms include ads, Android, Chrome, devices, Gmail, Google Drive, Google Maps, Google Photos, Google Play, Search, and YouTube, with broad and growing adoption by users around the world.

Our products and services have come a long way since the company was founded more than 25 years ago. While Google Search started as a way to find web pages, organized into ten blue links, we have driven technical advancements and product innovations that have transformed Google Search into a dynamic, multimodal experience. We first expanded from traditional desktop browsers into mobile web search, making it easier to navigate on smaller screens. As new types of content surfaced on the internet,

Universal Search made it possible to search multiple content types, like news, images, videos, and more, to deliver rich, relevant results. The introduction of new search modalities, like voice and visual search, made it easier for people to express their curiosity in natural and intuitive ways. We took that a step further with multisearch, which lets people search with text and images at the same time. Large language models like BERT and Multitask Unified Models, or MUMs, have made it possible to express more natural language queries, vastly improving the quality of results. Each advancement has made it easier and more natural for people to find what they are looking for.

This drive to make information more accessible and helpful has led us over the years to improve the discovery and creation of digital content both on the web and through platforms like Google Play and YouTube. People are consuming many forms of digital content, including watching videos, streaming TV, playing games, listening to music,

reading books, and using apps. Working with content creators and partners, we continue to build new ways for people around the world to create and find great digital content.

Fueling all of these great digital experiences are extraordinary platforms and devices. That is why we continue to invest in platforms like our Android mobile operating system, Chrome browser, and Chrome operating system, as well as growing our family of devices. We see tremendous potential for devices to be helpful and make people's lives easier by combining the best of our AI, software, and hardware. This potential is reflected in our latest generation of devices, such as the new Pixel 8 and Pixel 8 Pro, and the Pixel Watch 2. Creating products and services that people rely on every day is a journey that we are investing in for the long-term.

How We Make Money

We have built world-class advertising technologies for advertisers, agencies, and publishers to power their digital marketing businesses. Our advertising solutions help millions of companies grow their businesses through our wide range of products across devices and formats, and we aim to ensure positive user experiences by serving the right ads at the right time and by building deep partnerships with brands and agencies. AI has been foundational to our advertising business for more than a decade. Products like Performance Max and Product Studio use the full power of our AI to help advertisers find untapped and incremental conversion opportunities.

Google Services generates revenues primarily by delivering both performance and brand advertising that appears on Google Search & other properties, YouTube, and Google Network partners' properties ("Google Network properties"). We continue to invest in both performance and brand advertising and seek to improve the measurability of advertising so advertisers understand the effectiveness of their campaigns.

- **Performance advertising** creates and delivers relevant ads that users will click on leading to direct engagement with advertisers. Performance advertising lets our advertisers connect with users while driving measurable results. Our ads tools allow performance advertisers to create simple text-based ads.
- **Brand advertising** helps enhance users' awareness of and affinity for advertisers' products and services, through videos, text, images, and other interactive ads that run across various devices. We help brand advertisers deliver digital videos and other types of ads to specific audiences for their brand-building marketing campaigns.

We have allocated substantial resources to stopping bad advertising practices and protecting users on the web. We focus on creating the best advertising experiences for our users and advertisers in many ways, including filtering out invalid traffic, removing billions of bad ads from our systems every year, and closely monitoring the sites, apps, and videos where ads appear and blocklisting them when necessary to ensure that ads do not fund bad content.

In addition, Google Services increasingly generates revenues from products and services beyond advertising, including:

- **consumer subscriptions,** which primarily include revenues from YouTube services, such as YouTube TV, YouTube Music and Premium, and NFL Sunday Ticket, as well as Google One;
- **platforms**, which primarily include revenues from Google Play from the sales of apps and in-app purchases; and
- **devices,** which primarily include sales of the Pixel family of devices.

Google Cloud

Through our Google Cloud Platform and Google Workspace offerings, Google Cloud generates revenues primarily from consumption-based fees and subscriptions for infrastructure, platform, collaboration tools and other cloud services. Customers use five key capabilities from Google Cloud.

- **AI-optimized Infrastructure:** provides open, reliable, and scalable compute, networking, and storage to enable customers to run workloads anywhere — on our Cloud, at the edge, or in their data centers. It can be used to migrate and modernize IT systems and to train and serve various types of AI models.
- **Cybersecurity:** helps customers detect, protect, and respond to a broad range of cybersecurity threats, with AI integrated to further strengthen security outcomes, prioritize which threats to investigate, and identify attack paths, as well as accelerate resolution of cybersecurity threats.
- **Databases and Analytics:** provides a variety of different types of databases — relational, key-value, in-memory — to store and manage data for different types of applications. Our Data Cloud also unifies data lakes, data warehouses, data governance, and advanced machine learning into a single platform that can analyze data across any cloud.

- **Collaboration Tools:** Google Workspace and Duet AI in Google Workspace provide easy-to-use, secure communication and collaboration tools, including apps like Gmail, Docs, Drive, Calendar, Meet, and more. These tools enable secure hybrid and remote work, boosting productivity and collaboration. AI has been used in Google Workspace for years to improve grammar, efficiency, security, and more with features like Smart Reply, Smart Compose, and malware and phishing protection in Gmail. Duet AI in Google Workspace helps users write, organize, visualize, accelerate workflows, and have richer meetings.
- **AI Platform and Duet AI for Google Cloud:** Our Vertex AI platform gives developers the ability to train, tune, augment, and deploy applications using generative AI models and services such as Enterprise Search and Conversations. Duet AI for Google Cloud provides pre-packaged AI agents that assist developers to write, test, document, and operate software.

Other Bets

Across Alphabet, we are also using technology to try to solve big problems that affect a wide variety of industries from improving transportation and health technology to exploring solutions to address climate change. Alphabet's investment in the portfolio of Other Bets includes businesses that are at various stages of development, ranging from those in the R&D phase to those that are in the beginning stages of commercialization. Our goal is for them to become thriving, successful businesses. Other Bets operate as independent companies and some of them have their own boards with independent members and outside investors. While these early-stage businesses naturally come with considerable uncertainty, some of them are already generating revenue and making important strides in their industries. Revenues from Other Bets are generated primarily from the sale of healthcare-related services and internet services.

Competition

Our business is characterized by rapid change as well as new and disruptive technologies. We face formidable competition in every aspect of our business, including, among others, from:

- general purpose search engines and information services;
- vertical search engines and e-commerce providers for queries related to travel, jobs, and health, which users may navigate directly to rather than go through Google;
- online advertising platforms and networks;
- other forms of advertising, such as billboards, magazines, newspapers, radio, and television as our advertisers typically advertise in multiple media, both online and offline;
- digital content and application platform providers;
- providers of enterprise cloud services;
- developers and providers of AI products and services;
- companies that design, manufacture, and market consumer hardware products, including businesses that have developed proprietary platforms;
- providers of digital video services;
- social networks, which users may rely on for product or service referrals, rather than seeking information through traditional search engines;
- providers of workspace communication and connectivity products; and
- digital assistant providers.

Competing successfully depends heavily on our ability to develop and distribute innovative products and technologies to the marketplace across our businesses. For example, for advertising, competing successfully depends on attracting and retaining:

- users, for whom other products and services are literally one click away, largely on the basis of the relevance of our advertising, as well as the general usefulness, security, and availability of our products and services;
- advertisers, primarily based on our ability to generate sales leads, and ultimately customers, and to deliver their advertisements in an efficient and effective manner across a variety of distribution channels; and
- content providers, primarily based on the quality of our advertiser base, our ability to help these partners generate revenues from advertising, and the terms of our agreements with them.

For additional information about competition, see Item 1A Risk Factors of this Annual Report on Form 10-K.

Ongoing Commitment to Sustainability

We believe that every business has the opportunity and obligation to protect our planet. Sustainability is one of our core values at Google, and we strive to build sustainability into everything we do. We have been a leader on sustainability and climate change since Google's founding more than 25 years ago.

Our sustainability work is focused on empowering individuals to take action, working together with our partners and customers, and working to reduce our carbon footprint across our operations and supply chain.

In 2020, we shared our aspiration to help individuals, cities, and other partners collectively reduce one gigaton of their carbon equivalent emissions annually by 2030. This is an ambitious vision that we have set to push us to contribute meaningfully to helping with climate solutions beyond our own operations and value chain.

In 2021, we set an ambitious goal to achieve net-zero emissions across all of our operations and value chain, by 2030. To accomplish this, we aim to reduce 50% of our combined Scope 1, Scope 2 (market-based), and Scope 3 absolute emissions (versus our 2019 baseline) before 2030, and plan to invest in nature-based and technology-based carbon removal solutions to neutralize our remaining emissions. We have formally committed to the Science Based Targets initiative to seek their validation of our absolute emissions reduction target.

One of the key levers for reducing emissions from our operations is transitioning to clean energy. Since 2017, we have matched 100% of the electricity consumption of our global operations with purchases of renewable energy on an annual basis. However, because of differences in the availability of renewable energy sources like solar and wind across the regions where we operate—and because of the variable supply of these resources—we still need to rely on carbon-emitting energy sources that power local grids. That is why we set a goal to run on 24/7 carbon-free energy (CFE) on every grid where we operate by 2030.

Achieving net-zero emissions and 24/7 CFE by 2030 are extremely ambitious goals. We also know that our path to net-zero emissions will not be easy or linear. Some of our plans may take years to deliver results, particularly where they involve building new large-scale infrastructure with long lead times. So as our business continues to evolve, we expect our emissions to rise before dropping towards our absolute emissions reduction target.

To benefit the people and places where we operate, we have set goals to replenish 120% of the freshwater volume we consume, on average, across our offices and data centers by 2030 and to help restore and improve the quality of water and health of ecosystems in the communities where we operate.

We also aim to maximize the reuse of finite resources across our operations, products, and supply chains. Our circularity principles focus on designing out waste from the start, keeping materials in use for as long as possible, and promoting healthy materials—for our data centers, workplaces, and products.

More information on our approach to sustainability can be found in our annual sustainability reports, including Google's Environmental Report. The contents of our sustainability reports are not incorporated by reference into this Annual Report on Form 10-K or in any other report or document we file with the SEC. For additional information about risks and uncertainties applicable to our commitments to attain certain sustainability goals, see Item 1A Risk Factors of this Annual Report on Form 10-K.

Culture and Workforce

We are a company of curious, talented, and passionate people. We embrace collaboration and creativity, and encourage the iteration of ideas to address complex challenges in technology and society.

Our people are critical for our continued success, so we work hard to create an environment where employees can have fulfilling careers, and be happy, healthy, and productive. We offer industry-leading benefits and programs to take care of the diverse needs of our employees and their families, including opportunities for career growth and development, resources to support their financial health, and access to excellent healthcare choices. Our competitive compensation programs help us to attract and retain top candidates, and we will continue to invest in recruiting talented people to technical and non-technical roles, and rewarding them well. We provide a variety of high quality training and support to managers to build and strengthen their capabilities-–ranging from courses for new managers, to learning resources that help them provide feedback and manage performance, to coaching and individual support.

At Alphabet, we are committed to making diversity, equity, and inclusion part of everything we do and to growing a workforce that is representative of the users we serve. More information on Google's approach to diversity can be found in our annual diversity reports, available publicly at diversity.google. The contents of our diversity reports are not incorporated by reference into this Annual Report on Form 10-K or in any other report or document we file with the SEC.

As of December 31, 2023, Alphabet had 182,502 employees. We have work councils and statutory employee representation obligations in certain countries, and we are committed to supporting protected labor rights, maintaining an open culture, and listening to all employees. Supporting healthy and open dialogue is central to how we work, and we communicate information about the company through multiple internal channels to our employees.

When necessary we contract with businesses around the world to provide specialized services where we do not have appropriate in-house expertise or resources, often in fields that require specialized training like cafe operations, content moderation, customer support, and physical security. We also contract with temporary staffing agencies when we need to cover short-term leaves, when we have spikes in business needs, or when we need to quickly incubate special projects. We choose our partners and staffing agencies carefully, and review their compliance with Google's Supplier Code of Conduct. We continually make improvements to promote a respectful and positive working environment for everyone — employees, vendors, and temporary staff alike.

Government Regulation

We are subject to numerous United States (U.S.) federal, state, and local, as well as foreign laws and regulations covering a wide variety of subjects, and the scope of this coverage continues to broaden with continuing new legal and regulatory developments in the U.S. and internationally. Like other companies in the technology industry, we face increasingly heightened scrutiny from both U.S. and foreign governments with respect to our compliance with laws and regulations. Many of these laws and regulations are evolving and their applicability and scope, as interpreted by the courts, remain uncertain. Particularly with regard to AI; climate change and sustainability; competition; consumer protection; content moderation; data privacy and security; news publications; and reporting on human capital and diversity, we have seen an increase in new and evolving laws and regulations, as well as related enforcement actions and investigations, being proposed and implemented in recent years by legislative and regulatory bodies around the world.

Our compliance with these laws and regulations may be onerous and could, individually or in the aggregate, increase our cost of doing business, make our products and services less useful, limit our ability to pursue certain business models, cause us to change our business practices, affect our competitive position relative to our peers, and/or otherwise harm our business, reputation, financial condition, and operating results.

For additional information about government regulation applicable to our business, see Item 1A Risk Factors; Trends in Our Business and Financial Effect in Part II, Item 7; and Legal Matters in Note 10 of the Notes to Consolidated Financial Statements included in Part II, Item 8 of this Annual Report on Form 10-K.

Intellectual Property

We rely on various intellectual property laws, confidentiality procedures and contractual provisions to protect our proprietary technology and our brand. We have registered, and applied for the registration of, U.S. and international trademarks, service marks, domain names, and copyrights. We have also filed patent applications in the U.S. and foreign countries covering certain of our technology, and acquired patent assets to supplement our portfolio. We have licensed in the past, and expect that we may license in the future, certain of our rights to other parties. For additional information, see Item 1A Risk Factors of this Annual Report on Form 10-K.

Available Information

Our website is located at www.abc.xyz, and our investor relations website is located at www.abc.xyz/investor. Access to our Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, and our Proxy Statements, and any amendments to these reports, is available on our investor relations website, free of charge, after we file or furnish them with the SEC and they are available on the SEC's website at www.sec.gov.

We webcast via our investor relations website our earnings calls and certain events we participate in or host with members of the investment community. Our investor relations website also provides notifications of news or announcements regarding our financial performance and other items that may be material or of interest to our investors, including SEC filings, investor events, press and earnings releases, and blogs. We also share Google news and product updates on Google's Keyword blog at https://www.blog.google/, which may be of interest or material to our investors. Further, corporate governance information, including our certificate of incorporation, bylaws, governance guidelines, board committee charters, and code of conduct, is also available on our investor relations website under the heading "Governance." The content of our websites are not incorporated by reference into this Annual Report on Form 10-K or in any other report or document we file with the SEC, and any references to our websites are intended to be inactive textual references only.

ITEM 1A. RISK FACTORS

Our operations and financial results are subject to various risks and uncertainties, including but not limited to those described below, which could harm our business, reputation, financial condition, and operating results, and affect the trading price of our Class A and Class C stock.

Risks Specific to our Company

We generate a significant portion of our revenues from advertising. Reduced spending by advertisers, a loss of partners, or new and existing technologies that block ads online and/or affect our ability to customize ads could harm our business.

We generated more than 75% of total revenues from online advertising in 2023. Many of our advertisers, companies that distribute our products and services, digital publishers, and content providers can terminate their contracts with us at any time. These partners may not continue to do business with us if we do not create more value (such as increased numbers of users or customers, new sales leads, increased brand awareness, or more effective monetization) than their available alternatives. Changes to our advertising policies and data privacy practices, such as our initiatives to phase out third-party cookies, as well as changes to other companies' advertising and/or data privacy practices have in the past, and may in the future, affect the advertising that we are able to provide. In addition, technologies have been developed that make customized ads more difficult, or that block the display of ads altogether, and some providers of online services have integrated these technologies that could potentially impair the availability and functionality of third-party digital advertising. Failing to provide superior value or deliver advertisements effectively and competitively could harm our business, reputation, financial condition, and operating results.

In addition, expenditures by advertisers tend to correlate with overall economic conditions. Adverse macroeconomic conditions have affected, and may in the future affect, the demand for advertising, resulting in fluctuations in the amounts our advertisers spend on advertising, which could harm our financial condition and operating results.

We face intense competition. If we do not continue to innovate and provide products and services that are useful to users, customers, and other partners, we may not remain competitive, which could harm our business, financial condition, and operating results.

Our business environment is rapidly evolving and intensely competitive. Our businesses face changing technologies, shifting user needs, and frequent introductions of rival products and services. To compete successfully, we must accurately anticipate technology developments and deliver innovative, relevant and useful products, services, and technologies in a timely manner. As our businesses evolve, the competitive pressure to innovate will encompass a wider range of products and services. We must continue to invest significant resources in technical infrastructure and R&D, including through acquisitions, in order to enhance our technology, products, and services.

We have many competitors in different industries. Our current and potential domestic and international competitors range from large and established companies to emerging start-ups. Some competitors have longer operating histories and well-established relationships in various sectors. They can use their experience and resources in ways that could affect our competitive position, including by making acquisitions and entering into other strategic arrangements; continuing to invest heavily in technical infrastructure, R&D, and in talent; initiating intellectual property and competition claims (whether or not meritorious); and continuing to compete for users, advertisers, customers, and content providers. Further, discrepancies in enforcement of existing laws may enable our lesser known competitors to aggressively interpret those laws without commensurate scrutiny, thereby affording them competitive advantages. Our competitors may also be able to innovate and provide products and services faster than we can or may foresee the need for products and services before we do.

We are expanding our investment in AI across the entire company. This includes generative AI and continuing to integrate AI capabilities into our products and services. AI technology and services are highly competitive, rapidly evolving, and require significant investment, including development and operational costs, to meet the changing needs and expectations of our existing users and attract new users. Our ability to deploy certain AI technologies critical for our products and services and for our business strategy may depend on the availability and pricing of third-party equipment and technical infrastructure. Additionally, other companies may develop AI products and technologies that are similar or superior to our technologies or more cost-effective to deploy. Other companies may also have (or in the future may obtain) patents or other proprietary rights that would prevent, limit, or interfere with our ability to make, use, or sell our own AI products and services.

Our financial condition and operating results may also suffer if our products and services are not responsive to the evolving needs and desires of our users, advertisers, publishers, customers, and content providers. As new and existing technologies continue to develop, competitors and new entrants may be able to offer experiences that are, or

that are seen to be, substantially similar to or better than ours. These technologies could reduce usage of our products and services, and force us to compete in different ways and expend significant resources to develop and operate equal or better products and services. Competitors' success in providing compelling products and services or in attracting and retaining users, advertisers, publishers, customers, and content providers could harm our financial condition and operating results.

Our ongoing investment in new businesses, products, services, and technologies is inherently risky, and could divert management attention and harm our business, financial condition, and operating results.

We have invested and expect to continue to invest in new businesses, products, services, and technologies in a wide range of industries beyond online advertising. The investments that we are making across our businesses, such as building AI capabilities into new and existing products and services, reflect our ongoing efforts to innovate and provide products and services that are helpful to users, advertisers, publishers, customers, and content providers. Our investments ultimately may not be commercially viable or may not result in an adequate return of capital and, in pursuing new strategies, we may incur unanticipated liabilities. Innovations in our products and services could also result in changes to user behavior and affect our revenue trends. These endeavors involve significant risks and uncertainties, including diversion of resources and management attention from current operations, different monetization models, and the use of alternative investment, governance, or compensation structures that may fail to adequately align incentives across the company or otherwise accomplish their objectives.

Within Google Services, we continue to invest heavily in devices, including our smartphones, home devices, and wearables, which is a highly competitive market with frequent introduction of new products and services, rapid adoption of technological advancements by competitors, increased market saturation in developed countries, short product life cycles, evolving industry standards, continual improvement in performance characteristics, and price and feature sensitivity on the part of consumers and businesses. There can be no assurance we will be able to provide devices that compete effectively.

Within Google Cloud, we devote significant resources to develop and deploy our enterprise-ready cloud services, including Google Cloud Platform and Google Workspace, and we are advancing our AI platforms and models to support these tools and technologies. We are incurring costs to build and maintain infrastructure to support cloud computing services, invest in cybersecurity, and hire talent, particularly to support and scale our sales force. At the same time, our competitors are rapidly developing and deploying cloud-based services. Pricing and delivery models are competitive and constantly evolving, and we may not attain sufficient scale and profitability to achieve our business objectives. Further, our business with public sector customers may present additional risks, including regulatory compliance risks. For instance, we may be subject to government audits and cost reviews, and any failure to comply or any deficiencies found may expose us to legal, financial, and/or reputational risks. Evolving laws and regulations may require us to make new capital investments, build new products, and seek partners to deliver localized services in other countries, and we may not be able to meet sovereign operating requirements.

Within Other Bets, we are investing significantly in the areas of health, life sciences, and transportation, among others. These investment areas face intense competition from large, experienced, and well-funded competitors, and our offerings, many of which involve the development of new and emerging technologies, may not be successful, or be able to compete effectively or operate at sufficient levels of profitability.

In addition, new and evolving products and services, including those that use AI, raise ethical, technological, legal, regulatory, and other challenges, which may negatively affect our brands and demand for our products and services. Because all of these investment areas are inherently risky, no assurance can be given that such strategies and offerings will be successful or will not harm our reputation, financial condition, and operating results.

Our revenue growth rate could decline over time, and we may experience downward pressure on our operating margin in the future.

Our revenue growth rate could decline over time as a result of a number of factors, including changes in the devices and modalities used to access our products and services; changes in geographic mix; deceleration or declines in advertiser spending; competition; customer usage and demand for our products; decreases in our pricing of our products and services; ongoing product and policy changes; and shifts to lower priced products and services.

In addition, we may experience downward pressure on our operating margin resulting from a variety of factors, such as an increase in the mix of lower-margin products and services, in particular from the continued expansion of our business into new fields, including products and services such as our devices, Google Cloud, and consumer subscription products, as well as significant investments in Other Bets, all of which may have margins lower than those we generate from advertising. In particular, margins on our devices have had, and may continue to have, an adverse effect on our consolidated margins due to pressures on pricing and higher cost of sales. We may also experience

downward pressure on our operating margins from increasing regulations, increasing competition, and increasing costs for many aspects of our business. Further, certain of our costs and expenses are generally less variable in nature and may not correlate to changes in revenue. We may also not be able to execute our efforts to re-engineer our cost base successfully or in a timely manner. Due to these factors and the evolving nature of our business, our historical revenue growth rate and historical operating margin may not be indicative of our future performance. For additional information, see Trends in Our Business and Financial Effect and Revenues and Monetization Metrics in Part II, Item 7 of this Annual Report on Form 10-K.

Our intellectual property rights are valuable, and any inability to protect them could reduce the value of our products, services, and brands as well as affect our ability to compete.

Our patents, trademarks, trade secrets, copyrights, and other intellectual property rights are important assets for us. Various events outside of our control pose a threat to our intellectual property rights, as well as to our products, services, and technologies. For example, effective intellectual property protection may not be available in every country in which our products and services are distributed or made available through the Internet. Also, the efforts we have taken to protect our proprietary rights may not be sufficient or effective. Although we seek to obtain patent protection for our innovations, it is possible we may not be able to protect some of these innovations. Moreover, we may not have adequate patent or copyright protection for certain innovations that later turn out to be important. There is always the possibility that the scope of the protection gained will be insufficient or that an issued patent may be deemed invalid or unenforceable.

We also seek to maintain certain intellectual property as trade secrets. The secrecy of such trade secrets and other sensitive information could be compromised, which could cause us to lose the competitive advantage resulting from these trade secrets. We also face risks associated with our trademarks. For example, there is a risk that the word “Google” could become so commonly used that it becomes synonymous with the word “search.” Some courts have ruled that "Google" is a protectable trademark, but it is possible that other courts, particularly those outside of the U.S., may reach a different determination. If this happens, we could lose protection for this trademark, which could result in other people using the word “Google” to refer to their own products, thus diminishing our brand.

Any significant impairment of our intellectual property rights could harm our business and our ability to compete. Also, protecting our intellectual property rights is costly and time consuming. Any increase in the unauthorized use of our intellectual property could make it more expensive to do business and harm our financial condition and operating results.

Our business depends on strong brands, and failing to maintain and enhance our brands would hurt our ability to expand our base of users, advertisers, customers, content providers, and other partners.

Our strong brands have significantly contributed to the success of our business. Maintaining and enhancing the brands within Google Services, Google Cloud, and Other Bets increases our ability to enter new categories and launch new and innovative products and services that better serve the needs of our users, advertisers, customers, content providers, and other partners. Our brands have been, and may in the future be, negatively affected by a number of factors, including, among others, reputational issues, third-party content shared on our platforms, data privacy and security issues and developments, and product or technical performance failures. For example, if we fail to respond appropriately to the sharing of misinformation or objectionable content on our services and/or products or objectionable practices by advertisers, or otherwise to adequately address user concerns, our users may lose confidence in our brands.

Furthermore, failure to maintain and enhance our brands could harm our business, reputation, financial condition, and operating results. Our success will depend largely on our ability to remain a

technology leader and continue to provide high-quality, trustworthy, innovative products and services that are truly useful and play a valuable role in a range of settings.

We face a number of manufacturing and supply chain risks that could harm our business, financial condition, and operating results.

We face a number of risks related to manufacturing and supply chain management, which could affect our ability to supply both our products and our services.

We rely on contract manufacturers to manufacture or assemble our devices and servers and networking equipment used in our technical infrastructure, and we may supply the contract manufacturers with components to assemble the devices and equipment. We also rely on other companies to participate in the supply of components and distribution of our products and services. Our business could be negatively affected if we are not able to engage these companies with the necessary capabilities or capacity on reasonable terms, or if those we engage fail to meet their

obligations (whether due to financial difficulties or other reasons), or make adverse changes in the pricing or other material terms of our arrangements with them.

We have experienced and/or may in the future experience supply shortages, price increases, quality issues, and/or longer lead times that could negatively affect our operations, driven by raw material, component availability, manufacturing capacity, labor shortages, industry allocations, logistics capacity, inflation, foreign currency exchange rates, tariffs, sanctions and export controls, trade disputes and barriers, forced labor concerns, sustainability sourcing requirements, geopolitical tensions, armed conflicts, natural disasters or pandemics, the effects of climate change (such as sea level rise, drought, flooding, heat waves, wildfires and resultant air quality effects and power shutdowns associated with wildfire prevention, and increased storm severity), power loss, and significant changes in the financial or business condition of our suppliers. Some of the components we use in our technical infrastructure and our devices are available from only one or limited sources, and we may not be able to find replacement vendors on favorable terms in the event of a supply chain disruption. A significant supply interruption that affects us or our vendors could delay critical data center upgrades or expansions and delay consumer product availability.

We may enter into long-term contracts for materials and products that commit us to significant terms and conditions. We may face costs for materials and products that are not consumed due to market demand, technological change, changed consumer preferences, quality, product recalls, and warranty issues. For instance, because certain of our hardware supply contracts have volume-based pricing or minimum purchase requirements, if the volume of sales of our devices decreases or does not reach projected targets, we could face increased materials and manufacturing costs or other financial liabilities that could make our products more costly per unit to manufacture and harm our financial condition and operating results. Furthermore, certain of our competitors may negotiate more favorable contractual terms based on volume and other commitments that may provide them with competitive advantages and may affect our supply.

Our devices have had, and in the future may have, quality issues resulting from design, manufacturing, or operations. Sometimes, these issues may be caused by components we purchase from other manufacturers or suppliers. If the quality of our products and services does not meet expectations or our products or services are defective or require a recall, it could harm our reputation, financial condition, and operating results.

We require our suppliers and business partners to comply with laws and, where applicable, our company policies and practices, such as the Google Supplier Code of Conduct, regarding workplace and employment practices, data security, environmental compliance, and intellectual property licensing, but we do not control them or their practices. Violations of law or unethical business practices could result in supply chain disruptions, canceled orders, harm to key relationships, and damage to our reputation. Their failure to procure necessary license rights to intellectual property could affect our ability to sell our products or services and expose us to litigation or financial claims.

Interruption to, interference with, or failure of our complex information technology and communications systems could hurt our ability to effectively provide our products and services, which could harm our reputation, financial condition, and operating results.

The availability of our products and services and fulfillment of our customer contracts depend on the continuing operation of our information technology and communications systems. Our systems are vulnerable to damage, interference, or interruption from modifications or upgrades, terrorist attacks, state-sponsored attacks, natural disasters or pandemics, geopolitical tensions or armed conflicts, export controls and sanctions, the effects of climate change (such as sea level rise, drought, flooding, heat waves, wildfires and resultant air quality effects and power shutdowns associated with wildfire prevention, and increased storm severity), power loss, utility outages, telecommunications failures, computer viruses, software bugs, ransomware attacks, supply-chain attacks, computer denial of service attacks, phishing

schemes, or other attempts to harm or access our systems. Some of our data centers are located in areas with a high risk of major earthquakes or other natural disasters. Our data centers are also subject to break-ins, sabotage, and intentional acts of vandalism, and, in some cases, to potential disruptions resulting from problems experienced by facility operators or disruptions as a result of geopolitical tensions and conflicts happening in the area. Some of our systems are not fully redundant, and disaster recovery planning cannot account for all eventualities. The occurrence of a natural disaster or pandemic, closure of a facility, or other unanticipated problems affecting our data centers could result in lengthy interruptions in our service. In addition, our products and services are highly technical and complex and have contained in the past, and may contain in the future, errors or vulnerabilities, which could result in interruptions in or failure of our services or systems. Any of these incidents could impede or prevent us from effectively offering products and providing services, which could harm our reputation, financial condition, and operating results.

Our international operations expose us to additional risks that could harm our business, financial condition, and operating results.

Our international operations are significant to our revenues and net income, and we plan to continue to grow internationally. International revenues accounted for approximately 53% of our consolidated revenues in 2023. In addition to risks described elsewhere in this section, our international operations expose us to other risks, including the following:

- restrictions on foreign ownership and investments, and stringent foreign exchange controls that might prevent us from repatriating cash earned in countries outside the U.S.;
- sanctions, import and export controls, other market access barriers, political unrest, geopolitical tensions, changes in regimes, or armed conflict (such as ongoing conflicts in the Middle East and Ukraine), any of which may affect our business continuity, increase our operating costs, limit demand for our products and services, limit our ability to source components or final products, or prevent or impede us from operating in certain jurisdictions, complying with local laws, or offering products or services;
- longer payment cycles in some countries, increased credit risk, and higher levels of payment fraud;
- an evolving foreign policy landscape that may adversely affect our revenues and could subject us to litigation, new regulatory costs and challenges (including new customer requirements), uncertainty regarding regulatory outcomes, and other liabilities under local laws that may not offer due process or clear legal precedent;
- anti-corruption laws, such as the U.S. Foreign Corrupt Practices Act, and other local laws prohibiting certain payments to government officials, violations of which could result in civil and criminal penalties; and
- different employee/employer relationships, existence of works councils and differing labor practices, and other challenges caused by distance, language, local expertise, and cultural differences, increasing the complexity of doing business in multiple jurisdictions.

Because we conduct business in currencies other than U.S. dollars but report our financial results in U.S. dollars, we have faced, and will continue to face, exposure to fluctuations in foreign currency exchange rates. Although we hedge a portion of our international currency exposure, significant fluctuations in exchange rates between the U.S. dollar and foreign currencies have in the past and may in the future adversely affect our revenues and earnings. Hedging programs are also inherently risky and could expose us to additional risks that could harm our financial condition and operating results.

We are exposed to fluctuations in the fair values of our investments and, in some instances, our financial statements incorporate inherently subjective valuation methodologies.

The fair value of our debt and equity investments may in the future be, and certain investments have been in the past, negatively affected by liquidity, credit deterioration or losses, performance and financial results of the underlying entities, foreign exchange rates, changes in interest rates, including changes that may result from the implementation of new benchmark rates, the effect of new or changing regulations, the stock market in general, or other factors.

We measure certain of our non-marketable equity and debt securities, certain other instruments including stock-based compensation awards settled in the stock of Other Bet companies, and certain assets and liabilities acquired in a business combination, at fair value on a nonrecurring basis, which is inherently subjective and requires management judgment and estimation. All gains and losses on non-marketable equity securities are recognized in OI&E, which increases the volatility of our OI&E. The unrealized gains and losses or impairments we record from fair value remeasurements in any particular period may differ significantly from the gains or losses we ultimately realize on such investments.

As a result of these factors, the value of our investments could decline, which could harm our financial condition and operating results.

Risks Related to our Industry

People access our products and services through a variety of platforms and devices that continue to evolve with the advancement of technology and user preferences. If manufacturers and users do not widely adopt versions of our products and services developed for these interfaces, our business could be harmed.

People access our products and services through a growing variety of devices such as desktop computers, mobile phones, smartphones, laptops and tablets, video game consoles, voice-activated speakers, wearables (including virtual reality and augmented reality devices), automobiles, and television-streaming devices. Our products and services may be less popular on some interfaces. Each manufacturer or distributor may establish unique technical standards for its devices, and our products and services may not be available or may only be available with limited functionality for our users or our advertisers on these devices as a result. Some manufacturers may also elect not to include our products on their devices. In addition, search queries may be undertaken via voice-activated search, apps,

social media or other platforms, which could harm our business. It is hard to predict the challenges we may encounter in adapting our products and services and developing competitive new products and services. We expect to continue to devote significant resources to creating and supporting products and services across multiple platforms and devices. Failing to attract and retain a substantial number of new device manufacturers, suppliers, distributors, developers, and users, or failing to develop products and technologies that work well on new devices and platforms, could harm our business, financial condition, and operating results and ability to capture future business opportunities.

Issues in the development and use of AI may result in reputational harm and increased liability exposure.

Our evolving AI-related efforts may give rise to risks related to harmful content, inaccuracies, discrimination, intellectual property infringement or misappropriation, defamation, data privacy, cybersecurity, and other issues. As a result of these and other challenges associated with innovative technologies, our implementation of AI systems could subject us to competitive harm, regulatory action, legal liability (including under new and proposed legislation and regulations), new applications of existing data protection, privacy, intellectual property, and other laws, and brand or reputational harm.

Some uses of AI will present ethical issues and may have broad effects on society. In order to implement AI responsibly and minimize unintended harmful effects, we have already devoted and will continue to invest significant resources to develop, test, and maintain our products and services, but we may not be able to identify or resolve all AI-related issues, deficiencies, and/or failures before they arise. Unintended consequences, uses, or customization of our AI tools and systems may negatively affect human rights, privacy, employment, or other social concerns, which may result in claims, lawsuits, brand or reputational harm, and increased regulatory scrutiny, any of which could harm our business, financial condition, and operating results.

Data privacy and security concerns relating to our technology and our practices could harm our reputation, cause us to incur significant liability, and deter current and potential users or customers from using our products and services. Computer viruses, software bugs or defects, security breaches, and attacks on our systems could result in the improper disclosure and use of user data and interference with our users' and customers' ability to use our products and services, harming our business and reputation.

Concerns about, including the adequacy of, our practices with regard to the collection, use, governance, disclosure, or security of personal data or other data-privacy-related matters, even if unfounded, could harm our business, reputation, financial condition, and operating results. Our policies and practices may change over time as expectations and regulations regarding privacy and data change.

Our products and services involve the storage, handling, and transmission of proprietary and other sensitive information. Software bugs, theft, misuse, defects, vulnerabilities in our products and services, and security breaches expose us to a risk of loss or improper use and disclosure of such information, which could result in litigation and other potential liabilities, including regulatory fines and penalties, as well as reputational harm. Additionally, our products incorporate highly technical and complex technologies, and thus our technologies and software have contained, and are likely in the future to contain, undetected errors, bugs, and/or vulnerabilities. We continue to add new features involving AI to our offerings and internal systems, and features that rely on AI may be susceptible to unanticipated security threats as our and the market's understanding of AI-centric security risks and protection methods continue to develop. We have in the past discovered, and may in the future discover, some errors in our software code only after we have released the code. Systems and control failures, security breaches, failure to comply with our privacy policies, and/or inadvertent disclosure of user data could result in government and legal exposure, seriously harm our reputation, brand, and business, and impair our ability to attract and retain users or customers. Such incidents have occurred in the past and may continue to occur due to the scale and nature of our products and services. While there is no guarantee that such incidents will not cause

significant damage, we expect to continue to expend significant resources to maintain security protections that limit the effect of bugs, theft, misuse, and security vulnerabilities or breaches.

We experience cyber attacks and other attempts to gain unauthorized access to our systems on a regular basis. Cyber attacks continue to evolve in sophistication and volume, and inherently may be difficult to detect for long periods of time. We have seen, and will continue to see, industry-wide software supply chain vulnerabilities, which could affect our or other parties' systems. We expect to continue to experience such incidents or vulnerabilities in the future. Our efforts to address undesirable activity on our platform may also increase the risk of retaliatory attack. In addition, we face the risk of cyber attacks by nation-states and state-sponsored actors. These attacks may target us or our customers, particularly our public sector customers (including federal, state, and local governments). Geopolitical tensions or armed conflicts, such as the ongoing conflict in the Middle East and Ukraine, may increase these risks.

We may experience security issues, whether due to employee or insider error or malfeasance, system errors, or vulnerabilities in our or other parties' systems. While we may not determine some of these issues to be material at the

time they occur and may remedy them quickly, there is no guarantee that these issues will not ultimately result in significant legal, financial, and reputational harm, including government inquiries, enforcement actions, litigation, and negative publicity. There is also no guarantee that a series of related issues may not be determined to be material at a later date in the aggregate, even if they may not be material individually at the time of their occurrence. Because the techniques used to obtain unauthorized access to, disable or degrade service provided by or otherwise sabotage systems change frequently and often are recognized only after being launched against a target, even taking all reasonable precautions, including those required by law, we have been unable in the past and may continue to be unable to anticipate or detect attacks or vulnerabilities or implement adequate preventative measures.

Further, if any partners with whom we share user or other customer information fail to implement adequate data-security practices, fail to comply with our terms and policies, or otherwise suffer a network or other security breach, our users' data may be improperly accessed, used, or disclosed. If an actual or perceived breach of our or our business partners' or service providers' security occurs, the market perception of the effectiveness of our security measures would be harmed, we could lose users and customers, our trade secrets or those of our business partners may be compromised, and we may be exposed to significant legal and financial risks, including legal claims (which may include class-action litigation) and regulatory actions, fines, and penalties. Any of the foregoing consequences could harm our business, reputation, financial condition, and operating results.

While we have dedicated significant resources to privacy and security incident response capabilities, including dedicated worldwide incident response teams, our response process, particularly during times of a natural disaster or pandemic, may not be adequate, may fail to accurately assess the severity of an incident, may not be fast enough to prevent or limit harm, or may fail to sufficiently remediate an incident. As a result, we may suffer significant legal, reputational, or financial exposure, which could harm our business, financial condition, and operating results.

For additional information, see also our risk factor on privacy and data protection regulations under 'Risks Related to Laws, Regulations, and Policies' below.

Our ongoing investments in safety, security, and content review will likely continue to identify abuse of our platforms and misuse of user data.

In addition to our efforts to prevent and mitigate cyber attacks, we are making significant investments in safety, security, and review efforts to combat misuse of our services and unauthorized access to user data by third parties, including investigation and review of platform applications that could access the information of users of our services. As a result of these efforts, we have in the past discovered, and may in the future discover, incidents of unnecessary access to or misuse of user data or other undesirable activity by third parties. However, we may not have discovered, and may in the future not discover, all such incidents or activity, whether as a result of our data limitations, including our lack of visibility over our encrypted services, the scale of activity on our platform, or other factors, including factors outside of our control such as a natural disaster or pandemic, and we may learn of such incidents or activity via third parties. Such incidents and activities may include the use of user data or our systems in a manner inconsistent with our terms, contracts or policies, the existence of false or undesirable user accounts, election interference, improper ad purchases, activities that threaten people's safety on- or off-line, or instances of spamming, scraping, or spreading disinformation. While we may not determine some of these incidents to be material at the time they occurred and we may remedy them quickly, there is no guarantee that these issues will not ultimately result in significant legal, financial, and reputational harm, including government inquiries and enforcement actions, litigation, and negative publicity. There is also no guarantee that a series of related issues may not be determined to be material at a later date in the aggregate, even if they may not be material individually at the time of their occurrence.

We may also be unsuccessful in our efforts to enforce our policies or otherwise prevent or remediate any such incidents. Any of the foregoing developments may negatively affect user trust and engagement,

harm our reputation and brands, require us to change our business practices in ways that harm our business operations, and adversely affect our business and financial results. Any such developments may also subject us to additional litigation and regulatory inquiries, which could result in monetary penalties and damages, divert management's time and attention, and lead to enhanced regulatory oversight.

Problematic content on our platforms, including low-quality user-generated content, web spam, content farms, and other violations of our guidelines could affect the quality of our services, which could harm our reputation and deter our current and potential users from using our products and services.

We, like others in the industry, face violations of our content guidelines across our platforms, including sophisticated attempts by bad actors to manipulate our hosting and advertising systems to fraudulently generate revenues, or to otherwise generate traffic that does not represent genuine user interest or intent. While we invest significantly in efforts to promote high-quality and relevant results and to detect and prevent low-quality content and invalid traffic, we have been unable and may continue to be unable to detect and prevent all such abuses or promote

uniformly high-quality content. Increased use of AI in our offerings and internal systems may create new avenues of abuse for bad actors.

Many websites violate or attempt to violate our guidelines, including by seeking to inappropriately rank higher in search results than our search engine's assessment of their relevance and utility would rank them. Such efforts have affected, and may continue to affect, the quality of content on our platforms and lead them to display false, misleading, or undesirable content. Although English-language web spam in our search results has been reduced, and web spam in most other languages is limited, we expect web spammers will continue to seek inappropriate ways to improve their rankings. Although we continue to invest in and deploy proprietary technology to detect and prevent web spam on our platforms, there is no guarantee that our technology will always be successful, and our users may have negative experiences on our platforms if our technology fails to work as intended, which may affect our users' decisions in continuing to use our platforms. We also face other challenges from low-quality and irrelevant content websites, including content farms, which are websites that generate large quantities of low-quality content to help them improve their search rankings. We are continually launching algorithmic changes designed to detect and prevent abuse from low-quality websites, but we may not always be successful. We also face other challenges on our platforms, including violations of our content guidelines involving incidents such as attempted election interference, activities that threaten the safety and/or well-being of our users on- or off-line, and the spreading of misinformation or disinformation.

If we fail to either detect and prevent an increase in problematic content or effectively promote high-quality content, it could hurt our reputation for delivering relevant information or reduce use of our platforms, harming our financial condition and operating results. It may also subject us to litigation and regulatory actions, which could result in monetary penalties and damages and divert management's time and attention.

Our business depends on continued and unimpeded access to the Internet by us and our users. Internet access providers may be able to restrict, block, degrade, or charge for access to certain of our products and services, which could lead to additional expenses and the loss of users and advertisers.

Our products and services depend on the ability of our users to access the Internet, and certain of our products require significant bandwidth to work effectively. Currently, this access is provided by companies that have significant market power in the broadband and internet access marketplace, including incumbent telephone companies, cable companies, mobile communications companies, and government-owned service providers. Some of these providers have taken, or have stated that they may take, measures that could degrade, disrupt, or increase the cost of user access to certain of our products by restricting or prohibiting the use of their infrastructure to support or facilitate our offerings, by charging increased fees to us or our users to provide our offerings, or by providing our competitors preferential access. Some jurisdictions have adopted regulations prohibiting certain forms of discrimination by internet access providers; however, substantial uncertainty exists in the U.S. and elsewhere regarding such protections. For example, in 2018 the U.S. Federal Communications Commission repealed net neutrality rules, which could permit internet access providers to restrict, block, degrade, or charge for access to certain of our products and services. In addition, in some jurisdictions, our products and services have been subject to government-initiated restrictions or blockages. These could harm existing key relationships, including with our users, customers, advertisers, and/or content providers, and impair our ability to attract new ones; harm our reputation; and increase costs, thereby negatively affecting our business.

Risks Related to Laws, Regulations, and Policies

We are subject to a variety of new, existing, and changing laws and regulations worldwide that could harm our business, and will likely be subject to an even broader scope of laws and regulations as we continue to expand our business.

We are subject to numerous U.S. and foreign laws and regulations covering a wide variety of subjects, and our introduction of new businesses, products, services, and technologies will likely continue to subject us to additional laws and regulations. In recent years, governments around the world have proposed and adopted a large number of new laws and regulations relevant to the digital economy, particularly in the areas of data privacy and security, competition, environmental, social and governance (ESG) requirements, AI, and online content. The costs of compliance with these measures are high and are likely to increase in the future.

New or changing laws and regulations, or new interpretations or applications of existing laws and regulations in a manner inconsistent with our practices, have resulted in, and may continue to result in, less useful products and services, altered business practices, limited ability to pursue certain business models or offer certain products and services, substantial costs, and civil or criminal liability. Examples include laws and regulations regarding:

- **Competition and technology platforms' business practices:** Laws and regulations focused on large technology platforms, including the Digital Markets Act in the European Union (EU); regulations and legal

settlements in the U.S., South Korea, and elsewhere that affect Google Play's billing policies, fees, and business model; as well as litigation and new regulations under consideration in a range of jurisdictions.

- **AI:** Laws and regulations focused on the development, use, and provision of AI technologies and other digital products and services, which could result in monetary penalties or other regulatory actions. For example, while legislative text has yet to be finalized and formally approved, provisional political agreement on a proposed EU AI Act was reached between co-legislators in December 2023, including that specific transparency and other requirements would be introduced for general purpose AI systems and the models on which those systems are based. In addition, the White House's Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence devises a framework for the U.S. government, among other things, to regulate private sector use and development of certain foundation models.
- **Data privacy, collection, and processing:** Laws and regulations further restricting the collection, processing, and/or sharing of user or advertising-related data, including privacy and data protection laws; laws affecting the processing of children's data (as discussed further below), data breach notification laws, and laws limiting data transfers (including data localization laws).
- **Copyright and other intellectual property**: Copyright and related laws, including the EU Directive on Copyright in the Digital Single Market and European Economic Area transpositions, which may introduce new licensing regimes, increase liability with respect to content uploaded by users or linked to from our platforms, or create property rights in news publications that could require payments to news agencies and publishers, which may result in other regulatory actions.
- **Content moderation**: Various laws covering content moderation and removal, and related disclosure obligations, such as the EU's Digital Services Act, Florida's Senate Bill 7072 and Texas' House Bill 20, and laws and proposed legislation in Singapore, Australia, and the United Kingdom that impose penalties for failure to remove certain types of content or require disclosure of information about the operation of our services and algorithms, which may make it harder for services like Google Search and YouTube to detect and deal with low-quality, deceptive, or harmful content.
- **Consumer protection**: Consumer protection laws, including the EU's New Deal for Consumers, which could result in monetary penalties and create a range of new compliance obligations.

In addition, the applicability and scope of these and other laws and regulations, as interpreted by courts, regulators, or administrative bodies, remain uncertain and could be interpreted in ways that harm our business. For example, we rely on statutory safe harbors, like those set forth in the Digital Millennium Copyright Act and Section 230 of the Communications Decency Act in the U.S. and the E-Commerce Directive in Europe, to protect against liability for various linking, caching, ranking, recommending, and hosting activities. Legislation or court rulings affecting these safe harbors may adversely affect us and may impose significant operational challenges. There are legislative proposals and pending litigation in the U.S., EU, and around the world that could diminish or eliminate safe harbor protection for websites and online platforms. Our development, use, and commercialization of AI products and services (including our implementation of AI in our offerings and internal systems) could subject us to regulatory action and legal liability, including under specific legislation regulating AI, as well as new applications of existing data protection, cybersecurity, privacy, intellectual property, and other laws.

We are and may continue to be subject to claims, lawsuits, regulatory and government investigations, enforcement actions, consent orders, and other forms of regulatory scrutiny and legal liability that could harm our business, reputation, financial condition, and operating results.

We are subject to claims, lawsuits, regulatory and government investigations, other proceedings, and orders involving competition, intellectual property, data privacy and security, tax and related compliance,

labor and employment, commercial disputes, content generated by our users, goods and services offered by advertisers or publishers using our platforms, personal injury, and other matters. We are also subject to a variety of claims including product warranty, product liability, and consumer protection claims related to product defects, among other litigation, and we may also be subject to claims involving health and safety, hazardous materials usage, other environmental effects, AI training, development, and commercialization, or service disruptions or failures. Claims have been brought, and we expect will continue to be brought, against us for defamation, negligence, breaches of contract, copyright and trademark infringement, unfair competition, unlawful activity, torts, privacy rights violations, fraud, or other legal theories based on the nature and content of information available on or via our services, the design and effect of our products and services, or due to our involvement in hosting, transmitting, marketing, branding, or providing access to content created by third parties.

For example, in December 2023, a California jury delivered a verdict in *Epic Games v. Google* finding that Google violated antitrust laws related to Google Play's billing practices. The presiding judge will determine remedies in 2024 and the range of potential remedies vary widely. We plan to appeal. In addition, the U.S. Department of Justice, various U.S. states, and other plaintiffs have filed several antitrust lawsuits about various aspects of our business, including our advertising technologies and practices, the operation and distribution of Google Search, and the operation and distribution of the Android operating system and Play Store. Other regulatory agencies in the U.S. and around the world, including competition enforcers, consumer protection agencies, and data protection authorities, have challenged and may continue to challenge our business practices and compliance with laws and regulations. We are cooperating with these investigations and defending litigation or appealing decisions where appropriate.

Various laws, regulations, investigations, enforcement lawsuits, and regulatory actions have involved in the past, and may in the future result in substantial fines and penalties, injunctive relief, ongoing monitoring and auditing obligations, changes to our products and services, alterations to our business models and operations, including divestiture, and collateral related civil litigation or other adverse consequences, all of which could harm our business, reputation, financial condition, and operating results.

Any of these legal proceedings could result in legal costs, diversion of management resources, negative publicity and other harms to our business. Estimating liabilities for our pending proceedings is a complex, fact-specific, and speculative process that requires significant judgment, and the amounts we are ultimately liable for may be less than or exceed our estimates. The resolution of one or more such proceedings has resulted in, and may in the future result in, additional substantial fines, penalties, injunctions, and other sanctions that could harm our business, reputation, financial condition, and operating results.

For additional information about the ongoing material legal proceedings to which we are subject, see Legal Proceedings in Part I, Item 3 of this Annual Report on Form 10-K.

Privacy, data protection, and data usage regulations are complex and rapidly evolving areas. Any failure or alleged failure to comply with these laws could harm our business, reputation, financial condition, and operating results.

Authorities around the world have adopted and are considering a number of legislative and regulatory proposals concerning data protection, data usage, and encryption of user data. Adverse legal rulings, legislation, or regulation have resulted in, and may continue to result in, fines and orders requiring that we change our practices, which have had and could continue to have an adverse effect on how we provide services, harming our business, reputation, financial condition, and operating results. These laws and regulations are evolving and subject to interpretation, and compliance obligations could cause us to incur substantial costs or harm the quality and operations of our products and services in ways that harm our business. Examples of these laws include:

- The General Data Protection Regulation and the United Kingdom General Data Protection Regulations, which apply to all of our activities conducted from an establishment in the EU or the United Kingdom, respectively, or related to products and services that we offer to EU or the United Kingdom users or customers, respectively, or the monitoring of their behavior in the EU or the UK, respectively.
- Various comprehensive U.S. state and foreign privacy laws, which give new data privacy rights to their respective residents (including, in California, a private right of action in the event of a data breach resulting from our failure to implement and maintain reasonable security procedures and practices) and impose significant obligations on controllers and processors of consumer data.

- State laws governing the processing of biometric information, such as the Illinois Biometric Information Privacy Act and the Texas Capture or Use of Biometric Identifier Act, which impose obligations on businesses that collect or disclose consumer biometric information.
- Various federal, state, and foreign laws governing how companies provide age appropriate experiences to children and minors, including the collection and processing of children and minor's data. These include the Children's Online Privacy Protection Act of 1998, and the United Kingdom Age-Appropriate Design Code, all of which address the use and disclosure of the personal data of children and minors and impose obligations on online services or products directed to or likely to be accessed by children.
- The California Internet of Things Security Law, which regulates the security of data used in connection with internet-connected devices.
- The EU's Digital Markets Act, which will require in-scope companies to obtain user consent for combining data across certain products and require search engines to share anonymized data with rival companies, among other changes.

Further, we are subject to evolving laws and regulations that dictate whether, how, and under what circumstances we can transfer, process and/or receive personal data, as well as ongoing enforcement actions from supervisory authorities related to cross-border transfers of personal data. The validity of various data transfer mechanisms we currently rely upon remains subject to legal, regulatory, and political developments in both Europe and the U.S., which may require us to adapt our existing arrangements.

We face, and may continue to face, intellectual property and other claims that could be costly to defend, result in significant damage awards or other costs (including indemnification awards), and limit our ability to use certain technologies.

We, like other internet, technology, and media companies, are frequently subject to litigation based on allegations of infringement or other violations of intellectual property rights, including patent, copyright, trade secrets, and trademarks. Parties have also sought broad injunctive relief against us by filing claims in U.S. and international courts and the U.S. International Trade Commission (ITC) for exclusion and cease-and-desist orders. In addition, patent-holding companies may frequently seek to generate income from patents they have obtained by bringing claims against us. As we continue to expand our business, the number of intellectual property claims against us has increased and may continue to increase as we develop and acquire new products, services, and technologies.

Adverse results in any of these lawsuits may include awards of monetary damages, costly royalty or licensing agreements (if licenses are available at all), or orders limiting our ability to sell our products and services in the U.S. or elsewhere, including by preventing us from offering certain features, functionalities, products, or services in certain jurisdictions. They may also cause us to change our business practices in ways that could result in a loss of revenues for us and otherwise harm our business.

Many of our agreements with our customers and partners, including certain suppliers, require us to defend against certain intellectual property infringement claims and in some cases indemnify them for certain intellectual property infringement claims against them, which could result in increased costs for defending such claims or significant damages if there were an adverse ruling in any such claims. Such customers and partners may also discontinue the use of our products, services, and technologies, as a result of injunctions or otherwise, which could result in loss of revenues and harm our business. Moreover, intellectual property indemnities provided to us by our suppliers, when obtainable, may not cover all damages and losses suffered by us and our customers arising from intellectual property infringement claims. Furthermore, in connection with our divestitures, we have agreed, and may in the future agree, to provide indemnification for certain potential liabilities, including those associated with intellectual property claims. Regardless of their merits, intellectual property claims are often time consuming and expensive to litigate or settle. To the extent such claims are successful, they could harm our business, including our product and service offerings, financial condition, and operating results.

Expectations relating to ESG considerations could expose us to potential liabilities, increased costs, and reputational harm.

We are subject to laws, regulations, and other measures that govern a wide range of topics, including those related to matters beyond our core products and services. For instance, new laws, regulations, policies, and international accords relating to ESG matters, including sustainability, climate change, human capital, and diversity, are being developed and formalized in Europe, the U.S., and elsewhere, which may entail specific, target-driven frameworks and/or disclosure requirements. We have implemented robust ESG programs, adopted reporting frameworks and principles, and announced a number of goals and initiatives. The implementation of these goals and initiatives may require considerable investments, and our goals, with all of their contingencies, dependencies, and in certain cases, reliance on third-party verification and/or performance, are complex and ambitious, and may change. We cannot guarantee that our goals and initiatives will be fully realized on the timelines we expect or at all, and projects that are completed as planned may not achieve the results we anticipate. Any failure, or perceived failure, by us to adhere to our public statements, comply fully with developing interpretations of ESG laws and regulations,

or meet evolving and varied stakeholder expectations and standards could harm our business, reputation, financial condition, and operating results.

We could be subject to changes in tax rates, the adoption of new U.S. or international tax legislation, or exposure to additional tax liabilities.

We are subject to a variety of taxes and tax collection obligations in the U.S. and numerous foreign jurisdictions. Our effective tax rates are affected by a variety of factors, including changes in the mix of earnings in jurisdictions with different statutory tax rates, net gains and losses on hedges and related transactions under our foreign exchange risk management program, changes in our stock price for shares issued as employee compensation, changes in the valuation of our deferred tax assets or liabilities, and the application of different provisions of tax laws or changes in tax laws, regulations, or accounting principles (including changes in the interpretation of existing laws). Further, if we are

unable or fail to collect taxes on behalf of customers, employees and partners as the withholding agent, we could become liable for taxes that are levied against third parties.

We are subject to regular review and audit by both domestic and foreign tax authorities. As a result, we have received, and may in the future receive, assessments in multiple jurisdictions, on various tax-related assertions, such as transfer-pricing adjustments or permanent-establishment claims. Any adverse outcome of such a review or audit could harm our financial condition and operating results, require adverse changes to our business practices, or subject us to additional litigation and regulatory inquiries. In addition, the determination of our worldwide provision for income taxes and other tax liabilities requires significant judgment and often involves uncertainty. Although we believe our estimates are reasonable, the ultimate tax outcome may differ from the amounts recorded in our financial statements and may affect our financial results in the period or periods for which such determination is made.

Furthermore, due to shifting economic and political conditions, tax policies, laws, or rates in various jurisdictions may be subject to significant changes in ways that could harm our financial condition and operating results. For example, various jurisdictions around the world have enacted or are considering revenue-based taxes such as digital services taxes and other targeted taxes, which could lead to inconsistent and potentially overlapping international tax regimes. The Organization for Economic Cooperation and Development (OECD) is coordinating negotiations among more than 140 countries with the goal of achieving consensus around substantial changes to international tax policies, including the implementation of a minimum global effective tax rate of 15%. Our effective tax rate and cash tax payments could increase in future years as a result of these changes.

Risks Related to Ownership of our Stock

We cannot guarantee that any share repurchase program will be fully consummated or will enhance long-term stockholder value, and share repurchases could increase the volatility of our stock prices and could diminish our cash reserves.

We engage in share repurchases of our Class A and Class C stock from time to time in accordance with authorizations from the Board of Directors of Alphabet. Our repurchase program does not have an expiration date and does not obligate Alphabet to repurchase any specific dollar amount or to acquire any specific number of shares. Further, our share repurchases could affect our share trading prices, increase their volatility, reduce our cash reserves and may be suspended or terminated at any time, which may result in a decrease in the trading prices of our stock.

The concentration of our stock ownership limits our stockholders' ability to influence corporate matters.

Our Class B stock has 10 votes per share, our Class A stock has one vote per share, and our Class C stock has no voting rights. As of December 31, 2023, Larry Page and Sergey Brin beneficially owned approximately 86.5% of our outstanding Class B stock, which represented approximately 51.5% of the voting power of our outstanding common stock. Through their stock ownership, Larry and Sergey have significant influence over all matters requiring stockholder approval, including the election of directors and significant corporate transactions, such as a merger or other sale of our company or our assets, for the foreseeable future. In addition, because our Class C stock carries no voting rights (except as required by applicable law), the issuance of the Class C stock, including in future stock-based acquisition transactions and to fund employee equity incentive programs, could continue Larry and Sergey's current relative voting power and their ability to elect all of our directors and to determine the outcome of most matters submitted to a vote of our stockholders. The share repurchases made pursuant to our repurchase program may also affect Larry and Sergey's relative voting power. This concentrated control limits or severely restricts other stockholders' ability to influence corporate matters and we may take actions that some of our stockholders do not view as beneficial, which could reduce the market price of our Class A stock and our Class C stock.

Provisions in our charter documents and under Delaware law could discourage a takeover that stockholders may consider favorable.

Provisions in Alphabet's certificate of incorporation and bylaws may have the effect of delaying or preventing a change of control or changes in our management. These provisions include the following:

- Our Board of Directors has the right to elect directors to fill a vacancy created by the expansion of the Board of Directors or the resignation, death, or removal of a director.
- Our stockholders may not act by written consent, which makes it difficult to take certain actions without holding a stockholders' meeting.
- Our certificate of incorporation prohibits cumulative voting in the election of directors. This limits the ability of minority stockholders to elect director candidates.

- Stockholders must provide advance notice to nominate individuals for election to the Board of Directors or to propose matters that can be acted upon at a stockholders' meeting. These provisions may discourage or deter a potential acquirer from conducting a solicitation of proxies to elect the acquirer's own slate of directors or otherwise attempting to obtain control of our company.
- Our Board of Directors may issue, without stockholder approval, shares of undesignated preferred stock, which makes it possible for our Board of Directors to issue preferred stock with voting or other rights or preferences that could impede the success of any attempt to acquire us.

As a Delaware corporation, we are also subject to certain Delaware anti-takeover provisions. Under Delaware law, a corporation may not engage in a business combination with any holder of 15% or more of its outstanding voting stock unless the holder has held the stock for three years or, among other things, the Board of Directors has approved the transaction. Our Board of Directors could rely on Delaware law to prevent or delay an acquisition of us.

The trading price for our Class A stock and non-voting Class C stock may continue to be volatile.

The trading price of our stock has at times experienced significant volatility and may continue to be volatile. In addition to the factors discussed in this report, the trading prices of our Class A stock and Class C stock have fluctuated, and may continue to fluctuate widely, in response to various factors, many of which are beyond our control, including, among others, the activities of our peers and changes in broader economic and political conditions around the world. These broad market and industry factors could harm the market price of our Class A stock and our Class C stock, regardless of our actual operating performance.

General Risks

Our operating results may fluctuate, which makes our results difficult to predict and could cause our results to fall short of expectations.

Our operating results have fluctuated, and may in the future fluctuate, as a result of a number of factors, many outside of our control, including the cyclical nature and seasonality in our business and geopolitical events. As a result, comparing our operating results (including our expenses as a percentage of our revenues) on a period-to-period basis may not be meaningful, and our past results should not be relied on as an indication of our future performance. Consequently, our operating results in future quarters may fall below expectations.

Acquisitions, joint ventures, investments, and divestitures could result in operating difficulties, dilution, and other consequences that could harm our business, financial condition, and operating results.

Acquisitions, joint ventures, investments, and divestitures are important elements of our overall corporate strategy and use of capital, and these transactions could be material to our financial condition and operating results. We expect to continue to evaluate and enter into discussions regarding a wide array of such potential strategic arrangements, which could create unforeseen operating difficulties and expenditures. Some of the areas where we face risks include:

- diversion of management time and focus from operating our business to challenges related to acquisitions and other strategic arrangements;
- failure to obtain required approvals on a timely basis, if at all, from governmental authorities, or conditions placed upon approval that could, among other things, delay or prevent us from completing a transaction, or otherwise restrict our ability to realize the expected financial or strategic goals of a transaction;

- failure to successfully integrate the acquired operations, technologies, services, and personnel (including cultural integration and retention of employees) and further develop the acquired business or technology;
- implementation of controls (or remediation of control deficiencies), procedures, and policies at the acquired company;
- integration of the acquired company's accounting and other administrative systems, and the coordination of product, engineering, and sales and marketing functions;
- transition of operations, users, and customers onto our existing platforms;
- in the case of foreign acquisitions, the need to integrate operations across different cultures and languages and to address the particular economic, currency, political, and regulatory risks associated with specific countries;
- failure to accomplish commercial, strategic or financial objectives with respect to investments, joint ventures, and other strategic arrangements;
- failure to realize the value of investments and joint ventures due to a lack of liquidity;

- liability for activities of the acquired company before the acquisition, including patent and trademark infringement claims, data privacy and security issues, violations of laws, commercial disputes, tax liabilities, warranty claims, product liabilities, and other known and unknown liabilities; and
- litigation or other claims in connection with the acquired company, including claims from terminated employees, customers, former stockholders, or other third parties.

Our failure to address these risks or other problems encountered in connection with our past or future acquisitions and other strategic arrangements could cause us to fail to realize their anticipated benefits, incur unanticipated liabilities, and harm our business generally.

Our acquisitions and other strategic arrangements could also result in dilutive issuances of our equity securities, the incurrence of debt, contingent liabilities, or amortization expenses, or impairment of goodwill and/or purchased long-lived assets, and restructuring charges, any of which could harm our financial condition and operating results. Also, the anticipated benefits or value of our acquisitions and other strategic arrangements may not materialize. In connection with our divestitures, we have agreed, and may in the future agree, to provide indemnification for certain potential liabilities, which could harm our financial condition and operating results.

We rely on highly skilled personnel and, if we are unable to retain or motivate key personnel, hire qualified personnel, or maintain and continue to adapt our corporate culture, we may not be able to grow or operate effectively.

Our performance and future success depends in large part upon the continued service of key technical leads as well as members of our senior management team. For instance, Sundar Pichai is critical to the overall management of Alphabet and its subsidiaries and plays an important role in the development of our technology, maintaining our culture, and setting our strategic direction.

Our ability to compete effectively and our future success depend on our continuing to identify, hire, develop, motivate, and retain highly skilled personnel for all areas of our organization. Competition in our industry for qualified employees is intense, and certain of our competitors have directly targeted, and may continue to target, our employees. In addition, our compensation arrangements, such as our equity award programs, may not always be successful in attracting new employees and retaining and motivating our existing employees. Restrictive immigration policy and regulatory changes may also affect our ability to hire, mobilize, or retain some of our global talent. All of our executive officers and key employees are at-will employees, and we do not maintain any key-person life insurance policies.

In addition, we believe that our corporate culture fosters innovation, creativity, and teamwork. As our organization grows and evolves, we may need to adapt our corporate culture and work environments to ever-changing circumstances, such as during times of a natural disaster or pandemic, and these changes could affect our ability to compete effectively or have an adverse effect on our corporate culture. Under our hybrid work models, we may experience increased costs and/or disruption, in addition to potential effects on our ability to operate effectively and maintain our corporate culture.

ITEM 1B. UNRESOLVED STAFF COMMENTS

Not applicable.

ITEM 1C. CYBERSECURITY

We maintain a comprehensive process for identifying, assessing, and managing material risks from cybersecurity threats as part of our broader risk management system and processes. We obtain input, as appropriate, for our cybersecurity risk management program on the security industry and threat trends from multiple external experts and internal threat intelligence teams. Teams of dedicated privacy, safety, and security professionals oversee cybersecurity risk management and mitigation, incident prevention, detection, and remediation. Leadership for these teams are professionals with deep cybersecurity

expertise across multiple industries, including our Vice President of Privacy, Safety, and Security Engineering. Our executive leadership team, along with input from the above teams, are responsible for our overall enterprise risk management system and processes and regularly consider cybersecurity risks in the context of other material risks to the company.

As part of our cybersecurity risk management system, our incident management teams track and log privacy and security incidents across Alphabet, our vendors, and other third-party service providers to remediate and resolve any such incidents. Significant incidents are reviewed regularly by a cross-functional working group to determine whether further escalation is appropriate. Any incident assessed as potentially being or potentially becoming material is immediately escalated for further assessment, and then reported to designated members of our senior management. We consult with outside counsel as appropriate, including on materiality analysis and disclosure matters, and our

senior management makes the final materiality determinations and disclosure and other compliance decisions. Our management apprises Alphabet's independent public accounting firm of matters and any relevant developments.

The Audit and Compliance Committee has oversight responsibility for risks and incidents relating to cybersecurity threats, including compliance with disclosure requirements, cooperation with law enforcement, and related effects on financial and other risks, and it reports any findings and recommendations, as appropriate, to the full Board for consideration. Senior management regularly discusses cyber risks and trends and, should they arise, any material incidents with the Audit and Compliance Committee. Internal Audit maintains a dedicated cybersecurity auditing team that independently tests our cybersecurity controls.

Our business strategy, results of operations and financial condition have not been materially affected by risks from cybersecurity threats, including as a result of previously identified cybersecurity incidents, but we cannot provide assurance that they will not be materially affected in the future by such risks or any future material incidents. For more information on our cybersecurity related risks, see Item 1A Risk Factors of this Annual Report on Form 10-K.

ITEM 2. PROPERTIES

Our headquarters are located in Mountain View, California. We own and lease office facilities and data centers around the world, primarily in Asia, Europe, and North America. We believe our existing facilities are in good condition and suitable for the conduct of our business.

ITEM 3. LEGAL PROCEEDINGS

For a description of our material pending legal proceedings, see Legal Matters in Note 10 of the Notes to Consolidated Financial Statements included in Part II, Item 8 of this Annual Report on Form 10-K, which is incorporated herein by reference.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS, AND ISSUER PURCHASES OF EQUITY SECURITIES

As of October 2, 2015, Alphabet Inc. became the successor issuer of Google Inc. pursuant to Rule 12g-3(a) under the Exchange Act. Our Class A stock has been listed on the Nasdaq Global Select Market under the symbol "GOOG" since August 19, 2004, and under the symbol "GOOGL" since April 3, 2014. Prior to August 19, 2004, there was no public market for our stock. Our Class B stock is neither listed nor traded. Our Class C stock has been listed on the Nasdaq Global Select Market under the symbol "GOOG" since April 3, 2014.

Holders of Record

As of December 31, 2023, there were approximately 7,305 and 1,757 stockholders of record of our Class A stock and Class C stock, respectively. Because many of our shares of Class A stock and Class C stock are held by brokers and other institutions on behalf of stockholders, we are unable to estimate the total number of stockholders represented by these record holders. As of December 31, 2023, there were approximately 59 stockholders of record of our Class B stock.

Dividend Policy

We have never declared or paid any cash dividend on our common or capital stock. The primary use of capital continues to be to invest for the long-term growth of the business. We regularly evaluate our cash and capital structure, including the size, pace, and form of capital return to stockholders.

Issuer Purchases of Equity Securities

The following table presents information with respect to Alphabet's repurchases of Class A and Class C stock during the quarter ended December 31, 2023:

Period	Total Number of Class A Shares Purchased (in thousands) [1]	Total Number of Class C Shares Purchased (in thousands) [1]	Average Price Paid per Class A Share[2]	Average Price Paid per Class C Share[2]	Total Number of Shares Purchased as Part of Publicly Announced Programs (in thousands) [1]	Approximate Dollar Value of Shares that May Yet Be Purchased Under the Program (in millions)
October 1 - 31	9,923	38,687	$ 134.66	$ 135.65	48,610	$ 45,736
November 1 - 30	9,197	28,198	$ 134.53	$ 135.16	37,395	$ 40,725
December 1 - 31	7,502	24,760	$ 135.76	$ 136.37	32,262	$ 36,347
Total	26,622	91,645			118,267	

(1) Repurchases are being executed from time to time, subject to general business and market conditions and other investment opportunities, through open market purchases or privately negotiated transactions, including through Rule 10b5-1 plans. The repurchase program does not have an expiration date. For additional information related to share repurchases, see Note 11 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

(2) Average price paid per share includes costs associated with the repurchases.

Stock Performance Graphs

The graph below matches Alphabet Inc. Class A's cumulative five-year total stockholder return on common stock with the cumulative total returns of the S&P 500 index, the NASDAQ Composite index, and the RDG Internet Composite index. The graph tracks the performance of a $100 investment in our common stock and in each index (with the reinvestment of all dividends) from December 31, 2018, to December 31, 2023. The returns shown are based on historical results and are not intended to suggest future performance.

COMPARISON OF CUMULATIVE 5-YEAR TOTAL RETURN*

ALPHABET INC. CLASS A COMMON STOCK

Among Alphabet Inc., the S&P 500 Index, the

NASDAQ Composite Index, and the RDG Internet Composite Index

2832

*$100 invested on December 31, 2018, in stock or index, including reinvestment of dividends.

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The graph below matches Alphabet Inc. Class C's cumulative five-year total stockholder return on capital stock with the cumulative total returns of the S&P 500 index, the NASDAQ Composite index, and the RDG Internet Composite index. The graph tracks the performance of a $100 investment in our Class C capital stock and in each index (with the reinvestment of all dividends) from December 31, 2018, to December 31, 2023. The returns shown are based on historical results and are not intended to suggest future performance.

COMPARISON OF CUMULATIVE 5-YEAR TOTAL RETURN*

ALPHABET INC. CLASS C CAPITAL STOCK

Among Alphabet Inc., the S&P 500 Index, the

NASDAQ Composite Index, and the RDG Internet Composite Index

3673

*$100 invested on December 31, 2018, in stock or in index, including reinvestment of dividends.

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ITEM 6. [Reserved]

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

Please read the following discussion and analysis of our financial condition and results of operations together with "Note about Forward-Looking Statements," Part I, Item 1 "Business," Part I, Item 1A "Risk Factors," and our consolidated financial statements and related notes included under Item 8 of this Annual Report on Form 10-K.

The following section generally discusses 2023 results compared to 2022 results. Discussion of 2022 results compared to 2021 results to the extent not included in this report can be found in Item 7 of our 2022 Annual Report on Form 10-K.

Understanding Alphabet's Financial Results

Alphabet is a collection of businesses — the largest of which is Google. We report Google in two segments, Google Services and Google Cloud; we also report all non-Google businesses collectively as Other Bets. For additional information on our segments, see Part I, Item 1 Business and Note 15 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Trends in Our Business and Financial Effect

The following long-term trends have contributed to the results of our consolidated operations, and we anticipate that they will continue to affect our future results:

- **Users' behaviors and advertising continue to shift online as the digital economy evolves.**

The continuing evolution of the online world has contributed to the growth of our business and our revenues since inception. We expect that this evolution will continue to benefit our business and our revenues, although at a slower pace than we have experienced historically, in particular after the outsized growth in our advertising revenues during the COVID-19 pandemic. In addition, we face increasing competition for user engagement and advertisers, which may affect our revenues.

- **Users continue to access our products and services using diverse devices and modalities, which allows for new advertising formats that may benefit our revenues but adversely affect our margins.**

Our users are accessing our products and services via diverse devices and modalities, such as smartphones, wearables, connected TVs, and smart home devices, and want to be able to be connected no matter where they are or what they are doing. We are focused on expanding our products and services to stay in front of these trends in order to maintain and grow our business.

We benefit from advertising revenues generated from different channels, including mobile, and newer advertising formats. The margins from these channels and newer products have generally been lower than those from traditional desktop search. Additionally, as the market for a particular device type or modality matures, our advertising revenues may be affected. For example, changing dynamics within the global smartphone market, such as increased market saturation in developed countries, can affect our mobile advertising revenues.

We expect TAC paid to our distribution partners and Google Network partners to increase as our revenues grow and TAC as a percentage of our advertising revenues ("TAC rate") to be affected by changes in device mix; geographic mix; partner agreement terms; partner mix; the percentage of queries channeled through paid access points; product mix; the relative revenue growth rates of advertising revenues from different channels; and revenue share terms.

We expect these trends to continue to affect our revenues and put pressure on our margins.

- **As online advertising evolves, we continue to expand our product offerings, which may affect our monetization.**

As interactions between users and advertisers change, and as online user behavior evolves, we continue to expand our product offerings to serve these changing needs, which may affect our monetization. For example, revenues from ads on YouTube and Google Play monetize at a lower rate than our traditional search ads. We also expect to continue to incorporate AI innovations into our products, such as AI in Search, that could affect our monetization trends. When developing new products and services we generally focus first on user experience and then on monetization.

- **As users in developing economies increasingly come online, our revenues from international markets continue to increase, and may require continued investments. In addition, movements in foreign exchange rates affect such revenues.**

The shift to online, as well as the advent of the multi-device world, has brought opportunities outside of the U.S., including in emerging markets, such as India. We continue to invest heavily and develop localized versions of our products and advertising programs relevant to our users in these markets. This has led to a trend of increased revenues from emerging markets. We expect that our results will continue to be affected by our performance in these markets, particularly as low-cost mobile devices become more available. This trend could affect our revenues as developing markets initially monetize at a lower rate than more mature markets.

International revenues represent a significant portion of our revenues and are subject to fluctuations in foreign currency exchange rates relative to the U.S. dollar. While we have a foreign exchange risk management program designed to reduce our exposure to these fluctuations, this program does not fully offset their effect on our revenues and earnings.

- **The revenues that we derive beyond advertising are increasing and may adversely affect our margins.**

Revenues from cloud, consumer subscriptions, platforms, and devices, which may have differing characteristics than our advertising revenues, have grown over time, and we expect this trend to continue as we focus on expanding our products and services. The margins on these revenues vary significantly and are generally lower than the margins on our advertising revenues. For example, sales of our devices adversely affect our consolidated margins due to pressures on pricing and higher cost of sales.

- **As we continue to serve our users and expand our businesses, we will invest heavily in operating and capital expenditures.**

We continue to make significant research and development investments in areas of strategic focus as we seek to develop new, innovative offerings, improve our existing offerings, and rapidly and responsibly deploy AI across our businesses. We also expect to increase, relative to 2023, our investment in our technical infrastructure, including servers, network equipment, and data centers, to support the growth of our business and our long-term initiatives, in particular in support of AI products and services. In addition, acquisitions and strategic investments contribute to the breadth and depth of our offerings, expand our expertise in engineering and other functional areas, and build strong partnerships around strategic initiatives.

- **We continue to face an evolving regulatory environment, and we are subject to claims, lawsuits, investigations, and other forms of potential legal liability, which could affect our business practices and financial results.**

Changes in social, political, economic, tax, and regulatory conditions or in laws and policies governing a wide range of topics and related legal matters, including investigations, lawsuits, and regulatory actions, have resulted in fines and caused us to change our business practices. As these global trends continue, our cost of doing business may increase, our ability to pursue certain business models or offer certain products or services may be limited, and we may need to change our business practices to comply with evolving regulatory and legal matters. Examples include the antitrust complaints filed by the U.S. Department of Justice and a number of state Attorneys General; legislative proposals and pending litigation in the U.S., EU, and around the world that could diminish or eliminate safe harbor protection for websites and online platforms; and the Digital Markets Act and Digital Services Act in Europe and various legislative proposals in the U.S. focused on large technology platforms. For additional information, see Item 1A Risk Factors and Legal Matters in Note 10 of the Notes to Consolidated Financial Statements included in Part II, Item 8 of this Annual Report on Form 10-K.

- **Our employees are critical to our success and we expect to continue investing in them.**

Our employees are among our best assets and are critical for our continued success. We expect to continue hiring talented employees around the globe and to provide competitive compensation programs.

For additional information, see Culture and Workforce in Part I, Item 1 Business of this Annual Report on Form 10-K.

Revenues and Monetization Metrics

We generate revenues by delivering relevant, cost-effective online advertising; cloud-based solutions that provide enterprise customers of all sizes with infrastructure and platform services as well as communication and collaboration tools; sales of other products and services, such as apps and in-app purchases, and devices; and fees received for consumer subscription-based products. For additional information on how we recognize revenue, see Note 1 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

In addition to the long-term trends and their financial effect on our business discussed above, fluctuations in our revenues have been, and may continue to be, affected by a combination of general factors, including:

- changes in foreign currency exchange rates;

- changes in pricing, such as those resulting from changes in fee structures, discounts, and customer incentives;
- general economic conditions and various external dynamics, including geopolitical events, regulations, and other measures and their effect on advertiser, consumer, and enterprise spending;
- new product and service launches; and
- seasonality.

Additionally, fluctuations in our revenues generated from advertising ("Google advertising"), revenues from other sources ("Google subscriptions, platforms, and devices revenues"), Google Cloud, and Other Bets revenues have been, and may continue to be, affected by other factors unique to each set of revenues, as described below.

Google Services

Google Services revenues consist of Google advertising as well as Google subscriptions, platforms, and devices revenues.

Google Advertising

Google advertising revenues are comprised of the following:

- Google Search & other, which includes revenues generated on Google search properties (including revenues from traffic generated by search distribution partners who use Google.com as their default search in browsers, toolbars, etc.), and other Google owned and operated properties like Gmail, Google Maps, and Google Play;
- YouTube ads, which includes revenues generated on YouTube properties; and
- Google Network, which includes revenues generated on Google Network properties participating in AdMob, AdSense, and Google Ad Manager.

We use certain metrics to track how well traffic across various properties is monetized as it relates to our advertising revenues: paid clicks and cost-per-click pertain to traffic on Google Search & other properties, while impressions and cost-per-impression pertain to traffic on our Google Network properties.

Paid clicks represent engagement by users and include clicks on advertisements by end-users on Google search properties and other Google owned and operated properties including Gmail, Google Maps, and Google Play. Cost-per-click is defined as click-driven revenues divided by our total number of paid clicks and represents the average amount we charge advertisers for each engagement by users.

Impressions include impressions displayed to users on Google Network properties participating primarily in AdMob, AdSense, and Google Ad Manager. Cost-per-impression is defined as impression-based and click-based revenues divided by our total number of impressions, and represents the average amount we charge advertisers for each impression displayed to users.

As our business evolves, we periodically review, refine, and update our methodologies for monitoring, gathering, and counting the number of paid clicks and the number of impressions, and for identifying the revenues generated by the corresponding click and impression activity.

Fluctuations in our advertising revenues, as well as the change in paid clicks and cost-per-click on Google Search & other properties and the change in impressions and cost-per-impression on Google Network properties and the correlation between these items have been, and may continue to be, affected by factors in addition to the general factors described above, such as:

- advertiser competition for keywords;

- changes in advertising quality, formats, delivery or policy;
- changes in device mix;
- seasonal fluctuations in internet usage, advertising expenditures, and underlying business trends, such as traditional retail seasonality; and
- traffic growth in emerging markets compared to more mature markets and across various verticals and channels.

Google Subscriptions, Platforms, and Devices

Google subscriptions, platforms, and devices revenues are comprised of the following:

- consumer subscriptions, which primarily include revenues from YouTube services, such YouTube TV, YouTube Music and Premium, and NFL Sunday Ticket, as well as Google One;
- platforms, which primarily include revenues from Google Play from the sales of apps and in-app purchases;
- devices, which primarily include sales of the Pixel family of devices; and
- other products and services.

Fluctuations in our Google subscriptions, platforms, and devices revenues have been, and may continue to be, affected by factors in addition to the general factors described above, such as changes in customer usage and demand, number of subscribers, and fluctuations in the timing of product launches.

Google Cloud

Google Cloud revenues are comprised of the following:

- Google Cloud Platform, which generates consumption-based fees and subscriptions for infrastructure, platform, and other services. These services provide access to solutions such as cybersecurity, databases, analytics, and AI offerings including our AI infrastructure, Vertex AI platform, and Duet AI for Google Cloud;
- Google Workspace, which includes subscriptions for cloud-based communication and collaboration tools for enterprises, such as Calendar, Gmail, Docs, Drive, and Meet, with integrated features like Duet AI in Google Workspace; and
- other enterprise services.

Fluctuations in our Google Cloud revenues have been, and may continue to be, affected by factors in addition to the general factors described above, such as customer usage.

Other Bets

Revenues from Other Bets are generated primarily from the sale of healthcare-related services and internet services.

Costs and Expenses

Our cost structure has two components: cost of revenues and operating expenses. Our operating expenses include costs related to R&D, sales and marketing, and general and administrative functions. Certain of our costs and expenses, including those associated with the operation of our technical infrastructure as well as components of our operating expenses, are generally less variable in nature and may not correlate to changes in revenue. Additionally, fluctuations in compensation expenses may not directly correlate with changes in headcount, in particular due to annual stock-based compensation (SBC) awards that generally vest over four years.

Cost of Revenues

Cost of revenues is comprised of TAC and other costs of revenues.

- TAC includes:
    - amounts paid to our distribution partners who make available our search access points and services. Our distribution partners include browser providers, mobile carriers, original equipment manufacturers, and software developers; and
    - amounts paid to Google Network partners primarily for ads displayed on their properties.

- Other cost of revenues primarily includes:
  - compensation expense related to our data centers and other operations such as content review and customer and product support;
  - content acquisition costs, which are payments to content providers from whom we license video and other content for distribution on YouTube and Google Play (we pay fees to these content providers based on revenues generated or a flat fee);
  - depreciation expense related to our technical infrastructure; and
  - inventory and other costs related to the devices we sell.

TAC as a percentage of revenues generated from ads placed on Google Network properties are significantly higher than TAC as a percentage of revenues generated from ads placed on Google Search & other properties, because most of the advertiser revenues from ads served on Google Network properties are paid as TAC to our Google Network partners.

Operating Expenses

Operating expenses are generally incurred during our normal course of business, which we categorize as either R&D, sales and marketing, or general and administrative.

The main components of our R&D expenses are:

- compensation expenses for engineering and technical employees responsible for R&D related to our existing and new products and services;
- depreciation; and
- third-party services fees primarily relating to consulting and outsourced services in support of our engineering and product development efforts.

The main components of our sales and marketing expenses are:

- compensation expenses for employees engaged in sales and marketing, sales support, and certain customer service functions; and
- spending relating to our advertising and promotional activities in support of our products and services.

The main components of our general and administrative expenses are:

- compensation expenses for employees in finance, human resources, information technology, legal, and other administrative support functions;
- expenses relating to legal matters, including certain fines and settlements; and
- third-party services fees, including audit, consulting, outside legal, and other outsourced administrative services.

Other Income (Expense), Net

OI&E, net primarily consists of interest income (expense), the effect of foreign currency exchange gains (losses), net gains (losses) and impairment on our marketable and non-marketable securities, performance fees, and income (loss) and impairment from our equity method investments.

For additional information, including how we account for our investments and factors that can drive fluctuations in the value of our investments, see Note 1 and Note 3 of the Notes to Consolidated Financial Statements included in Part II, Item 8 as well as Item 7A Quantitative and Qualitative Disclosures About Market Risk of this Annual Report on Form 10-K.

Provision for Income Taxes

Provision for income taxes represents the estimated amount of federal, state, and foreign income taxes incurred in the U.S. and the many jurisdictions in which we operate. The provision includes the effect of reserve provisions and changes to reserves that are considered appropriate as well as the related net interest and penalties.

For additional information, including a reconciliation of the U.S. federal statutory rate to our effective tax rate, see Note 14 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Executive Overview

The following table summarizes our consolidated financial results (in millions, except for per share information and percentages):

	Year Ended December 31,			
	2022	2023	$ Change	% Change
Consolidated revenues	$ 282,836	$ 307,394	$ 24,558	9 %
Change in consolidated constant currency revenues[1]				10 %
Cost of revenues	$ 126,203	$ 133,332	$ 7,129	6 %
Operating expenses	$ 81,791	$ 89,769	$ 7,978	10 %
Operating income	$ 74,842	$ 84,293	$ 9,451	13 %
Operating margin	26 %	27 %		1 %
Other income (expense), net	$ (3,514)	$ 1,424	$ 4,938	NM
Net income	$ 59,972	$ 73,795	$ 13,823	23 %
Diluted EPS	$ 4.56	$ 5.80	$ 1.24	27 %

NM = Not Meaningful

(1) See "Use of Non-GAAP Constant Currency Information" below for details relating to our use of constant currency information.

- Revenues were $307.4 billion, an increase of 9% year over year, primarily driven by an increase in Google Services revenues of $19.0 billion, or 8%, and an increase in Google Cloud revenues of $6.8 billion, or 26%.
- Total constant currency revenues, which exclude the effect of hedging, increased 10% year over year.
- Cost of revenues was $133.3 billion, an increase of 6% year over year, primarily driven by increases in content acquisition costs, compensation expenses, and TAC. The increase in compensation expenses included charges related to employee severance associated with the reduction in our workforce. Additionally, cost of revenues benefited from a reduction in depreciation due to the change in estimated useful lives of our servers and network equipment.
- Operating expenses were $89.8 billion, an increase of 10% year over year, primarily driven by an increase in compensation expenses and charges related to our office space optimization efforts. The increase in compensation expenses was largely the result of charges related to employee severance associated with the reduction in our workforce and an increase in SBC expense. Operating expenses benefited from the change in the estimated useful lives of our servers and certain network equipment.

Other Information:

- In January 2023, we announced a reduction of our workforce, and as a result we recorded employee severance and related charges of $2.1 billion for the year ended December 31, 2023. In addition, we are taking actions to optimize our global office space. As a result, exit charges recorded during the year ended December 31, 2023, were $1.8 billion. In addition to these exit charges, for the

year ended December 31, 2023, we incurred $269 million in accelerated rent and accelerated depreciation. For additional information, see Note 8 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

- In January 2023, we completed an assessment of the useful lives of our servers and network equipment, resulting in a change in the estimated useful life of our servers and certain network equipment to six years. The effect of this change was a reduction in depreciation expense of $3.9 billion for the year ended December 31, 2023, recognized primarily in cost of revenues and R&D expenses. For additional information, see Note 1 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

- On July 21, 2023, the IRS announced a rule change allowing taxpayers to temporarily apply the regulations in effect prior to 2022 related to U.S. federal foreign tax credits. This announcement applies to foreign taxes paid or accrued in the fiscal years 2022 and 2023. A cumulative one-time adjustment applicable to the prior period for this tax rule change was recorded in 2023 and is reflected in our effective tax rate of 13.9% for the year ended December 31, 2023.
- Repurchases of Class A and Class C shares were $62.2 billion for the year ended December 31, 2023. For additional information, see Note 11 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.
- Operating cash flow was $101.7 billion for the year ended December 31, 2023.
- Capital expenditures, which primarily reflected investments in technical infrastructure, were $32.3 billion for the year ended December 31, 2023.
- As of December 31, 2023, we had 182,502 employees.

Financial Results

Revenues

The following table presents revenues by type (in millions):

	Year Ended December 31,	
	2022	2023
Google Search & other	$ 162,450	$ 175,033
YouTube ads	29,243	31,510
Google Network	32,780	31,312
Google advertising	224,473	237,855
Google subscriptions, platforms, and devices	29,055	34,688
Google Services total	253,528	272,543
Google Cloud	26,280	33,088
Other Bets	1,068	1,527
Hedging gains (losses)	1,960	236
Total revenues	$ 282,836	$ 307,394

Google Services

Google advertising revenues

Google Search & other

Google Search & other revenues increased $12.6 billion from 2022 to 2023. The overall growth was driven by interrelated factors including increases in search queries resulting from growth in user adoption and usage on mobile devices; growth in advertiser spending; and improvements we have made in ad formats and delivery.

YouTube ads

YouTube ads revenues increased $2.3 billion from 2022 to 2023. The growth was driven by our brand and direct response advertising products, both of which benefited from increased spending by our advertisers.

Google Network

Google Network revenues decreased $1.5 billion from 2022 to 2023, primarily driven by a decrease in Google Ad Manager and AdSense revenues.

Monetization Metrics

The following table presents changes in monetization metrics for Google Search & other revenues (paid clicks and cost-per-click) and Google Network revenues (impressions and cost-per-impression), expressed as a percentage, from 2022 to 2023:

Google Search & other	
Paid clicks change	7 %
Cost-per-click change	1 %
Google Network	
Impressions change	(5)%
Cost-per-impression change	0 %

Changes in paid clicks and impressions are driven by a number of interrelated factors, including changes in advertiser spending; ongoing product and policy changes; and, as it relates to paid clicks, fluctuations in search queries resulting from changes in user adoption and usage, primarily on mobile devices.

Changes in cost-per-click and cost-per-impression are driven by a number of interrelated factors including changes in device mix, geographic mix, advertiser spending, ongoing product and policy changes, product mix, property mix, and changes in foreign currency exchange rates.

Google subscriptions, platforms, and devices

Google subscriptions, platforms, and devices revenues increased $5.6 billion from 2022 to 2023 primarily driven by growth in subscriptions, largely for YouTube services. The growth in YouTube services was primarily due to an increase in paid subscribers.

Google subscriptions, platforms, and devices revenues increased $1.0 billion from 2021 to 2022 primarily driven by growth in subscription and device revenues, partially offset by a decrease in platform revenues. The growth in subscriptions was largely for YouTube services, primarily due to an increase in paid subscribers. The growth in device revenues was primarily driven by increased sales of Pixel devices. The decrease in platform revenues was primarily due to Google Play, driven by the fee structure changes we announced in 2021 as well as a decrease in buyer spending. Additionally, the overall increase in Google subscriptions, platforms, and devices revenues was adversely affected by the unfavorable effect of foreign currency exchange rates.

Google Cloud

Google Cloud revenues increased $6.8 billion from 2022 to 2023. Growth was primarily driven by Google Cloud Platform followed by Google Workspace offerings. Google Cloud's infrastructure and platform services were the largest drivers of growth in Google Cloud Platform.

Revenues by Geography

The following table presents revenues by geography as a percentage of revenues, determined based on the addresses of our customers:

	Year Ended December 31,	
	2022	2023
United States	48 %	47 %
EMEA	29 %	30 %
APAC	16 %	17 %
Other Americas	6 %	6 %
Hedging gains (losses)	1 %	0 %

For additional information, see Note 2 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Use of Non-GAAP Constant Currency Information

International revenues, which represent a significant portion of our revenues, are generally transacted in multiple currencies and therefore are affected by fluctuations in foreign currency exchange rates.

The effect of currency exchange rates on our business is an important factor in understanding period-to-period comparisons. We use non-GAAP constant currency revenues ("constant currency revenues") and non-GAAP percentage change in constant currency revenues ("percentage change in constant currency revenues") for financial and operational decision-making and as a means to evaluate period-to-period comparisons. We believe the presentation of results on a constant currency basis in addition to U.S. Generally Accepted Accounting Principles (GAAP) results helps improve the ability to understand our performance, because it excludes the effects of foreign currency volatility that are not indicative of our core operating results.

Constant currency information compares results between periods as if exchange rates had remained constant period over period. We define constant currency revenues as revenues excluding the effect of foreign currency exchange rate movements ("FX Effect") as well as hedging activities, which are recognized at the consolidated level. We use constant currency revenues to determine the constant currency revenue percentage change on a year-on-year basis. Constant currency revenues are calculated by translating current period revenues using prior year comparable period exchange rates, as well as excluding any hedging effects realized in the current period.

Constant currency revenue percentage change is calculated by determining the change in current period revenues over prior year comparable period revenues where current period foreign currency revenues are translated using prior year comparable period exchange rates and hedging effects are excluded from revenues of both periods.

These results should be considered in addition to, not as a substitute for, results reported in accordance with GAAP. Results on a constant currency basis, as we present them, may not be comparable to similarly titled measures used by other companies and are not a measure of performance presented in accordance with GAAP.

The following table presents the foreign currency exchange effect on international revenues and total revenues (in millions, except percentages):

	Year Ended December 31,		Year Ended December 31, 2023					
					% Change from Prior Period			
	2022	2023	Less FX Effect	Constant Currency Revenues	As Reported	Less Hedging Effect	Less FX Effect	Constant Currency Revenues
United States	$ 134,814	$ 146,286	$ 0	$ 146,286	9 %		0 %	9 %
EMEA	82,062	91,038	460	90,578	11 %		1 %	10 %
APAC	47,024	51,514	(1,759)	53,273	10 %		(3)%	13 %
Other Americas	16,976	18,320	(654)	18,974	8 %		(4)%	12 %
Revenues, excluding hedging effect	280,876	307,158	(1,953)	309,111	9 %		(1)%	10 %
Hedging gains (losses)	1,960	236						
Total revenues[1]	$ 282,836	$ 307,394		$ 309,111	9 %	0 %	(1)%	10 %

(1) Total constant currency revenues of $309.1 billion for 2023 increased $28.2 billion compared to $280.9 billion in revenues, excluding hedging effect, for 2022.

EMEA revenue growth was favorably affected by changes in foreign currency exchange rates, primarily due to the U.S. dollar weakening relative to the Euro, partially offset by the U.S. dollar strengthening relative to the Turkish lira.

APAC revenue growth was unfavorably affected by changes in foreign currency exchange rates, primarily due to the U.S. dollar strengthening relative to the Japanese yen.

Other Americas revenue growth was unfavorably affected by changes in foreign currency exchange rates, primarily due to the U.S. dollar strengthening relative to the Argentine peso.

Costs and Expenses

Cost of Revenues

The following table presents cost of revenues, including TAC (in millions, except percentages):

	Year Ended December 31,		
	2021	2022	2023
TAC	$ 45,566	$ 48,955	$ 50,886
Other cost of revenues	65,373	77,248	82,446
Total cost of revenues	$ 110,939	$ 126,203	$ 133,332
Total cost of revenues as a percentage of revenues	43 %	45 %	43 %

Cost of revenues increased $7.1 billion from 2022 to 2023 due to an increase in other cost of revenues and TAC of $5.2 billion and $1.9 billion, respectively.

The increase in TAC from 2022 to 2023 was largely due to an increase in TAC paid to distribution partners, primarily driven by growth in revenues subject to TAC. The TAC rate decreased from 21.8% to 21.4% from 2022 to 2023 primarily due to a revenue mix shift from Google Network properties to Google Search & other properties. The TAC rate on Google Search & other revenues and the TAC rate on Google Network revenues were both substantially consistent from 2022 to 2023.

The increase in other cost of revenues from 2022 to 2023 was primarily due to increases in content acquisition costs, largely for YouTube, and compensation expenses, which included $479 million of charges related to employee severance associated with the reduction in our workforce. Additionally, other cost of revenues benefited from a reduction in depreciation expense due to the change in estimated useful lives of our servers and network equipment.

The increase in other cost of revenues of $11.9 billion from 2021 to 2022 was primarily due to increases in device costs, compensation expenses, depreciation, and equipment-related expenses.

Research and Development

The following table presents R&D expenses (in millions, except percentages):

	Year Ended December 31,	
	2022	2023
Research and development expenses	$ 39,500	$ 45,427
Research and development expenses as a percentage of revenues	14 %	15 %

R&D expenses increased $5.9 billion from 2022 to 2023 primarily driven by an increase in compensation expenses of $2.9 billion, $870 million in charges related to our office space optimization efforts, and an increase in depreciation expense of $722 million. The $2.9 billion increase in compensation expenses was largely the result of a 4% increase in average headcount, after adjusting for roles affected by the reduction in our workforce, and an increase in SBC expense. Additionally, the increase in compensation expenses included $848 million in employee severance charges associated with the reduction in our workforce. The $722 million increase in depreciation expense reflected an offsetting benefit of the change in the estimated useful lives of our servers and network equipment.

Sales and Marketing

The following table presents sales and marketing expenses (in millions, except percentages):

	Year Ended December 31,	
	2022	2023
Sales and marketing expenses	$ 26,567	$ 27,917
Sales and marketing expenses as a percentage of revenues	9 %	9 %

Sales and marketing expenses increased $1.4 billion from 2022 to 2023, primarily driven by an increase in compensation expenses of $1.6 billion, partially offset by a decrease in advertising and promotional activities of $441 million. The $1.6 billion increase in compensation expenses was largely the result of $497 million in employee severance charges associated with the reduction in our workforce in addition to a combination of other factors, none of which were individually significant.

General and Administrative

The following table presents general and administrative expenses (in millions, except percentages):

	Year Ended December 31,	
	2022	2023
General and administrative expenses	$ 15,724	$ 16,425
General and administrative expenses as a percentage of revenues	6 %	5 %

General and administrative expenses increased $701 million from 2022 to 2023, primarily driven by an increase in compensation expenses of $416 million, which was largely the result of $264 million in employee severance charges associated with the reduction in our workforce in addition to a combination of other factors, none of which were individually significant.

Segment Profitability

The following table presents segment operating income (loss) (in millions).

	Year Ended December 31,	
	2022	2023
Operating income (loss):		
Google Services	$ 82,699	$ 95,858
Google Cloud	(1,922)	1,716
Other Bets	(4,636)	(4,095)
Alphabet-level activities[1]	(1,299)	(9,186)
Total income from operations	$ 74,842	$ 84,293

(1) In addition to the costs included in Alphabet-level activities, hedging gains (losses) related to revenue were $2.0 billion and $236 million in 2022 and 2023, respectively. For the year ended December 31, 2023, Alphabet-level activities include charges related to the reduction in force and our office space optimization efforts totaling $3.9 billion. In addition, for the year ended December 31, 2023, we incurred $269 million in accelerated rent and accelerated depreciation. For additional information relating to our workforce reduction and other initiatives, see Note 8 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K. For additional information relating to our segments, see Note 15 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Google Services

Google Services operating income increased $13.2 billion from 2022 to 2023. The increase in operating income was primarily driven by an increase in revenues, partially offset by an increase in content acquisition costs and compensation expenses including an increase in SBC expense. Additionally, operating income benefited from a reduction in costs driven by the change in the estimated useful lives of our servers and certain network equipment.

Google Cloud

Google Cloud operating income of $1.7 billion for 2023 compared to an operating loss of $1.9 billion for 2022 represents an increase of $3.6 billion. The increase in operating income was primarily driven by an increase in revenues, partially offset by an increase in compensation expenses largely driven by headcount growth. Additionally, operating income benefited from a reduction in costs driven by the change in the estimated useful lives of our servers and certain network equipment.

Other Bets

Other Bets operating loss decreased $541 million from 2022 to 2023 primarily due to growth in revenues as well as a reduction in valuation-based compensation liabilities related to Other Bet companies.

Other Income (Expense), Net

The following table presents OI&E, (in millions):

	Year Ended December 31,	
	2022	2023
Interest income	$ 2,174	$ 3,865
Interest expense	(357)	(308)
Foreign currency exchange gain (loss), net	(654)	(1,238)
Gain (loss) on debt securities, net	(2,064)	(1,215)
Gain (loss) on equity securities, net	(3,455)	392
Performance fees	798	257
Income (loss) and impairment from equity method investments, net	(337)	(628)
Other	381	299
Other income (expense), net	$ (3,514)	$ 1,424

OI&E, net increased $4.9 billion from 2022 to 2023. The increase was primarily due to fluctuations in the value of equity securities reflecting market driven changes in the value of our marketable equity securities, investment specific event driven changes in our non-marketable equity securities, and increased interest income due to interest rates.

For additional information, see Note 7 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Provision for Income Taxes

The following table presents provision for income taxes (in millions, except for effective tax rate):

	Year Ended December 31,	
	2022	2023
Income before provision for income taxes	$ 71,328	$ 85,717
Provision for income taxes	$ 11,356	$ 11,922
Effective tax rate	15.9 %	13.9 %

In 2023, the Internal Revenue Services (IRS) issued a rule change allowing taxpayers to temporarily apply the regulations in effect prior to 2022 related to U.S. federal foreign tax credits, as well as a separate rule change with interim guidance on the capitalization and amortization of R&D expenses. A cumulative one-time adjustment applicable to the prior period for these tax rule changes was recorded in 2023.

The effective tax rate decreased from 2022 to 2023, reflecting the effect of the two tax rule changes described above, particularly the change related to foreign tax credits. The effect of these tax rule changes was partially offset by changes in uncertain tax benefits and a decrease in the U.S. federal Foreign Derived Intangible Income tax deduction.

The OECD is coordinating negotiations among more than 140 countries with the goal of achieving consensus around substantial changes to international tax policies, including the implementation of a minimum global effective tax rate of 15%. While various countries have implemented the legislation as of January 1, 2024, we do not expect a resulting material change to our income tax provision for the 2024 fiscal year. As additional jurisdictions enact such legislation, we expect our effective tax rate and cash tax payments could increase in future years.

Financial Condition

Cash, Cash Equivalents, and Marketable Securities

As of December 31, 2023, we had $110.9 billion in cash, cash equivalents, and short-term marketable securities. Cash equivalents and marketable securities are comprised of time deposits, money market funds, highly liquid government bonds, corporate debt securities, mortgage-backed and asset-backed securities, and marketable equity securities.

Sources, Uses of Cash and Related Trends

Our principal sources of liquidity are cash, cash equivalents, and marketable securities, as well as the cash flow that we generate from operations. The primary use of capital continues to be to invest for the long-term growth of the business. We regularly evaluate our cash and capital structure, including the size, pace, and form of capital return to stockholders.

The following table presents our cash flows (in millions):

	Year Ended December 31,	
	2022	2023
Net cash provided by operating activities	$ 91,495	$ 101,746
Net cash used in investing activities	$ (20,298)	$ (27,063)
Net cash used in financing activities	$ (69,757)	$ (72,093)

Cash Provided by Operating Activities

Our largest source of cash provided by operations are advertising revenues generated by Google Search & other properties, Google Network properties, and YouTube properties. In Google Services, we also generate cash through consumer subscriptions and the sale of apps and in-app purchases and devices. In Google Cloud we generate cash through consumption-based fees and subscriptions for infrastructure, platform, collaboration tools, and other cloud services.

Our primary uses of cash from operating activities include payments to distribution and Google Network partners, to employees for compensation, and to content providers. Other uses of cash from operating activities include payments to suppliers for devices, to tax authorities for income taxes, and other general corporate expenditures.

Net cash provided by operating activities increased from 2022 to 2023 due to the increase in cash received from customers, partially offset by increases in cash paid for cost of revenues and operating expenses.

Cash Used in Investing Activities

Cash provided by investing activities consists primarily of maturities and sales of investments in marketable and non-marketable securities. Cash used in investing activities consists primarily of purchases of marketable and non-marketable securities, purchases of property and equipment, and payments for acquisitions.

Net cash used in investing activities increased from 2022 to 2023 due to a decrease in maturities and sales of marketable securities, partially offset by a decrease in payments for acquisitions.

Cash Used in Financing Activities

Cash provided by financing activities consists primarily of proceeds from issuance of debt and proceeds from the sale of interests in consolidated entities. Cash used in financing activities consists primarily of repurchases of stock, net payments related to stock-based award activities, and repayments of debt.

Net cash used in financing activities increased from 2022 to 2023 due to an increase in repurchases of stock.

Liquidity and Material Cash Requirements

We expect existing cash, cash equivalents, short-term marketable securities, cash flows from operations and financing activities to continue to be sufficient to fund our operating activities and cash commitments for investing and financing activities for at least the next 12 months and thereafter for the foreseeable future.

Capital Expenditures and Leases

We make investments in land and buildings for data centers and offices and information technology assets through purchases of property and equipment and lease arrangements to provide capacity for the growth of our services and products.

Capital Expenditures

Our capital investments in property and equipment consist primarily of the following major categories:

- technical infrastructure, which consists of our investments in servers and network equipment for computing, storage, and networking requirements for ongoing business activities, including AI, (collectively referred to as our information technology assets) and data center land and building construction; and
- office facilities, ground-up development projects, and building improvements (also referred to as "fit-outs").

Construction in progress consists primarily of technical infrastructure and office facilities which have not yet been placed in service. The time frame from date of purchase to placement in service of these assets may extend from months to years. For example, our data center construction projects are generally multi-year projects with multiple phases, where we acquire land and buildings, construct buildings, and secure and install information technology assets.

During the years ended December 31, 2022 and 2023, we spent $31.5 billion and $32.3 billion on capital expenditures, respectively. We expect to increase, relative to 2023, our investment in our technical infrastructure, including servers, network equipment, and data centers, to support the growth of our business and our long-term initiatives, in particular in support of AI products and services. Depreciation of our property and equipment commences when the deployment of such assets are completed and are ready for our intended use. Land is not depreciated. For the years ended December 31, 2022 and 2023, our depreciation on property and equipment were $13.5 billion and $11.9 billion, respectively.

Leases

For the years ended December 31, 2022 and 2023, we recognized total operating lease assets of $4.4 billion and $2.9 billion, respectively. As of December 31, 2023, the amount of total future lease payments under operating leases, which had a weighted average remaining lease term of eight years, was $17.7 billion, of which $3.2 billion is short-term. As of December 31, 2023, we have entered into leases that have not yet commenced with future short-term and long-term lease payments of $657 million and $3.3 billion, that are not yet recorded on our Consolidated Balance Sheets. These leases will commence between 2024 and 2026 with non-cancelable lease terms of one to 25 years.

For the years ended December 31, 2022 and 2023, our operating lease expenses (including variable lease costs) were $3.7 billion and $4.5 billion, respectively. Finance lease costs were not material for the years ended December 31, 2022 and 2023. For additional information, see Note 4 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Financing

We have a short-term debt financing program of up to $10.0 billion through the issuance of commercial paper. Net proceeds from this program are used for general corporate purposes. As of December 31, 2023, we had no commercial paper outstanding.

As of December 31, 2023, we had $10.0 billion of revolving credit facilities, $4.0 billion expiring in April 2024 and $6.0 billion expiring in April 2028. The interest rates for all credit facilities are determined based on a formula using certain market rates, as well as our progress toward the achievement of certain sustainability goals. No amounts have been borrowed under the credit facilities.

As of December 31, 2023, we had senior unsecured notes outstanding with a total carrying value of $12.9 billion with short-term and long-term future interest payments of $214 million and $3.6 billion, respectively. For additional information, see Note 6 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

We primarily utilize contract manufacturers for the assembly of our servers used in our technical infrastructure and devices we sell. We have agreements where we may purchase components directly from suppliers and then supply these components to contract manufacturers for use in the assembly of the servers and devices. Certain of these arrangements result in a portion of the cash received from and paid to the contract manufacturers to be presented as financing activities in the Consolidated Statements of Cash Flows included in Item 8 of this Annual Report on Form 10-K.

Share Repurchase Program

During 2023 we repurchased and subsequently retired 528 million shares for $62.2 billion.

In April 2023, the Board of Directors of Alphabet authorized the company to repurchase up to an additional $70.0 billion of its Class A and Class C shares. As of December 31, 2023, $36.3 billion remains available for Class A and Class C share repurchases.

The following table presents Class A and Class C shares repurchased and subsequently retired (in millions):

	Year Ended December 31, 2022		Year Ended December 31, 2023	
	Shares	Amount	Shares	Amount
Class A share repurchases	61	$ 6,719	78	$ 9,316
Class C share repurchases	469	52,577	450	52,868
Total share repurchases[1]	530	$ 59,296	528	$ 62,184

[1] Shares repurchased include unsettled repurchases as of December 31, 2023.

For additional information, see Note 11 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

European Commission Fines

In 2017, 2018 and 2019, the EC announced decisions that certain actions taken by Google infringed European competition law and imposed fines of €2.4 billion ($2.7 billion as of June 27, 2017), €4.3 billion ($5.1 billion as of June 30, 2018), and €1.5 billion ($1.7 billion as of March 20, 2019), respectively. On September 14, 2022, the General Court reduced the 2018 fine from €4.3 billion to €4.1 billion. We subsequently filed an appeal to the European Court of Justice.

While each EC decision is under appeal, we included the fines in accrued expenses and other current liabilities on our Consolidated Balance Sheets as we provided bank guarantees (in lieu of a cash payment) for the fines. For additional information, see Note 10 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Taxes

As of December 31, 2023, we had income taxes payable of $4.2 billion, of which $2.1 billion was short-term, related to a one-time transition tax payable incurred as a result of the U.S. Tax Cuts and Jobs Act ("Tax Act"). As permitted by the Tax Act, we will pay the transition tax in annual interest-free installments through 2025. We also have long-term taxes payable of $6.3 billion primarily related to uncertain tax positions as of December 31, 2023.

Purchase Commitments and Other Contractual Obligations

As of December 31, 2023, we had material purchase commitments and other contractual obligations of $45.9 billion, of which $31.6 billion was short-term. These amounts primarily consist of purchase orders for certain technical infrastructure as well as the non-cancelable portion or the minimum cancellation fee in certain agreements related to commitments to purchase licenses, including content licenses, inventory and network capacity. For those agreements with variable terms, we do not estimate the non-cancelable obligation beyond any minimum quantities and/or pricing as of December 31, 2023. In certain instances, the amount of our contractual obligations may change based on the expected timing of order fulfillment from our suppliers. For more information related to our content licenses, see Note 10 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

In addition, we regularly enter into multi-year, non-cancellable agreements to purchase renewable energy and energy attributes, such as renewable energy certificates. These agreements do not include a minimum dollar commitment. The amounts to be paid under these agreements are based on the actual volumes to be generated and are not readily determinable.

Critical Accounting Estimates

We prepare our consolidated financial statements in accordance with GAAP. In doing so, we have to make estimates and assumptions. Our critical accounting estimates are those estimates that involve a significant level of uncertainty at the time the estimate was made, and changes in them have had or are reasonably likely to have a material effect on our financial condition or results of operations. Accordingly, actual results could differ materially from our estimates. We base our estimates on past experience and other assumptions that we believe are reasonable under the circumstances, and we evaluate these estimates on an ongoing basis. We have reviewed our critical accounting estimates with the Audit and Compliance Committee of our Board of Directors.

For a summary of significant accounting policies and the effect on our financial statements, see Note 1 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Fair Value Measurements of Non-Marketable Equity Securities

We measure certain financial instruments at fair value on a nonrecurring basis, consisting primarily of our non-marketable equity securities. These investments are accounted for under the measurement

alternative method ("the measurement alternative") and are measured at cost, less impairment, subject to upward and downward adjustments resulting from observable price changes for identical or similar investments of the same issuer. These adjustments require quantitative assessments of the fair value of our securities, which may require the use of unobservable inputs. Adjustments are determined primarily based on a market approach as of the transaction date and involve the use of estimates using the best information available, which may include cash flow projections or other available market data.

Non-marketable equity securities are also evaluated for impairment, based on qualitative factors including the companies' financial and liquidity position and access to capital resources, among others. When indicators of impairment exist, we prepare quantitative measurements of the fair value of our equity investments using a market approach or an income approach, which requires judgment and the use of unobservable inputs, including discount rates, investee revenues and costs, and comparable market data of private and public companies, among others.

When the quantitative remeasurements of fair value indicate an impairment exists, we write down the investment to its current fair value.

We also have compensation arrangements with payouts based on realized returns from certain investments, i.e. performance fees. We record compensation expense based on the estimated payouts on an ongoing basis, which may result in expense recognized before investment returns are realized and compensation is paid and may require the use of unobservable inputs.

Property and Equipment

We assess the reasonableness of the useful lives of our property and equipment periodically as well as when other changes occur, such as when there are changes to ongoing business operations, changes in the planned use and utilization of assets, or technological advancements, that could indicate a change in the period over which we expect to benefit from the assets.

Income Taxes

We are subject to income taxes in the U.S. and foreign jurisdictions. Significant judgment is required in evaluating our uncertain tax positions and determining our provision for income taxes.

Recording an uncertain tax position involves various qualitative considerations, including evaluation of comparable and resolved tax exposures, applicability of tax laws, and likelihood of settlement. We evaluate uncertain tax positions periodically, considering changes in facts and circumstances, such as new regulations or recent judicial opinions, as well as the status of audit activities by taxing authorities. Although we believe we have adequately reserved for our uncertain tax positions, no assurance can be given that the final tax outcome of these matters will not be different. To the extent that the final tax outcome of these matters is different than the amounts recorded, such differences will affect the provision for income taxes and the effective tax rate in the period in which such determination is made.

The provision for income taxes includes the effect of reserve provisions and changes to reserves as well as the related net interest and penalties. In addition, we are subject to the continuous examination of our income tax returns by the IRS and other tax authorities which may assert assessments against us. We regularly assess the likelihood of adverse outcomes resulting from these examinations and assessments to determine the adequacy of our provision for income taxes.

Loss Contingencies

We are regularly subject to claims, lawsuits, regulatory and government investigations, other proceedings, and consent orders involving competition, intellectual property, privacy, data security, tax and related compliance, labor and employment, commercial disputes, content generated by our users, goods and services offered by advertisers or publishers using our platforms, personal injury consumer protection, and other matters. Certain of these matters include speculative claims for substantial or indeterminate amounts of damages. We record a liability when we believe that it is probable that a loss has been incurred and the amount can be reasonably estimated. If we determine that a loss is reasonably possible and the loss or range of loss can be estimated, we disclose the possible loss in Note 10 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

We evaluate, on a regular basis, developments in our legal matters that could affect the amount of liability that has been previously accrued, and the matters and related reasonably possible losses disclosed, and make adjustments and changes to our disclosures. Significant judgment is required to determine both the likelihood and the estimated amount of a loss related to such matters. Until the final resolution of such matters, there may be an exposure to loss in excess of the amount recorded, and such amounts could be material.

Change in Accounting Estimate

In January 2023, we completed an assessment of the useful lives of our servers and network equipment resulting in a change in the estimated useful life of our servers and certain network equipment to six years. This change in accounting estimate was effective beginning fiscal year 2023. For additional information, see Note 1 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

We are exposed to financial market risks, including changes in foreign currency exchange rates, interest rates, and equity investment risks.

Foreign Currency Exchange Risk

We transact business globally in multiple currencies. International revenues, as well as costs and expenses denominated in foreign currencies, expose us to the risk of fluctuations in foreign currency exchange rates against the U.S. dollar. As discussed below, we enter into derivative instruments to hedge foreign currency risk. Principal currencies hedged included the Australian dollar, British pound, Canadian dollar, Euro, and Japanese yen. For the purpose of analyzing foreign currency exchange risk, we considered the historical trends in foreign currency exchange rates and determined that it was reasonably possible that adverse changes in exchange rates of 10% could be experienced.

We use foreign currency forward and option contracts to offset the foreign exchange risk on assets and liabilities denominated in currencies other than the functional currency of the subsidiary. These forward and option contracts reduce, but do not entirely eliminate, the effect of foreign currency exchange rate movements on our assets and liabilities. The foreign currency gains and losses on these assets and liabilities are recorded in OI&E, which are offset by the gains and losses on the forward and option contracts.

If an adverse 10% foreign currency exchange rate change was applied to total monetary assets, liabilities, and commitments denominated in currencies other than the functional currencies at the balance sheet date, it would have resulted in an adverse effect on income before income taxes of approximately $136 million and $503 million as of December 31, 2022 and 2023, respectively, after consideration of the effect of foreign exchange contracts in place for the years ended December 31, 2022 and 2023.

We use foreign currency forward and option contracts, including collars (an option strategy comprised of a combination of purchased and written options) to protect forecasted U.S. dollar-equivalent earnings from changes in foreign currency exchange rates. When the U.S. dollar strengthens, gains from foreign currency forward and option contacts reduce the foreign currency losses related to our earnings. When the U.S. dollar weakens, losses from foreign currency forward and option contracts offset the foreign currency gains related to our earnings. These hedging contracts reduce, but do not entirely eliminate, the effect of foreign currency exchange rate movements. We designate these contracts as cash flow hedges for accounting purposes. We reflect the gains or losses of foreign currency spot rate changes as a component of accumulated other comprehensive income (AOCI) and subsequently reclassify them into revenues to offset the hedged exposures as they occur.

If the U.S. dollar weakened by 10% as of December 31, 2022 and 2023, the amount recorded in AOCI related to our cash flow hedges before tax effect would have been approximately $1.3 billion and $1.5 billion lower as of December 31, 2022 and 2023, respectively. The change in the value recorded in AOCI would be expected to offset a corresponding foreign currency change in forecasted hedged revenues when recognized.

We use foreign exchange forward contracts designated as net investment hedges to hedge the foreign currency risks related to investment in foreign subsidiaries. These forward contracts serve to offset the foreign currency translation risk from our foreign operations.

If the U.S. dollar weakened by 10%, the amount recorded in cumulative translation adjustment (CTA) within AOCI related to our net investment hedges before tax effect would have been approximately $903 million and $946 million lower as of December 31, 2022 and 2023, respectively. The change in value recorded in CTA would be expected to offset a corresponding foreign currency translation gain or loss from our investment in foreign subsidiaries.

Interest Rate Risk

Our Corporate Treasury investment strategy is to achieve a return that will allow us to preserve capital and maintain liquidity. We invest primarily in debt securities, including government bonds, corporate debt securities, mortgage-backed and asset-backed securities, money market and other funds, time deposits, and interest rate derivatives. By policy, we limit the amount of credit exposure to any one issuer. Our investments in both fixed rate and floating rate interest earning securities carry a degree of interest rate risk. Fixed rate securities may have their fair market value adversely affected due to a rise in interest rates, while floating rate securities may produce less income than predicted if interest rates fall. Unrealized gains or losses on our marketable debt securities are primarily due to interest rate fluctuations as compared to interest rates at the time of purchase. For certain fixed and variable rate debt securities, we have elected the fair value option for which changes in fair value are recorded in OI&E. We measure securities for which we have not elected the fair value option at fair value with gains and losses recorded in AOCI until the securities are sold, less any expected credit losses.

We use value-at-risk (VaR) analysis to determine the potential effect of fluctuations in interest rates on the value of our marketable debt security portfolio. The VaR is the expected loss in fair value, for a given confidence interval, for our investment portfolio due to adverse movements in interest rates. We use a variance/covariance VaR model with 95% confidence interval. The estimated one-day loss in fair value of marketable debt securities as of December 31, 2022 and 2023 are shown below (in millions):

	As of December 31,		12-Month Average As of December 31,	
	2022	2023	2022	2023
Risk category - interest rate	$ 256	$ 296	$ 198	$ 271

Actual future gains and losses associated with our marketable debt security portfolio may differ materially from the sensitivity analyses performed as of December 31, 2022 and 2023 due to the inherent limitations associated with predicting the timing and amount of changes in interest rates and our actual exposures and positions. VaR analysis is not intended to represent actual losses but is used as a risk estimation.

Equity Investment Risk

Our marketable and non-marketable equity securities are subject to a wide variety of market-related risks that could substantially reduce or increase the fair value of our holdings.

Our marketable equity securities are publicly traded stocks or funds and our non-marketable equity securities are investments in privately held companies, some of which are in the startup or development stages.

We record marketable equity securities not accounted for under the equity method at fair value based on readily determinable market values, of which publicly traded stocks and mutual funds are subject to market price volatility, and represent $5.2 billion and $6.0 billion of our investments as of December 31, 2022 and 2023, respectively. A hypothetical adverse price change of 10% on our December 31, 2023 balance would decrease the fair value of marketable equity securities by $597 million. From time to time, we may enter into derivatives to hedge the market price risk on certain of our marketable equity securities.

Our non-marketable equity securities not accounted for under the equity method are adjusted to fair value for observable transactions for identical or similar investments of the same issuer or impairment (referred to as the measurement alternative). The fair value measured at the time of the observable transaction is not necessarily an indication of the current fair value as of the balance sheet date. These investments, especially those that are in the early stages, are inherently risky because the technologies or products these companies have under development are typically in the early phases and may never materialize, and they may experience a decline in financial condition, which could result in a loss of a substantial part of our investment in these companies. Valuations of our equity investments in private companies are inherently more complex due to the lack of readily available market data and observable transactions at lower valuations could result in significant losses. In addition, global economic conditions could result in additional volatility. The success of our investment in any private company is also typically dependent on the likelihood of our ability to realize appreciation in the value of investments through liquidity events such as public offerings, acquisitions, private sales or other market events. Changes in the valuation of non-marketable equity securities may not directly correlate with changes in valuation of marketable equity securities. As of December 31, 2022 and 2023, the carrying value of our non-marketable equity securities, which were accounted for under the measurement alternative, was $28.5 billion and $28.8 billion, respectively.

The carrying values of our equity method investments, which totaled approximately $1.7 billion as of December 31, 2022 and 2023, generally do not fluctuate based on market price changes. However, these investments could be impaired if the carrying value exceeds the fair value and is not expected to recover.

For additional information about our equity investments, see Note 1 and Note 3 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

Alphabet Inc.

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REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and the Board of Directors of Alphabet Inc.

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of Alphabet Inc. (the Company) as of December 31, 2022 and 2023, the related consolidated statements of income, comprehensive income, stockholders' equity and cash flows for each of the three years in the period ended December 31, 2023, and the related notes and financial statement schedule listed in the Index at Item 15 (collectively referred to as the “consolidated financial statements”). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2022 and 2023, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2023, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework), and our report dated January 30, 2024 expressed an unqualified opinion thereon.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company’s financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective or complex judgments. The communication of the critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the account or disclosure to which it relates.

Loss Contingencies

Description of the Matter	The Company is regularly subject to claims, lawsuits, regulatory and government investigations, other proceedings, and consent orders involving competition, intellectual property, data privacy and security, tax and related compliance, labor and employment, commercial disputes, content generated by its users, goods and services offered by advertisers or publishers using their platforms, personal injury, consumer protection, and other matters. As described in Note 10 to the consolidated financial statements "Commitments and contingencies" such claims, lawsuits, regulatory and government investigations, other proceedings, and consent orders could result in adverse consequences. Significant judgment is required to determine both the likelihood, and the estimated amount, of a loss related to such matters. Auditing management's accounting for and disclosure of loss contingencies from these matters involved challenging and subjective auditor judgment in assessing the Company's evaluation of the probability of a loss, and the estimated amount or range of loss.
How We Addressed the Matter in Our Audit	We tested relevant controls over the identified risks associated with management's accounting for and disclosure of these matters. This included controls over management's assessment of the probability of incurrence of a loss and whether the loss or range of loss was reasonably estimable and the development of related disclosures. Our audit procedures included gaining an understanding of previous rulings and the status of ongoing lawsuits, reviewing letters addressing the matters from internal and external legal counsel, meeting with internal legal counsel to discuss the allegations, and obtaining a representation letter from management on these matters. We also evaluated the Company's disclosures in relation to these matters.

/s/ Ernst & Young LLP

We have served as the Company's auditor since 1999.

San Jose, California
January 30, 2024

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and the Board of Directors of Alphabet Inc.

Opinion on Internal Control Over Financial Reporting

We have audited Alphabet Inc.'s internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, Alphabet Inc. (the Company) maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the 2023 consolidated financial statements of the Company and our report dated January 30, 2024 expressed an unqualified opinion thereon.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Management's Report on Internal Control over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Ernst & Young LLP

San Jose, California
January 30, 2024

Alphabet Inc.
CONSOLIDATED BALANCE SHEETS

(in millions, except par value per share amounts)

	As of December 31,	
	2022	2023
Assets		
Current assets:		
Cash and cash equivalents	$ 21,879	$ 24,048
Marketable securities	91,883	86,868
Total cash, cash equivalents, and marketable securities	113,762	110,916
Accounts receivable, net	40,258	47,964
Other current assets	10,775	12,650
Total current assets	164,795	171,530
Non-marketable securities	30,492	31,008
Deferred income taxes	5,261	12,169
Property and equipment, net	112,668	134,345
Operating lease assets	14,381	14,091
Goodwill	28,960	29,198
Other non-current assets	8,707	10,051
Total assets	$ 365,264	$ 402,392
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	$ 5,128	$ 7,493
Accrued compensation and benefits	14,028	15,140
Accrued expenses and other current liabilities	37,866	46,168
Accrued revenue share	8,370	8,876
Deferred revenue	3,908	4,137
Total current liabilities	69,300	81,814
Long-term debt	14,701	13,253
Deferred revenue, non-current	599	911
Income taxes payable, non-current	9,258	8,474
Deferred income taxes	514	485
Operating lease liabilities	12,501	12,460
Other long-term liabilities	2,247	1,616
Total liabilities	109,120	119,013
Commitments and Contingencies (Note 10)		
Stockholders' equity:		
Preferred stock, $0.001 par value per share, 100 shares authorized; no shares issued and outstanding	0	0
Class A, Class B, and Class C stock and additional paid-in capital, $0.001 par value per share: 300,000 shares authorized (Class A 180,000, Class B 60,000, Class C 60,000); 12,849 (Class A 5,964, Class B 883, Class C 6,002) and 12,460 (Class A 5,899, Class B 870, Class C 5,691) shares issued and outstanding	68,184	76,534
Accumulated other comprehensive income (loss)	(7,603)	(4,402)
Retained earnings	195,563	211,247
Total stockholders' equity	256,144	283,379
Total liabilities and stockholders' equity	$ 365,264	$ 402,392

See accompanying notes.

Alphabet Inc.
CONSOLIDATED STATEMENTS OF INCOME
(in millions, except per share amounts)

	Year Ended December 31,		
	2021	2022	2023
Revenues	$ 257,637	$ 282,836	$ 307,394
Costs and expenses:			
Cost of revenues	110,939	126,203	133,332
Research and development	31,562	39,500	45,427
Sales and marketing	22,912	26,567	27,917
General and administrative	13,510	15,724	16,425
Total costs and expenses	178,923	207,994	223,101
Income from operations	78,714	74,842	84,293
Other income (expense), net	12,020	(3,514)	1,424
Income before income taxes	90,734	71,328	85,717
Provision for income taxes	14,701	11,356	11,922
Net income	$ 76,033	$ 59,972	$ 73,795
Basic net income per share of Class A, Class B, and Class C stock	$ 5.69	$ 4.59	$ 5.84
Diluted net income per share of Class A, Class B, and Class C stock	$ 5.61	$ 4.56	$ 5.80

See accompanying notes.

Alphabet Inc.
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME
(in millions)

	Year Ended December 31,		
	2021	2022	2023
Net income	$ 76,033	$ 59,972	$ 73,795
Other comprehensive income (loss):			
Change in foreign currency translation adjustment	(1,442)	(1,836)	735
Available-for-sale investments:			
Change in net unrealized gains (losses)	(1,312)	(4,720)	1,344
Less: reclassification adjustment for net (gains) losses included in net income	(64)	1,007	1,168
Net change, net of income tax benefit (expense) of $394, $1,056, and $(698)	(1,376)	(3,713)	2,512
Cash flow hedges:			
Change in net unrealized gains (losses)	716	1,275	168
Less: reclassification adjustment for net (gains) losses included in net income	(154)	(1,706)	(214)
Net change, net of income tax benefit (expense) of $ (122), $110, and $2	562	(431)	(46)
Other comprehensive income (loss)	(2,256)	(5,980)	3,201
Comprehensive income	$ 73,777	$ 53,992	$ 76,996

See accompanying notes.

Alphabet Inc.
CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY

(in millions)

	Class A, Class B, Class C Stock and Additional Paid-In Capital				
	Shares	Amount	Accumulated Other Comprehensive Income (Loss)	Retained Earnings	Total Stockholders' Equity
Balance as of December 31, 2020	13,504	$ 58,510	$ 633	$ 163,401	$ 222,544
Stock issued	145	12	0	0	12
Stock-based compensation expense	0	15,539	0	0	15,539
Tax withholding related to vesting of restricted stock units and other	0	(10,273)	0	0	(10,273)
Repurchases of stock	(407)	(2,324)	0	(47,950)	(50,274)
Sale of interest in consolidated entities	0	310	0	0	310
Net income	0	0	0	76,033	76,033
Other comprehensive income (loss)	0	0	(2,256)	0	(2,256)
Balance as of December 31, 2021	13,242	61,774	(1,623)	191,484	251,635
Stock issued	137	8	0	0	8
Stock-based compensation expense	0	19,525	0	0	19,525
Tax withholding related to vesting of restricted stock units and other	0	(9,754)	0	(1)	(9,755)
Repurchases of stock	(530)	(3,404)	0	(55,892)	(59,296)
Sale of interest in consolidated entities	0	35	0	0	35
Net income	0	0	0	59,972	59,972
Other comprehensive income (loss)	0	0	(5,980)	0	(5,980)
Balance as of December 31, 2022	12,849	68,184	(7,603)	195,563	256,144
Stock issued	139	0	0	0	0
Stock-based compensation expense	0	22,578	0	0	22,578
Tax withholding related to vesting of restricted stock units and other	0	(10,164)	0	9	(10,155)
Repurchases of stock	(528)	(4,064)	0	(58,120)	(62,184)
Net income	0	0	0	73,795	73,795
Other comprehensive income (loss)	0	0	3,201	0	3,201
Balance as of December 31, 2023	12,460	$ 76,534	$ (4,402)	$ 211,247	$ 283,379

See accompanying notes.

Alphabet Inc.
CONSOLIDATED STATEMENTS OF CASH FLOWS

(in millions)

	Year Ended December 31,		
	2021	2022	2023
Operating activities			
Net income	$ 76,033	$ 59,972	$ 73,795
Adjustments:			
Depreciation of property and equipment	10,273	13,475	11,946
Stock-based compensation expense	15,376	19,362	22,460
Deferred income taxes	1,808	(8,081)	(7,763)
(Gain) loss on debt and equity securities, net	(12,270)	5,519	823
Other	1,955	3,483	4,330
Changes in assets and liabilities, net of effects of acquisitions:			
Accounts receivable, net	(9,095)	(2,317)	(7,833)
Income taxes, net	(625)	584	523
Other assets	(1,846)	(5,046)	(2,143)
Accounts payable	283	707	664
Accrued expenses and other liabilities	7,304	3,915	3,937
Accrued revenue share	1,682	(445)	482
Deferred revenue	774	367	525
Net cash provided by operating activities	91,652	91,495	101,746
Investing activities			
Purchases of property and equipment	(24,640)	(31,485)	(32,251)
Purchases of marketable securities	(135,196)	(78,874)	(77,858)
Maturities and sales of marketable securities	128,294	97,822	86,672
Purchases of non-marketable securities	(2,838)	(2,531)	(3,027)
Maturities and sales of non-marketable securities	934	150	947
Acquisitions, net of cash acquired, and purchases of intangible assets	(2,618)	(6,969)	(495)
Other investing activities	541	1,589	(1,051)
Net cash used in investing activities	(35,523)	(20,298)	(27,063)
Financing activities			
Net payments related to stock-based award activities	(10,162)	(9,300)	(9,837)
Repurchases of stock	(50,274)	(59,296)	(61,504)
Proceeds from issuance of debt, net of costs	20,199	52,872	10,790
Repayments of debt	(21,435)	(54,068)	(11,550)
Proceeds from sale of interest in consolidated entities, net	310	35	8
Net cash used in financing activities	(61,362)	(69,757)	(72,093)
Effect of exchange rate changes on cash and cash equivalents	(287)	(506)	(421)
Net increase (decrease) in cash and cash equivalents	(5,520)	934	2,169
Cash and cash equivalents at beginning of period	26,465	20,945	21,879
Cash and cash equivalents at end of period	$ 20,945	$ 21,879	$ 24,048
Supplemental disclosures of cash flow information			
Cash paid for income taxes, net of refunds	$ 13,412	$ 18,892	$ 19,164

See accompanying notes.

Alphabet Inc.
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

Note 1. Summary of Significant Accounting Policies

Nature of Operations

Google was incorporated in California in September 1998 and re-incorporated in the State of Delaware in August 2003. In 2015, we implemented a holding company reorganization, and as a result, Alphabet Inc. ("Alphabet") became the successor issuer to Google.

We generate revenues by delivering relevant, cost-effective online advertising; cloud-based solutions that provide enterprise customers with infrastructure and platform services as well as communication and collaboration tools; sales of other products and services, such as fees received for consumer subscription-based products, apps and in-app purchases, and devices.

Basis of Consolidation

The consolidated financial statements of Alphabet include the accounts of Alphabet and entities consolidated under the variable interest and voting models. Intercompany balances and transactions have been eliminated.

Use of Estimates

Preparation of consolidated financial statements in conformity with GAAP requires us to make estimates and assumptions that affect the amounts reported and disclosed in the financial statements and the accompanying notes. Actual results could differ materially from these estimates due to uncertainties. On an ongoing basis, we evaluate our estimates, including those related to the allowance for credit losses; content licenses; contingent liabilities; fair values of financial instruments and goodwill; income taxes; inventory; and useful lives of property and equipment, among others. We base our estimates on assumptions, both historical and forward looking, that are believed to be reasonable, and the results of which form the basis for making judgments about the carrying values of assets and liabilities.

In January 2023, we completed an assessment of the useful lives of our servers and network equipment and adjusted the estimated useful life of our servers from four years to six years and the estimated useful life of certain network equipment from five years to six years. This change in accounting estimate was effective beginning in fiscal year 2023. Based on the carrying value of servers and certain network equipment as of December 31, 2022, and those placed in service during the year ended December 31, 2023, the effect of this change in estimate was a reduction in depreciation expense of $3.9 billion and an increase in net income of $3.0 billion, or $0.24 per basic and $0.24 per diluted share, for the year ended December 31, 2023.

Revenue Recognition

Revenues are recognized when control of the promised goods or services is transferred to our customers, and the collectibility of an amount that we expect in exchange for those goods or services is probable. Sales and other similar taxes are excluded from revenues.

Advertising Revenues

We generate advertising revenues primarily by delivering advertising on:

- Google Search and other properties, including revenues from traffic generated by search distribution partners who use Google.com as their default search in browsers, toolbars, etc. and other Google owned and operated properties like Gmail, Google Maps, and Google Play;
- YouTube properties; and

- Google Network properties, including revenues from Google Network properties participating in AdMob, AdSense, and Google Ad Manager.

Our customers generally purchase advertising inventory through Google Ads, Google Ad Manager, Google Display & Video 360, and Google Marketing Platform, among others.

We offer advertising by delivering both performance and brand advertising. We recognize revenues for performance advertising when a user engages with the advertisement. For brand advertising, we recognize revenues when the ad is displayed, or a user views the ad.

For ads placed on Google Network properties, we evaluate whether we are the principal (i.e., report revenues on a gross basis) or agent (i.e., report revenues on a net basis). Generally, we report advertising revenues for ads placed on Google Network properties on a gross basis, that is, the amounts billed to our customers are recorded as revenues,

and amounts paid to Google Network partners are recorded as cost of revenues. Where we are the principal, we control the advertising inventory before it is transferred to our customers. Our control is evidenced by our sole ability to monetize the advertising inventory before it is transferred to our customers and is further supported by us being primarily responsible to our customers and having a level of discretion in establishing pricing.

Google Subscriptions, Platforms, and Devices

Google subscriptions, platforms, and devices revenues consist of revenues from:

- consumer subscriptions, which primarily include revenues from YouTube services, such YouTube TV, YouTube Music and Premium, and NFL Sunday Ticket, as well as Google One;
- platforms, which primarily include revenues from Google Play from the sales of apps and in-app purchases;
- devices, which primarily include sales of the Pixel family of devices; and
- other products and services.

Subscription revenues are recognized ratably over the period of the subscription, primarily monthly. We report revenues from Google Play app sales and in-app purchases on a net basis, because our performance obligation is to facilitate a transaction between app developers and end users, for which we earn a service fee.

Google Cloud Revenues

Google Cloud revenues consist of revenues from:

- Google Cloud Platform, which generates consumption-based fees and subscriptions for infrastructure, platform, and other services. These services provide access to solutions such as cybersecurity, databases, analytics, and AI offerings including our AI infrastructure, Vertex AI platform, and Duet AI for Google Cloud;
- Google Workspace, which includes subscriptions for cloud-based communication and collaboration tools for enterprises, such as Calendar, Gmail, Docs, Drive, and Meet, with integrated features like Duet AI in Google Workspace; and
- other enterprise services.

Our cloud services are generally provided on either a consumption or subscription basis and may have contract terms longer than a year. Revenues related to cloud services provided on a consumption basis are recognized when the customer utilizes the services, based on the quantity of services consumed. Revenues related to cloud services provided on a subscription basis are recognized ratably over the contract term as the customer receives and consumes the benefits of the cloud services.

Arrangements with Multiple Performance Obligations

Our contracts with customers may include multiple performance obligations. For such arrangements, we allocate revenues to each performance obligation based on its relative standalone selling price. We generally determine standalone selling prices based on the prices charged to customers.

Customer Incentives and Credits

Certain customers receive cash-based incentives or credits, which are accounted for as variable consideration. We estimate these amounts based on the expected amount to be provided to customers and reduce revenues. We believe that there will not be significant changes to our estimates of variable consideration.

Sales Commissions

We expense sales commissions when incurred when the period of the expected benefit is one year or less. We recognize an asset for certain sales commissions and amortize if the expected benefit period is greater than one year. These costs are recorded within sales and marketing expenses.

Cost of Revenues

Cost of revenues consists of TAC and other costs of revenues.

- TAC includes:
  - amounts paid to our distribution partners who make available our search access points and services. Our distribution partners include browser providers, mobile carriers, original equipment manufacturers, and software developers; and
  - amounts paid to Google Network partners primarily for ads displayed on their properties.

- Other cost of revenues includes:
  - compensation expense related to our data centers and other operations such as content review and customer and product support;
  - content acquisition costs, which are payments to content providers from whom we license video and other content for distribution on YouTube and Google Play (we pay fees to these content providers based on revenues generated or a flat fee);
  - depreciation expense related to our technical infrastructure; and
  - inventory and other costs related to the devices we sell.

Software Development Costs

We expense software development costs, including costs to develop software products or the software component of products to be sold, leased, or marketed to external users, before technological feasibility is reached. Technological feasibility is typically reached shortly before the release of such products. As a result, development costs that meet the criteria for capitalization were not material for the periods presented.

Software development costs also include costs to develop software to be used solely to meet internal needs and cloud-based applications used to deliver our services. We capitalize development costs related to these software applications once the preliminary project stage is complete and it is probable that the project will be completed and the software will be used to perform the function intended. Costs capitalized for developing such software applications were not material for the periods presented.

Stock-based Compensation

Stock-based compensation (SBC) primarily consists of Alphabet restricted stock units (RSUs). RSUs are equity classified and measured at the fair market value of the underlying stock at the grant date. We recognize RSU expense using the straight-line attribution method over the requisite service period and account for forfeitures as they occur.

For RSUs, shares are issued on the vesting dates net of the applicable statutory income tax withholding to be paid by us on behalf of our employees. As a result, fewer shares are issued than the number of RSUs outstanding, and the income tax withholding is recorded as a reduction to additional paid-in capital.

Additionally, SBC includes other stock-based awards, such as performance stock units (PSUs) that include market conditions and awards that may be settled in cash or the stock of certain Other Bet companies. PSUs and certain awards granted by Other Bet companies are equity classified and expense is recognized over the requisite service period. Certain awards granted by Other Bet companies are liability classified and remeasured at fair value through settlement. The fair value of awards granted by Other Bet companies is based on the equity valuation of the respective Other Bet company.

Advertising and Promotional Expenses

We expense advertising and promotional costs in the period in which they are incurred. For the years ended December 31, 2021, 2022, and 2023, advertising and promotional expenses totaled approximately $7.9 billion, $9.2 billion, and $8.7 billion, respectively.

Performance Fees

Performance fees refer to compensation arrangements with payouts based on realized returns from certain investments. We record compensation expense based on the estimated payouts on an ongoing basis, which may result in expense recognized before investment returns are realized and compensation is paid and may require the use of unobservable inputs. Performance fees are recorded as a component of OI&E.

Fair Value Measurements

Fair value is an exit price, representing the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants. As such, fair value is a market-based measurement that is determined based on assumptions that market participants would use in pricing an asset or a liability. Assets and liabilities recorded at fair value are measured and classified in accordance with a three-tier fair value hierarchy based on the observability of the inputs available in the market used to measure fair value:

Level 1 - Observable inputs that reflect quoted prices (unadjusted) for identical assets or liabilities in active markets.

Level 2 - Inputs that are based upon quoted prices for similar instruments in active markets, quoted prices for identical or similar instruments in markets that are not active, and model-based valuation techniques for which all significant inputs are observable in the market or can be derived from observable market data. Where applicable, these models project future cash flows and discount the future amounts to a present value using market-based observable inputs including interest rate curves, foreign exchange rates, and credit ratings.

Level 3 - Unobservable inputs that are supported by little or no market activities.

The fair value hierarchy requires an entity to maximize the use of observable inputs and minimize the use of unobservable inputs when measuring fair value. The determination of fair value involves the use of appropriate valuation methods and relevant inputs into valuation models.

Our financial assets and liabilities that are measured at fair value on a recurring basis include cash equivalents, marketable securities, derivative financial instruments, and certain non-marketable debt securities. Our financial assets measured at fair value on a nonrecurring basis include non-marketable equity securities. Other financial assets and liabilities are carried at cost with fair value disclosed, if required.

We measure certain other instruments, including SBC awards settled in the stock of Other Bet companies, and certain assets and liabilities acquired in a business combination, also at fair value on a nonrecurring basis.

Financial Instruments

Our financial instruments include cash, cash equivalents, marketable and non-marketable securities, derivative financial instruments and accounts receivable.

Credit Risks

We are subject to credit risk primarily from cash equivalents, marketable debt securities, derivative financial instruments, including foreign exchange contracts, and accounts receivable. We manage our credit risk exposure through timely assessment of our counterparty creditworthiness, credit limits and use of collateral management. Foreign exchange contracts are transacted with various financial institutions with high credit standing. Accounts receivable are typically unsecured and are derived from revenues earned from customers located around the world. We manage our credit risk exposure by performing ongoing evaluations to determine customer credit and we limit the amount of credit we extend. We generally do not require collateral from our customers.

Cash Equivalents

We invest excess cash primarily in government bonds, corporate debt securities, mortgage-backed and asset-backed securities, time deposits, and money market funds.

Marketable Securities

We classify all marketable debt securities that have effective maturities of three months or less from the date of purchase as cash equivalents and those with effective maturities of greater than three months as marketable securities on our Consolidated Balance Sheets. We determine the appropriate classification of our investments in marketable debt securities at the time of purchase and reevaluate such designation at each balance sheet date. We have classified and accounted for our marketable debt securities as available-for-sale. After consideration of our risk versus reward objectives, as well as our liquidity requirements, we may sell these debt securities prior to their effective maturities. As we view these securities as available to support current operations, we classify highly liquid securities with maturities beyond 12 months as current assets under the caption marketable securities on the Consolidated Balance Sheets. We carry these securities at fair value, and report the unrealized gains and losses, net of taxes, as a component of stockholders' equity, except for the changes in allowance for expected credit losses, which

are recorded in OI&E. For certain marketable debt securities we have elected the fair value option, for which changes in fair value are recorded in OI&E. We determine any realized gains or losses on the sale of marketable debt securities on a specific identification method, and we record such gains and losses as a component of OI&E.

Our investments in marketable equity securities are measured at fair value with the related gains and losses, including unrealized, recognized in OI&E. We classify our marketable equity securities subject to long-term lock-up restrictions beyond 12 months as other non-current assets on the Consolidated Balance Sheets.

Non-Marketable Securities

Non-marketable securities primarily consist of equity securities. We account for non-marketable equity securities through which we exercise significant influence but do not have control over the investee under the equity method. All other non-marketable equity securities that we hold are primarily accounted for under the measurement alternative. Under the measurement alternative, the carrying value is measured at cost, less any impairment, plus or minus

changes resulting from observable price changes in orderly transactions for identical or similar investments of the same issuer. Adjustments are determined primarily based on a market approach as of the transaction date and are recorded as a component of OI&E.

Non-marketable securities that do not have effective contractual maturity dates are classified as other non-current assets on the Consolidated Balance Sheets.

Derivative Financial Instruments

See Note 3 for the accounting policy pertaining to derivative financial instruments.

Accounts Receivable

Our payment terms for accounts receivable vary by the types and locations of our customers and the products or services offered. The term between invoicing and when payment is due is not significant. For certain products or services and customers, we require payment before the products or services are delivered to the customer. Additionally, accounts receivable includes amounts for services performed in advance of the right to invoice the customer.

We maintain an allowance for credit losses for accounts receivable, which is recorded as an offset to accounts receivable, and changes in such are classified as general and administrative expense in the Consolidated Statements of Income. We assess collectibility by reviewing accounts receivable on a collective basis where similar characteristics exist and on an individual basis when we identify specific customers with known disputes or collectibility issues. In determining the amount of the allowance for credit losses, we consider historical collectibility based on past due status and make judgments about the creditworthiness of customers based on ongoing credit evaluations. We also consider customer-specific information, current market conditions, and reasonable and supportable forecasts of future economic conditions.

Other

Our financial instruments also include debt and equity investments in companies with which we also entered into commercial arrangements at or near the same time. For these transactions, judgment is required in assessing the substance of the arrangements, including assessing whether the components of the arrangements should be accounted for as separate transactions under the applicable GAAP, and determining the value of the components of the arrangements, including the fair value of the investments. Additionally, if our investment in such companies becomes impaired, any remaining performance obligations would be reassessed and may be reduced.

Impairment of Investments

We periodically review our debt and non-marketable equity securities for impairment.

For debt securities in an unrealized loss position, we determine whether a credit loss exists. The credit loss is estimated by considering available information relevant to the collectibility of the security and information about past events, current conditions, and reasonable and supportable forecasts. Any credit loss is recorded as a charge to OI&E, not to exceed the amount of the unrealized loss. Unrealized losses other than the credit loss are recognized in AOCI. If we have an intent to sell, or if it is more likely than not that we will be required to sell a debt security in an unrealized loss position before recovery of its amortized cost basis, we will write down the security to its fair value and record the corresponding charge as a component of OI&E.

For non-marketable equity securities, including equity method investments, we consider whether impairment indicators exist by evaluating the companies' financial and liquidity position and access to capital resources, among other indicators. If the assessment indicates that the investment is impaired, we write down the investment to its fair value by recording the corresponding charge as a component of

OI&E. We prepare quantitative measurements of the fair value of our equity investments using a market approach or an income approach.

Inventory

Inventory consists primarily of finished goods and is stated at the lower of cost and net realizable value. Cost is computed using the first-in, first-out method.

Variable Interest Entities

We determine at the inception of each arrangement whether an entity in which we have made an investment or in which we have other variable interests is considered a variable interest entity (VIE). We consolidate VIEs when we are the primary beneficiary. We are the primary beneficiary of a VIE when we have the power to direct activities that most significantly affect the economic performance of the VIE and have the obligation to absorb the majority of their losses

or benefits. If we are not the primary beneficiary in a VIE, we account for the investment or other variable interests in a VIE in accordance with applicable GAAP.

Periodically, we assess whether any changes in our interest or relationship with the entity affect our determination of whether the entity is a VIE and, if so, whether we are the primary beneficiary.

Property and Equipment

Property and equipment includes the following categories: land and buildings, information technology assets, construction in progress, leasehold improvements, and furniture and fixtures. Land and buildings include land, offices, data centers, and related building improvements. Information technology assets include servers and network equipment. Construction in progress is the construction or development of property and equipment that have not yet been placed in service.

Property and equipment are stated at cost less accumulated depreciation. Depreciation is recorded using the straight-line method over the estimated useful lives of the assets, which we regularly evaluate. Land is not depreciated. We depreciate buildings over periods of seven to 25 years. We depreciate information technology assets generally over a period of six years for servers and network equipment. We depreciate leasehold improvements over the shorter of the remaining lease term or the estimated useful lives of the assets. Depreciation for buildings, information technology assets, leasehold improvements, and furniture and fixtures commences once they are ready for our intended use.

Goodwill

We allocate goodwill to reporting units based on the expected benefit from the business combination. We evaluate our reporting units periodically, as well as when changes in our operating segments occur. For changes in reporting units, we reassign goodwill using a relative fair value allocation approach. We test our goodwill for impairment at least annually, or more frequently if events or changes in circumstances indicate that the asset may be impaired. Goodwill impairments were not material for the periods presented.

Leases

We determine if an arrangement is a lease at inception. Our lease agreements generally contain lease and non-lease components. Payments under our lease arrangements are primarily fixed. Non-lease components primarily include payments for maintenance and utilities. We combine fixed payments for non-lease components with lease payments and account for them together as a single lease component which increases the amount of our lease assets and liabilities.

Certain lease agreements contain variable payments, which are expensed as incurred and not included in the lease assets and liabilities. These amounts primarily include payments affected by the Consumer Price Index, and payments for maintenance and utilities.

Lease assets and liabilities are recognized at the present value of the future lease payments at the lease commencement date. The interest rate used to determine the present value of the future lease payments is our incremental borrowing rate, because the interest rate implicit in our leases is not readily determinable. Our incremental borrowing rate is estimated to approximate the interest rate on a collateralized basis with similar terms and payments, and in economic environments where the leased asset is located. Our lease terms include periods under options to extend or terminate the lease when it is reasonably certain that we will exercise that option. We generally use the base, non-cancelable, lease term when determining the lease assets and liabilities. Lease assets also include any prepaid lease payments and lease incentives.

Operating lease assets and liabilities are included on our Consolidated Balance Sheets. The current portion of our operating lease liabilities is included in accrued expenses and other current liabilities, and the long-term portion is included in operating lease liabilities. Finance lease assets are included in

property and equipment, net. Finance lease liabilities are included in accrued expenses and other current liabilities or long-term debt.

Operating lease expense (excluding variable lease costs) is recognized on a straight-line basis over the lease term.

Impairment of Long-Lived Assets

We review leases, property and equipment, and intangible assets, excluding goodwill, for impairment when events or changes in circumstances indicate the carrying amount may not be recoverable. The evaluation is performed at the lowest level of identifiable cash flows independent of other assets. We measure recoverability of these assets by comparing the carrying amounts to the future undiscounted cash flows that the assets or the asset group are expected

to generate. If the carrying value of the assets or asset group is not recoverable, the impairment recognized is measured as the amount by which the carrying value exceeds its fair value.

Income Taxes

We account for income taxes using the asset and liability method, under which we recognize the amount of taxes payable or refundable for the current year and deferred tax assets and liabilities for the future tax consequences of events that have been recognized in our financial statements or tax returns. We measure current and deferred tax assets and liabilities based on provisions of enacted tax law. We evaluate the likelihood of future realization of our deferred tax assets based on all available evidence and establish a valuation allowance to reduce deferred tax assets when it is more likely than not that they will not be realized or release a valuation allowance to increase deferred tax assets when it is more likely than not that they will be realized. We have elected to account for the tax effects of the global intangible low tax Income provision as a current period expense.

We recognize the financial statement effects of a tax position when it is more likely than not that, based on technical merits, the position will be sustained upon examination. The tax benefits of the position recognized in the financial statements are then measured based on the largest amount of benefit that is greater than 50% likely to be realized upon settlement with a taxing authority. In addition, we recognize interest and penalties related to unrecognized tax benefits as a component of the income tax provision.

Business Combinations

We include the results of operations of the businesses that we acquire as of the acquisition date. We allocate the purchase price of the acquisitions to the assets acquired and liabilities assumed based on their estimated fair values, except for revenue contracts acquired, which are recognized in accordance with our revenue recognition policy. The excess of the purchase price over the fair values of identifiable assets and liabilities is recorded as goodwill. Acquisition-related expenses are recognized separately from the business combination and are expensed as incurred.

Foreign Currency

We translate the financial statements of our international subsidiaries to U.S. dollars using month-end exchange rates for assets and liabilities, and average rates for the annual period derived from month-end exchange rates for revenues, costs, and expenses. We record translation gains and losses in AOCI as a component of stockholders' equity. We reflect net foreign exchange transaction gains and losses resulting from the conversion of the transaction currency to functional currency as a component of foreign currency exchange gain (loss) in OI&E.

Recent Accounting Pronouncements

In November 2023, the Financial Standards Accounting Board (FASB) issued Accounting Standards Update (ASU) 2023-07 "Segment Reporting (Topic 280):Improvements to Reportable Segment Disclosures" which expands annual and interim disclosure requirements for reportable segments, primarily through enhanced disclosures about significant segment expenses. ASU 2023-07 is effective for our annual periods beginning January 1, 2024, and for interim periods beginning January 1, 2025, with early adoption permitted. We are currently evaluating the potential effect that the updated standard will have on our financial statement disclosures.

In December 2023, the FASB issued ASU 2023-09 "Income Taxes (Topics 740): Improvements to Income Tax Disclosures" to expand the disclosure requirements for income taxes, specifically related to the rate reconciliation and income taxes paid. ASU 2023-09 is effective for our annual periods beginning January 1, 2025, with early adoption permitted. We are currently evaluating the potential effect that the updated standard will have on our financial statement disclosures.

Prior Period Reclassifications

Certain amounts in prior periods have been reclassified to conform with current period presentation.

Note 2. Revenues

Disaggregated Revenues

The following table presents revenues disaggregated by type (in millions):

	Year Ended December 31,		
	2021	2022	2023
Google Search & other	$ 148,951	$ 162,450	$ 175,033
YouTube ads	28,845	29,243	31,510
Google Network	31,701	32,780	31,312
Google advertising	209,497	224,473	237,855
Google subscriptions, platforms, and devices	28,032	29,055	34,688
Google Services total	237,529	253,528	272,543
Google Cloud	19,206	26,280	33,088
Other Bets	753	1,068	1,527
Hedging gains (losses)	149	1,960	236
Total revenues	$ 257,637	$ 282,836	$ 307,394

No individual customer or groups of affiliated customers represented more than 10% of our revenues in 2021, 2022, or 2023.

The following table presents revenues disaggregated by geography, based on the addresses of our customers (in millions):

	Year Ended December 31,					
	2021		2022		2023	
United States	$ 117,854	46 %	$ 134,814	48 %	$ 146,286	47 %
EMEA[1]	79,107	31	82,062	29	91,038	30
APAC[1]	46,123	18	47,024	16	51,514	17
Other Americas[1]	14,404	5	16,976	6	18,320	6
Hedging gains (losses)	149	0	1,960	1	236	0
Total revenues	$ 257,637	100 %	$ 282,836	100 %	$ 307,394	100 %

(1) Regions represent Europe, the Middle East, and Africa (EMEA); Asia-Pacific (APAC); and Canada and Latin America ("Other Americas").

Revenue Backlog

As of December 31, 2023, we had $74.1 billion of remaining performance obligations ("revenue backlog"), primarily related to Google Cloud. Our revenue backlog represents commitments in customer contracts for future services that have not yet been recognized as revenue. The estimated revenue backlog and timing of revenue recognition for these commitments is largely driven by our ability to deliver in accordance with relevant contract terms and when our customers utilize services. We expect to recognize approximately half of the revenue backlog as revenues over the next 24 months with the remaining to be recognized thereafter. Revenue backlog includes related deferred revenue currently recorded as well as amounts that will be invoiced in future periods, and excludes contracts with an original expected term of one year or less and cancellable contracts.

Deferred Revenues

We record deferred revenues when cash payments are received or due in advance of our performance, including amounts which are refundable. Deferred revenues primarily relate to Google Cloud and Google subscriptions, platforms, and devices. Total deferred revenue as of December 31, 2022 was $4.5 billion, of which $2.4 billion was recognized as revenues for the year ended December 31, 2023.

Note 3. Financial Instruments

Fair Value Measurements

Investments Measured at Fair Value on a Recurring Basis

Cash, cash equivalents, and marketable equity securities are measured at fair value and classified within Level 1

and Level 2 in the fair value hierarchy, because we use quoted prices for identical assets in active markets or inputs that are based upon quoted prices for similar instruments in active markets.

Debt securities are measured at fair value and classified within Level 2 in the fair value hierarchy, because we use quoted market prices to the extent available or alternative pricing sources and models utilizing market observable inputs to determine fair value. For certain marketable debt securities, we have elected the fair value option for which changes in fair value are recorded in OI&E. The fair value option was elected for these securities to align with the unrealized gains and losses from related derivative contracts.

The following tables summarize our cash, cash equivalents, and marketable securities measured at fair value on a recurring basis (in millions):

		As of December 31, 2022					
	Fair Value Hierarchy	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Marketable Securities
Fair value changes recorded in other comprehensive income							
Time deposits	Level 2	$ 5,297	$ 0	$ 0	$ 5,297	$ 5,293	$ 4
Government bonds	Level 2	41,036	64	(2,045)	39,055	283	38,772
Corporate debt securities	Level 2	28,578	8	(1,569)	27,017	1	27,016
Mortgage-backed and asset-backed securities	Level 2	16,176	5	(1,242)	14,939	0	14,939
Total investments with fair value change reflected in other comprehensive income[1]		$ 91,087	$ 77	$ (4,856)	$ 86,308	$ 5,577	$ 80,731
Fair value adjustments recorded in net income							
Money market funds	Level 1				$ 7,234	$ 7,234	$ 0
Current marketable equity securities[2]	Level 1				4,013	0	4,013
Mutual funds	Level 2				339	0	339
Government bonds	Level 2				1,877	440	1,437
Corporate debt securities	Level 2				3,744	65	3,679
Mortgage-backed and asset-backed securities	Level 2				1,686	2	1,684
Total investments with fair value change recorded in net income					$ 18,893	$ 7,741	$ 11,152
Cash					0	8,561	0
Total		$ 91,087	$ 77	$ (4,856)	$ 105,201	$ 21,879	$ 91,883

(1) Represents gross unrealized gains and losses for debt securities recorded to AOCI.

(2) The long-term portion of marketable equity securities (subject to long-term lock-up restrictions) of $803 million as of December 31, 2022 is included within other non-current assets.

		As of December 31, 2023					
	Fair Value Hierarchy	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Marketable Securities
Fair value changes recorded in other comprehensive income							
Time deposits	Level 2	$ 2,628	$ 0	$ 0	$ 2,628	$ 2,628	$ 0
Government bonds	Level 2	38,106	233	(679)	37,660	1,993	35,667
Corporate debt securities	Level 2	22,457	112	(637)	21,932	0	21,932
Mortgage-backed and asset-backed securities	Level 2	17,243	88	(634)	16,697	0	16,697
Total investments with fair value change reflected in other comprehensive income[1]		$ 80,434	$ 433	$ (1,950)	$ 78,917	$ 4,621	$ 74,296
Fair value adjustments recorded in net income							
Money market funds	Level 1				$ 6,480	$ 6,480	$ 0
Current marketable equity securities[2]	Level 1				4,282	0	4,282
Mutual funds	Level 2				311	0	311
Government bonds	Level 2				1,952	347	1,605
Corporate debt securities	Level 2				3,782	91	3,691
Mortgage-backed and asset-backed securities	Level 2				2,683	0	2,683
Total investments with fair value change recorded in net income					$ 19,490	$ 6,918	$ 12,572
Cash					0	12,509	0
Total		$ 80,434	$ 433	$ (1,950)	$ 98,407	$ 24,048	$ 86,868

(1) Represents gross unrealized gains and losses for debt securities recorded to AOCI.

(2) The long-term portion of marketable equity securities (subject to long-term lock-up restrictions) of $1.4 billion as of December 31, 2023 is included within other non-current assets.

Investments Measured at Fair Value on a Nonrecurring Basis

Our non-marketable equity securities are investments in privately held companies without readily determinable market values. The carrying value of our non-marketable equity securities is adjusted to fair value upon observable transactions for identical or similar investments of the same issuer or impairment. Non-marketable equity securities that have been remeasured during the period based on observable transactions are classified within Level 2 or Level 3 in the fair value hierarchy because we estimate the value based on valuation methods, including option pricing models, market comparable approach, and common stock equivalent method, which may include a combination of the observable transaction price at the transaction date and other unobservable inputs including volatility, expected time to exit, risk free rate, and the rights, and obligations of the securities we hold. These inputs significantly vary based on investment type. The fair value of non-marketable equity securities that have been remeasured due to impairment are classified within Level 3.

As of December 31, 2023, the carrying value of our non-marketable equity securities was $28.8 billion, of which $13.7 billion were remeasured at fair value during the year ended December 31, 2023, and primarily classified within Level 2 of the fair value hierarchy at the time of measurement.

Debt Securities

The following table summarizes the estimated fair value of investments in available-for-sale marketable debt securities by effective contractual maturity dates (in millions):

	As of December 31, 2023
Due in one year or less	$ 11,231
Due in one year through five years	41,477
Due in five years through 10 years	15,351
Due after 10 years	14,216
Total	$ 82,275

The following tables present fair values and gross unrealized losses recorded to AOCI, aggregated by investment category and the length of time that individual securities have been in a continuous loss position (in millions):

	As of December 31, 2022					
	Less than 12 Months		12 Months or Greater		Total	
	Fair Value	Unrealized Loss	Fair Value	Unrealized Loss	Fair Value	Unrealized Loss
Government bonds	$ 21,039	$ (1,004)	$ 13,438	$ (1,041)	$ 34,477	$ (2,045)
Corporate debt securities	11,228	(440)	15,125	(1,052)	26,353	(1,492)
Mortgage-backed and asset-backed securities	7,725	(585)	6,964	(657)	14,689	(1,242)
Total	$ 39,992	$ (2,029)	$ 35,527	$ (2,750)	$ 75,519	$ (4,779)

	As of December 31, 2023					
	Less than 12 Months		12 Months or Greater		Total	
	Fair Value	Unrealized Loss	Fair Value	Unrealized Loss	Fair Value	Unrealized Loss
Government bonds	$ 1,456	$ (22)	$ 13,897	$ (657)	$ 15,353	$ (679)
Corporate debt securities	827	(5)	15,367	(592)	16,194	(597)
Mortgage-backed and asset-backed securities	2,945	(26)	7,916	(608)	10,861	(634)
Total	$ 5,228	$ (53)	$ 37,180	$ (1,857)	$ 42,408	$ (1,910)

We determine realized gains or losses on the sale or extinguishment of debt securities on a specific identification method.The following table summarizes gains and losses for debt securities, reflected as a component of OI&E (in millions):

	Year Ended December 31,		
	2021	2022	2023
Unrealized gain (loss) on fair value option debt securities	$ (122)	$ (557)	$ 386
Gross realized gain on debt securities	432	103	182
Gross realized loss on debt securities	(329)	(1,588)	(1,833)
(Increase) decrease in allowance for credit losses	(91)	(22)	50
Total gain (loss) on debt securities recognized in other income (expense), net	$ (110)	$ (2,064)	$ (1,215)

Equity Investments

The carrying value of equity securities is measured as the total initial cost plus the cumulative net gain (loss). Gains and losses, including impairments, are included as a component of OI&E in the Consolidated Statements of Income. See Note 7 for further details on OI&E.

The carrying values for marketable and non-marketable equity securities are summarized below (in millions):

	As of December 31, 2022			As of December 31, 2023		
	Marketable Equity Securities	Non-Marketable Equity Securities	Total	Marketable Equity Securities	Non-Marketable Equity Securities	Total
Total initial cost	$ 5,764	$ 16,157	$ 21,921	$ 5,418	$ 17,616	$ 23,034
Cumulative net gain (loss)[1]	(608)	12,372	11,764	555	11,150	11,705
Carrying value	$ 5,156	$ 28,529	$ 33,685	$ 5,973	$ 28,766	$ 34,739

(1) Non-marketable equity securities cumulative net gain (loss) is comprised of $16.8 billion gains and $4.5 billion losses (including impairments) as of December 31, 2022 and $18.1 billion gains and $6.9 billion losses (including impairments) as of December 31, 2023.

Gains and Losses on Marketable and Non-marketable Equity Securities

Gains and losses (including impairments), net, for marketable and non-marketable equity securities included in OI&E are summarized below (in millions):

	Year Ended December 31,		
	2021	2022	2023
Realized net gain (loss) on equity securities sold during the period	$ 1,196	$ (442)	$ 690
Unrealized net gain (loss) on marketable equity securities	1,335	(3,242)	790
Unrealized net gain (loss) on non-marketable equity securities[1]	9,849	229	(1,088)
Total gain (loss) on equity securities in other income (expense), net	$ 12,380	$ (3,455)	$ 392

(1) Unrealized gain (loss) on non-marketable equity securities accounted for under the measurement alternative is comprised of $10.0 billion, $3.3 billion, and $1.8 billion of upward adjustments as of December 31, 2021, 2022, and 2023, respectively, and $122 million, $3.0 billion, and $2.9 billion of downward adjustments (including impairments) as of December 31, 2021, 2022, and 2023, respectively.

In the table above, realized net gain (loss) on equity securities sold during the period reflects the difference between the sale proceeds and the carrying value of the equity securities at the beginning of the period or the purchase date, if later.

Cumulative net gains (losses) on equity securities sold during the period, which is summarized in the following table (in millions), represents the total net gains (losses) recognized after the initial purchase date of the equity security sold during the period. While these net gains (losses) may have been reflected

in periods prior to the period of sale, we believe they are important supplemental information as they reflect the economic net gains (losses) on the securities sold during the period. Cumulative net gains (losses) are calculated as the difference between the sale price and the initial purchase price for the equity security sold during the period.

	Equity Securities Sold During the Year Ended December 31,	
	2022	2023
Total sale price	$ 1,784	$ 1,981
Total initial cost	937	1,512
Cumulative net gains (losses)	$ 847	$ 469

Equity Securities Accounted for Under the Equity Method

As of December 31, 2022 and 2023, equity securities accounted for under the equity method had a carrying value of approximately $1.5 billion and $1.7 billion, respectively. Our share of gains and losses, including impairments, are included as a component of OI&E, in the Consolidated Statements of Income. See Note 7 for further details on OI&E.

Derivative Financial Instruments

We use derivative instruments to manage risks relating to our ongoing business operations. The primary risk managed is foreign exchange risk. We use foreign currency contracts to reduce the risk that our cash flows, earnings, and investment in foreign subsidiaries will be adversely affected by foreign currency exchange rate fluctuations. We also enter into derivative instruments to partially offset our exposure to other risks and enhance investment returns.

We recognize derivative instruments in the Consolidated Balance Sheets at fair value and classify the derivatives primarily within Level 2 in the fair value hierarchy. We present our collar contracts (an option strategy comprised of a combination of purchased and written options) at net fair values and present all other derivatives at gross fair values. The accounting treatment for derivatives is based on the intended use and hedge designation.

Cash Flow Hedges

We designate foreign currency forward and option contracts (including collars) as cash flow hedges to hedge certain forecasted revenue transactions denominated in currencies other than the U.S. dollar. These contracts have maturities of 24 months or less.

Cash flow hedge amounts included in the assessment of hedge effectiveness are deferred in AOCI and subsequently reclassified to revenue when the hedged item is recognized in earnings. We exclude forward points and time value from our assessment of hedge effectiveness and amortize them on a straight-line basis over the life of the hedging instrument in revenues. The difference between fair value changes of the excluded component and the amount amortized to revenues is recorded in AOCI.

As of December 31, 2023, the net accumulated gain on our foreign currency cash flow hedges before tax effect was $47 million, which is expected to be reclassified from AOCI into revenues within the next 12 months.

Fair Value Hedges

We designate foreign currency forward contracts as fair value hedges to hedge foreign currency risks for our marketable securities denominated in currencies other than the U.S. dollar. Fair value hedge amounts included in the assessment of hedge effectiveness are recognized in OI&E, along with the offsetting gains and losses of the related hedged items. We exclude forward points from the assessment of hedge effectiveness and recognize changes in the excluded component in OI&E.

Net Investment Hedges

We designate foreign currency forward contracts as net investment hedges to hedge the foreign currency risks related to our investment in foreign subsidiaries. Net investment hedge amounts included in the assessment of hedge effectiveness are recognized in AOCI along with the foreign currency translation adjustment. We exclude forward points from the assessment of hedge effectiveness and recognize changes in the excluded component in OI&E.

Other Derivatives

We enter into foreign currency forward and option contracts that are not designated as hedging instruments to hedge intercompany transactions and other monetary assets or liabilities denominated in currencies other than the functional currency of a subsidiary. Gains and losses on these derivatives that are not designated as accounting hedges are primarily recorded in OI&E along with the foreign currency gains and losses on monetary assets and liabilities.

We also use derivatives not designated as hedging instruments to manage risks relating to interest rates, commodity prices, credit exposures, and to enhance investment returns. From time to time, we enter into derivatives to hedge the market price risk on certain of our marketable equity securities. Gains

and losses arising from other derivatives are primarily reflected within the “other” component of OI&E. See Note 7 for further details.

The gross notional amounts of outstanding derivative instruments were as follows (in millions):

	As of December 31,	
	2022	2023
Derivatives designated as hedging instruments:		
Foreign exchange contracts		
Cash flow hedges	$ 15,972	$ 18,039
Fair value hedges	$ 2,117	$ 2,065
Net investment hedges	$ 8,751	$ 9,472
Derivatives not designated as hedging instruments:		
Foreign exchange contracts	$ 34,979	$ 39,722
Other contracts	$ 7,932	$ 10,818

The fair values of outstanding derivative instruments were as follows (in millions):

	As of December 31, 2022		As of December 31, 2023	
	Assets[1]	Liabilities[2]	Assets[1]	Liabilities[2]
Derivatives designated as hedging instruments:				
Foreign exchange contracts	$ 271	$ 556	$ 205	$ 242
Derivatives not designated as hedging instruments:				
Foreign exchange contracts	365	207	134	156
Other contracts	40	47	114	47
Total derivatives not designated as hedging instruments	405	254	248	203
Total	$ 676	$ 810	$ 453	$ 445

(1) Derivative assets are recorded as other current and non-current assets in the Consolidated Balance Sheets.

(2) Derivative liabilities are recorded as accrued expenses and other liabilities, current and non-current in the Consolidated Balance Sheets.

The gains (losses) on derivatives in cash flow hedging and net investment hedging relationships recognized in other comprehensive income (OCI) are summarized below (in millions):

	Year Ended December 31,		
	2021	2022	2023
Derivatives in cash flow hedging relationship:			
Foreign exchange contracts			
Amount included in the assessment of effectiveness	$ 806	$ 1,699	$ 90
Amount excluded from the assessment of effectiveness	48	(188)	84
Derivatives in net investment hedging relationship:			
Foreign exchange contracts			
Amount included in the assessment of effectiveness	754	608	(287)
Total	$ 1,608	$ 2,119	$ (113)

The table below presents the gains (losses) of our derivatives on the Consolidated Statements of Income: (in millions):

	Year Ended December 31,					
	2021		2022		2023	
	Revenues	Other income (expense), net	Revenues	Other income (expense), net	Revenues	Other income (expense), net
Total amounts in the Consolidated Statements of Income	$ 257,637	$ 12,020	$ 282,836	$ (3,514)	$ 307,394	$ 1,424
Effect of cash flow hedges:						
Foreign exchange contracts						
Amount reclassified from AOCI to income	$ 165	$ 0	$ 2,046	$ 0	$ 213	$ 0
Amount excluded from the assessment of effectiveness (amortized)	(16)	0	(85)	0	24	0
Effect of fair value hedges:						
Foreign exchange contracts						
Hedged items	0	(95)	0	(162)	0	59
Derivatives designated as hedging instruments	0	95	0	163	0	(59)
Amount excluded from the assessment of effectiveness	0	8	0	16	0	15
Effect of net investment hedges:						
Foreign exchange contracts						
Amount excluded from the assessment of effectiveness	0	82	0	171	0	187
Effect of non designated hedges:						
Foreign exchange contracts	0	(860)	0	(395)	0	7
Other contracts	0	101	0	144	0	53
Total gains (losses)	$ 149	$ (669)	$ 1,961	$ (63)	$ 237	$ 262

Offsetting of Derivatives

We enter into master netting arrangements and collateral security arrangements to reduce credit risk. Cash collateral received related to derivative instruments under our collateral security arrangements are included in other current assets with a corresponding liability. Cash and non-cash collateral pledged related to derivative instruments under our collateral security arrangements are included in other current assets.

The gross amounts of derivative instruments subject to master netting arrangements with various counterparties, and cash and non-cash collateral received and pledged under such agreements were as follows (in millions):

	As of December 31, 2022					
				Gross Amounts Not Offset in the Consolidated Balance Sheets, but Have Legal Rights to Offset		
	Gross Amounts Recognized	Gross Amounts Offset in the Consolidated Balance Sheets	Net Amounts Presented in the Consolidated Balance Sheets	Financial Instruments[1]	Cash and Non-Cash Collateral Received or Pledged	Net Amounts
Derivatives assets	$ 760	$ (84)	$ 676	$ (463)	$ (132)	$ 81
Derivatives liabilities	$ 894	$ (84)	$ 810	$ (463)	$ (28)	$ 319

	As of December 31, 2023					
				Gross Amounts Not Offset in the Consolidated Balance Sheets, but Have Legal Rights to Offset		
	Gross Amounts Recognized	Gross Amounts Offset in the Consolidated Balance Sheets	Net Amounts Presented in the Consolidated Balance Sheets	Financial Instruments[1]	Cash and Non-Cash Collateral Received or Pledged	Net Amounts
Derivatives assets	$ 535	$ (82)	$ 453	$ (213)	$ (75)	$ 165
Derivatives liabilities	$ 527	$ (82)	$ 445	$ (213)	$ (16)	$ 216

(1) The balances as of December 31, 2022 and 2023 were related to derivatives allowed to be net settled in accordance with our master netting agreements.

Note 4. Leases

We have entered into operating lease agreements primarily for data centers, land, and offices throughout the world with lease periods expiring between 2024 and 2063.

Components of operating lease expense were as follows (in millions):

	Year Ended December 31,		
	2021	2022	2023
Operating lease cost	$ 2,699	$ 2,900	$ 3,362
Variable lease cost	726	838	1,182
Total operating lease cost	$ 3,425	$ 3,738	$ 4,544

Supplemental information related to operating leases was as follows (in millions):

	Year Ended December 31,		
	2021	2022	2023
Cash payments for operating leases	$ 2,489	$ 2,722	$ 3,173
New operating lease assets obtained in exchange for operating lease liabilities	$ 2,951	$ 4,383	$ 2,877

As of December 31, 2023, our operating leases had a weighted average remaining lease term of 8.1 years and a weighted average discount rate of 3.1%. Future lease payments under operating leases as of December 31, 2023 were as follows (in millions):

2024	$	3,179
2025		2,929
2026		2,450
2027		1,951
2028		1,488
Thereafter		5,685
Total future lease payments		17,682
Less imputed interest		(2,431)
Total lease liability balance	$	15,251

As of December 31, 2023, we have entered into leases that have not yet commenced with short-term and long-term future lease payments of $657 million and $3.3 billion that are not yet recorded on our Consolidated Balance Sheets. These leases will commence between 2024 and 2026 with non-cancelable lease terms between one and 25 years.

Note 5. Variable Interest Entities

Consolidated Variable Interest Entities

We consolidate VIEs in which we hold a variable interest and are the primary beneficiary. The results of operations and financial position of these VIEs are included in our consolidated financial statements.

For certain consolidated VIEs, their assets are not available to us and their creditors do not have recourse to us. As of December 31, 2022 and 2023, assets that can only be used to settle obligations of these VIEs were $4.1 billion and $4.9 billion, respectively, and the liabilities for which creditors only have recourse to the VIEs were $2.6 billion and $2.5 billion, respectively. We may continue to fund ongoing operations of certain VIEs that are included within Other Bets.

Total noncontrolling interests (NCI) in our consolidated subsidiaries were $3.8 billion and $3.4 billion as of December 31, 2022 and 2023, respectively, of which $1.1 billion is redeemable noncontrolling interest (RNCI) for both periods. NCI and RNCI are included within additional paid-in capital. Net loss attributable to noncontrolling interests was not material for any period presented and is included within the "other" component of OI&E. See Note 7 for further details on OI&E.

Unconsolidated Variable Interest Entities

We have investments in VIEs in which we are not the primary beneficiary. These VIEs include private companies that are primarily early stage companies and certain renewable energy entities in which activities involve power generation using renewable sources.

We have determined that the governance structures of these entities do not allow us to direct the activities that would significantly affect their economic performance. Therefore, we are not the primary beneficiary, and the results of operations and financial position of these VIEs are not included in our consolidated financial statements. We account for these investments primarily as non-marketable equity securities or equity method investments.

The maximum exposure of these unconsolidated VIEs is generally based on the current carrying value of the investments and any future funding commitments. The maximum exposure and carrying value of these unconsolidated VIEs were $2.8 billion and $2.7 billion, respectively, as of December 31, 2022 and

$5.7 billion and $4.0 billion, respectively, as of December 31, 2023. The difference between the maximum exposure and the carrying value relates primarily to future funding commitments.

Note 6. Debt

Short-Term Debt

We have a debt financing program of up to $10.0 billion through the issuance of commercial paper. Net proceeds from this program are used for general corporate purposes. We had no commercial paper outstanding as of December 31, 2022 and 2023.

Our short-term debt balance also includes the current portion of certain long-term debt.

Long-Term Debt

Total outstanding debt is summarized below (in millions, except percentages):

	Maturity	Coupon Rate	Effective Interest Rate	As of December 31, 2022	As of December 31, 2023
Debt					
2014-2020 Notes issuances	2024 - 2060	0.45% - 3.38%	0.57% - 3.38%	$ 13,000	$ 13,000
Future finance lease payments, net and other [1]				2,142	1,746
Total debt				15,142	14,746
Unamortized discount and debt issuance costs				(143)	(130)
Less: Current portion of long-term notes[2]				0	(1,000)
Less: Current portion of future finance lease payments, net and other current debt[1][2]				(298)	(363)
Total long-term debt				$ 14,701	$ 13,253

(1) Future finance lease payments are net of imputed interest.

(2) Total current portion of long-term debt is included within other accrued expenses and current liabilities. See Note 7 for further details.

The notes in the table above are fixed-rate senior unsecured obligations and generally rank equally with each other. We may redeem the notes at any time in whole or in part at specified redemption prices. The effective interest rates are based on proceeds received with interest payable semi-annually.

The total estimated fair value of the outstanding notes was approximately $9.9 billion and $10.3 billion as of December 31, 2022 and December 31, 2023, respectively. The fair value was determined based on observable market prices of identical instruments in less active markets and is categorized accordingly as Level 2 in the fair value hierarchy.

As of December 31, 2023, the aggregate future principal payments for long-term debt, including finance lease liabilities, for each of the next five years and thereafter were as follows (in millions):

2024	$ 1,299
2025	1,163
2026	2,165
2027	1,143
2028	132
Thereafter	8,960
Total	$ 14,862

Credit Facility

As of December 31, 2023, we had $10.0 billion of revolving credit facilities, of which $4.0 billion expires in April 2024 and $6.0 billion expires in April 2028. The interest rates for all credit facilities are

determined based on a formula using certain market rates, as well as our progress toward the achievement of certain sustainability goals. No amounts were outstanding under the credit facilities as of December 31, 2022 and 2023.

Note 7. Supplemental Financial Statement Information

Accounts Receivable

The allowance for credit losses on accounts receivable was $754 million and $771 million as of December 31, 2022 and 2023, respectively.

Property and Equipment, Net

Property and equipment, net, consisted of the following (in millions):

	As of December 31,	
	2022	2023
Land and buildings	$ 66,897	$ 74,083
Information technology assets	66,267	80,594
Construction in progress	27,657	35,229
Leasehold improvements	10,575	11,425
Furniture and fixtures	314	472
Property and equipment, gross	171,710	201,803
Less: accumulated depreciation	(59,042)	(67,458)
Property and equipment, net	$ 112,668	$ 134,345

Our technical infrastructure is comprised of information technology assets, including servers and networking equipment, and data center land and buildings.

Accrued Expenses and Other Current Liabilities

Accrued expenses and other current liabilities consisted of the following (in millions):

	As of December 31,	
	2022	2023
European Commission fines[1]	$ 9,106	$ 9,525
Accrued purchases of property and equipment	3,019	4,679
Accrued customer liabilities	3,619	4,140
Current operating lease liabilities	2,477	2,791
Income taxes payable, net	1,632	2,748
Other accrued expenses and current liabilities	18,013	22,285
Accrued expenses and other current liabilities	$ 37,866	$ 46,168

[1] While each EC decision is under appeal, the fines are included in accrued expenses and other current liabilities on our Consolidated Balance Sheets, as we provided bank guarantees (in lieu of a cash payment) for the fines. Amounts include the effects of foreign exchange and interest. See Note 10 for further details.

Accumulated Other Comprehensive Income (Loss)

Components of AOCI, net of income tax, were as follows (in millions):

	Foreign Currency Translation Adjustments	Unrealized Gains (Losses) on Available-for-Sale Investments	Unrealized Gains (Losses) on Cash Flow Hedges	Total
Balance as of December 31, 2020	$ (864)	$ 1,612	$ (115)	$ 633
Other comprehensive income (loss) before reclassifications	(1,442)	(1,312)	668	(2,086)
Amounts excluded from the assessment of hedge effectiveness recorded in AOCI	0	0	48	48
Amounts reclassified from AOCI	0	(64)	(154)	(218)
Other comprehensive income (loss)	(1,442)	(1,376)	562	(2,256)
Balance as of December 31, 2021	(2,306)	236	447	(1,623)
Other comprehensive income (loss) before reclassifications	(1,836)	(4,720)	1,463	(5,093)
Amounts excluded from the assessment of hedge effectiveness recorded in AOCI	0	0	(188)	(188)
Amounts reclassified from AOCI	0	1,007	(1,706)	(699)
Other comprehensive income (loss)	(1,836)	(3,713)	(431)	(5,980)
Balance as of December 31, 2022	(4,142)	(3,477)	16	(7,603)
Other comprehensive income (loss) before reclassifications	735	1,344	84	2,163
Amounts excluded from the assessment of hedge effectiveness recorded in AOCI	0	0	84	84
Amounts reclassified from AOCI	0	1,168	(214)	954
Other comprehensive income (loss)	735	2,512	(46)	3,201
Balance as of December 31, 2023	$ (3,407)	$ (965)	$ (30)	$ (4,402)

The effects on net income of amounts reclassified from AOCI were as follows (in millions):

AOCI Components	Location	Gains (Losses) Reclassified from AOCI to the Consolidated Statements of Income		
		Year Ended December 31,		
		2021	2022	2023
Unrealized gains (losses) on available-for-sale investments				
	Other income (expense), net	$ 82	$ (1,291)	$ (1,497)
	Benefit (provision) for income taxes	(18)	284	329
	Net of income tax	64	(1,007)	(1,168)
Unrealized gains (losses) on cash flow hedges				
Foreign exchange contracts	Revenue	165	2,046	213
Interest rate contracts	Other income (expense), net	6	6	6
	Benefit (provision) for income taxes	(17)	(346)	(5)
	Net of income tax	154	1,706	214
Total amount reclassified, net of income tax		$ 218	$ 699	$ (954)

Other Income (Expense), Net

Components of OI&E were as follows (in millions):

	Year Ended December 31,		
	2021	2022	2023
Interest income	$ 1,499	$ 2,174	$ 3,865
Interest expense[1]	(346)	(357)	(308)
Foreign currency exchange gain (loss), net	(240)	(654)	(1,238)
Gain (loss) on debt securities, net	(110)	(2,064)	(1,215)
Gain (loss) on equity securities, net	12,380	(3,455)	392
Performance fees	(1,908)	798	257
Income (loss) and impairment from equity method investments, net	334	(337)	(628)
Other	411	381	299
Other income (expense), net	$ 12,020	$ (3,514)	$ 1,424

[1] Interest expense is net of interest capitalized of $163 million, $128 million, and $181 million for the years ended December 31, 2021, 2022, and 2023, respectively.

Note 8. Workforce Reduction and Other Initiatives

We have a company-wide effort underway to re-engineer our cost base. As part of this program, in January 2023, we announced a reduction of our workforce. As a result, total employee severance and related charges recorded during the year ended December 31, 2023 were $2.1 billion. Substantially all of the employees affected were no longer included in our headcount as of December 31, 2023.

In addition, we are taking actions to optimize our global office space. As a result, exit charges recorded during the year ended December 31, 2023, were $1.8 billion as reflected in the table below. In addition to these exit charges, for the year ended December 31, 2023, we incurred $269 million in accelerated rent and accelerated depreciation, which are not included in the table below.

Severance and office space exit charges are included within our consolidated statements of income as follows (in millions):

	Year Ended December 31, 2023		
	Severance and Related [1]	Office Space	Total
Cost of revenues	$ 479	$ 481	$ 960
Research and development	848	870	1,718
Sales and marketing	497	257	754
General and administrative	264	237	501
Total charges	$ 2,088	$ 1,845	$ 3,933

[1] Severance includes amounts to be settled in cash, accounted for as one-time involuntary employee termination benefits, and SBC.

For segment reporting, the substantial majority of these charges are included within Alphabet-level activities in our segment results.

For the year ended December 31, 2023, changes in liabilities resulting from the severance charges and related accruals were as follows (in millions):

	Severance and Related
Balance as of December 31, 2022	$ 0
Charges[1]	1,656
Cash payments	(1,579)
Balance as of December 31, 2023[2]	$ 77

(1) Excludes non-cash SBC of $432 million.

(2) Included in accrued compensation and benefits on the Consolidated Balance Sheets.

Note 9. Goodwill

Changes in the carrying amount of goodwill for the years ended December 31, 2022 and 2023 were as follows (in millions):

	Google Services	Google Cloud	Other Bets	Total
Balance as of December 31, 2021	$ 19,826	$ 2,337	$ 793	$ 22,956
Acquisitions	1,176	4,876	119	6,171
Foreign currency translation and other adjustments	(155)	(8)	(4)	(167)
Balance as of December 31, 2022	20,847	7,205	908	28,960
Acquisitions	240	3	0	243
Foreign currency translation and other adjustments	31	(9)	(27)	(5)
Balance as of December 31, 2023	$ 21,118	$ 7,199	$ 881	$ 29,198

Note 10. Commitments and Contingencies

Commitments

We have content licensing agreements with future fixed or minimum guaranteed commitments of $10.6 billion as of December 31, 2023, of which the majority is paid over seven years ending in the first quarter of 2030.

Indemnifications

In the normal course of business, including to facilitate transactions in our services and products and corporate activities, we indemnify certain parties, including advertisers, Google Network partners, distribution partners, customers of Google Cloud offerings, lessors, and service providers with respect to certain matters. We have agreed to defend and/or hold certain parties harmless against losses arising from a breach of representations or covenants, or out of intellectual property infringement or other claims made against certain parties. Several of these agreements limit the time within which an indemnification claim can be made and the amount of the claim. In addition, we have entered into indemnification agreements with our officers and directors, and our bylaws contain similar indemnification obligations to our agents.

It is not possible to make a reasonable estimate of the maximum potential amount under these indemnification agreements due to the unique facts and circumstances involved in each particular agreement. Additionally, the payments we have made under such agreements have not had a material adverse effect on our results of operations, cash flows, or financial position. However, to the extent that valid indemnification claims arise in the future, future payments by us could be significant and could have a material adverse effect on our results of operations or cash flows in a particular period.

As of December 31, 2023, we did not have any material indemnification claims that were probable or reasonably possible.

Legal Matters

We record a liability when we believe that it is probable that a loss has been incurred, and the amount can be reasonably estimated. If we determine that a loss is reasonably possible and the loss or range of loss can be estimated, we disclose the reasonably possible loss. We evaluate developments in our legal matters that could affect the amount of liability that has been previously accrued, and the matters and related reasonably possible losses disclosed, and make adjustments as appropriate.

Certain outstanding matters seek speculative, substantial or indeterminate monetary amounts, substantial changes to our business practices and products, or structural remedies. Significant judgment is required to determine both the likelihood of there being a loss and the estimated amount of a loss related to such matters, and we may be unable to estimate the reasonably possible loss or range of losses. The outcomes of outstanding legal matters are inherently unpredictable and subject to significant uncertainties, and could, either individually or in aggregate, have a material adverse effect.

We expense legal fees in the period in which they are incurred.

Antitrust Investigations

On November 30, 2010, the EC's Directorate General for Competition opened an investigation into various antitrust-related complaints against us.

On June 27, 2017, the EC announced its decision that certain actions taken by Google regarding its display and ranking of shopping search results and ads infringed European competition law. The EC decision imposed a €2.4 billion ($2.7 billion as of June 27, 2017) fine. On September 11, 2017, we appealed the EC decision to the General Court, and on September 27, 2017, we implemented product changes to bring shopping ads into compliance with the EC's decision. We recognized a charge of $2.7 billion for the fine in the second quarter of 2017. On November 10, 2021, the General Court rejected our appeal, and we subsequently filed an appeal with the European Court of Justice on January 20, 2022.

On July 18, 2018, the EC announced its decision that certain provisions in Google's Android-related distribution agreements infringed European competition law. The EC decision imposed a €4.3 billion ($5.1 billion as of June 30, 2018) fine and directed the termination of the conduct at issue. On October 9, 2018, we appealed the EC decision, and on October 29, 2018, we implemented changes to certain of our Android distribution practices. On September 14, 2022, the General Court reduced the fine from €4.3 billion to €4.1 billion. We subsequently filed an appeal with the European Court of Justice. In 2018, we recognized a charge of $5.1 billion for the fine, which we reduced by $217 million in 2022.

On March 20, 2019, the EC announced its decision that certain contractual provisions in agreements that Google had with AdSense for Search partners infringed European competition law. The EC decision imposed a fine of €1.5 billion ($1.7 billion as of March 20, 2019) and directed actions related to AdSense for Search partners' agreements, which we implemented prior to the decision. On June 4, 2019, we appealed the EC decision. We recognized a charge of $1.7 billion for the fine in the first quarter of 2019.

From time to time we are subject to formal and informal inquiries and investigations on various competition matters by regulatory authorities in the U.S., Europe, and other jurisdictions globally. Examples, for which given their nature we cannot estimate a possible loss, include:

- In August 2019, we began receiving civil investigative demands from the U.S. Department of Justice (DOJ) requesting information and documents relating to our prior antitrust investigations and certain aspects of our business. The DOJ and a number of state Attorneys General filed a lawsuit in the U.S. District Court for the District of Columbia on October 20, 2020 alleging that Google violated U.S. antitrust laws relating to Search and Search advertising. The trial ended on November 16, 2023, and we expect a decision in 2024. Further, in June 2022, the Australian Competition and Consumer Commission (ACCC) and the United Kingdom's Competition and Markets Authority (CMA) each opened an investigation into Search distribution practices.

- On December 16, 2020, a number of state Attorneys General filed an antitrust complaint in the U.S. District Court for the Eastern District of Texas, alleging that Google violated U.S. antitrust laws as well as state deceptive trade laws relating to its advertising technology, and a trial is scheduled for March 2025. Additionally, on January 24, 2023, the DOJ, along with a number of state Attorneys General, filed an antitrust complaint in the U.S. District Court for the Eastern District of Virginia alleging that Google's digital advertising technology products violate U.S. antitrust laws, and on April 17, 2023, a number of additional state Attorneys General joined the complaint. The EC, the CMA, and the ACCC each opened a formal investigation into Google's advertising technology business practices on June 22, 2021, May 25, 2022, and June 29, 2022, respectively. On June 14, 2023, the EC issued a Statement of Objections (SO) informing Google of its preliminary view that Google violated European antitrust laws relating to its advertising technology. We responded to the SO on December 1, 2023.

- On July 7, 2021, a number of state Attorneys General filed an antitrust complaint in the U.S. District Court for the Northern District of California, alleging that Google’s operation of Android and Google Play violated U.S. antitrust laws and state antitrust and consumer protection laws. In September 2023, we reached a settlement in principle with 50 state Attorneys General and three territories. The U.S. District Court subsequently vacated the trial date with the states, and any final approval of the settlement is expected to occur in 2024. In May 2022, the EC and the CMA each opened investigations into Google Play’s business practices. Korean regulators are investigating Google Play's billing practices, including a formal review in May 2022 of Google's compliance with the new app store billing regulations.

We believe we have strong arguments against these claims and will defend ourselves vigorously. We continue to cooperate with federal and state regulators in the U.S., the EC, and other regulators around the world.

Privacy Matters

We are subject to a number of privacy-related laws and regulations, and we currently are party to a number of privacy investigations and lawsuits ongoing in multiple jurisdictions. For example, there are ongoing investigations and litigation in the U.S. and the EU, including those relating to our collection and use of location information and advertising practices, which could result in significant fines, judgments, and product changes.

Patent and Intellectual Property Claims

We have had patent, copyright, trade secret, and trademark infringement lawsuits filed against us claiming that certain of our products, services, and technologies infringe others' intellectual property rights. Adverse results in these lawsuits may include awards of substantial monetary damages, costly royalty or licensing agreements, or orders preventing us from offering certain features, functionalities, products, or services. As a result, we may have to change our business practices and develop non-infringing products or technologies, which could result in a loss of revenues for us and otherwise harm our business. In addition, the U.S. International Trade Commission (ITC) has increasingly become an important forum to litigate intellectual property disputes because an ultimate loss in an ITC action can result in a prohibition on importing infringing products into the U.S. Because the U.S. is an important market, a prohibition on importation could have an adverse effect on us, including preventing us from importing many important products into the U.S. or necessitating workarounds that may limit certain features of our products.

Furthermore, many of our agreements with our customers and partners require us to indemnify them against certain intellectual property infringement claims, which would increase our costs as a result of defending such claims, and may require that we pay significant damages if there were an adverse ruling in any such claims. In addition, our customers and partners may discontinue the use of our products, services, and technologies, as a result of injunctions or otherwise, which could result in loss of revenues and adversely affect our business.

Other

We are subject to claims, lawsuits, regulatory and government investigations, other proceedings, and consent orders involving competition, intellectual property, data security, tax and related compliance, labor and employment, commercial disputes, content generated by our users, goods and services offered by advertisers or publishers using our platforms, personal injury, consumer protection, and other matters. For example, in December 2023, a California jury delivered a verdict in *Epic Games v. Google* finding that Google violated antitrust laws related to Google Play's business. The presiding judge will determine remedies in 2024, and the range of potential remedies vary widely. We plan to appeal. We also periodically have data incidents that we report to relevant regulators as required by law.

These claims, consent orders, lawsuits, regulatory and government investigations, and other proceedings could result in substantial fines and penalties, injunctive relief, ongoing monitoring and auditing obligations, changes to our products and services, alterations to our business models and operations, and collateral related civil litigation or other adverse consequences, all of which could harm our business, reputation, financial condition, and operating results.

We have ongoing legal matters relating to Russia. For example, civil judgments that include compounding penalties have been imposed upon us in connection with disputes regarding the termination of accounts, including those of sanctioned parties. We do not believe these ongoing legal matters will have a material adverse effect.

Non-Income Taxes

We are under audit by various domestic and foreign tax authorities with regards to non-income tax matters. The subject matter of non-income tax audits primarily arises from disputes on the tax treatment and tax rate applied to the sale of our products and services in these jurisdictions and the tax treatment of

certain employee benefits. We accrue non-income taxes that may result from examinations by, or any negotiated agreements with, these tax authorities when a loss is probable and reasonably estimable. If we determine that a loss is reasonably possible and the loss or range of loss can be estimated, we disclose the reasonably possible loss. Due to the inherent complexity and uncertainty of these matters and judicial process in certain jurisdictions, the final outcome may be materially different from our expectations.

See, Note 14 for information regarding income tax contingencies.

Note 11. Stockholders' Equity

Class A and Class B Common Stock and Class C Capital Stock

Our Board of Directors has authorized three classes of stock, Class A and Class B common stock, and Class C capital stock. The rights of the holders of each class of our common and capital stock are identical, except with respect to voting. Each share of Class A common stock is entitled to one vote per share. Each share of Class B common stock is entitled to 10 votes per share. Class C capital stock has no voting rights, except as required by applicable law. Shares of Class B common stock may be converted at any time at the option of the stockholder and automatically convert upon sale or transfer to Class A common stock.

Share Repurchases

In the years ended December 31, 2021, 2022, and 2023, we repurchased $50.3 billion, $59.3 billion, and $62.2 billion, respectively, of Alphabet's Class A and Class C shares.

In April 2023, the Board of Directors of Alphabet authorized the company to repurchase up to an additional $70.0 billion of its Class A and Class C shares. As of December 31, 2023, $36.3 billion remains available for Class A and Class C share repurchases.

The following table presents Class A and Class C shares repurchased and subsequently retired (in millions):

	Year Ended December 31, 2022		Year Ended December 31, 2023	
	Shares	Amount	Shares	Amount
Class A share repurchases	61	$ 6,719	78	$ 9,316
Class C share repurchases	469	52,577	450	52,868
Total share repurchases[1]	530	$ 59,296	528	$ 62,184

[1] Shares repurchased include unsettled repurchases as of December 31, 2023.

Class A and Class C shares are repurchased in a manner deemed in the best interest of the company and its stockholders, taking into account the economic cost and prevailing market conditions, including the relative trading prices and volumes of the Class A and Class C shares. Repurchases are executed from time to time, subject to general business and market conditions and other investment opportunities, through open market purchases or privately negotiated transactions, including through Rule 10b5-1 plans. The repurchase program does not have an expiration date.

Note 12. Net Income Per Share

We compute net income per share of Class A, Class B, and Class C stock using the two-class method. Basic net income per share is computed using the weighted-average number of shares outstanding during the period. Diluted net income per share is computed using the weighted-average number of shares and the effect of potentially dilutive securities outstanding during the period. Potentially dilutive securities consist of RSUs and other contingently issuable shares. The dilutive effect of outstanding RSUs and other contingently issuable shares is reflected in diluted earnings per share by application of the treasury stock method. The computation of the diluted net income per share of Class A stock assumes the conversion of Class B stock, while the diluted net income per share of Class B stock does not assume the conversion of those shares.

The rights, including the liquidation and dividend rights, of the holders of our Class A, Class B, and Class C stock are identical, except with respect to voting. Furthermore, there are a number of safeguards built into our certificate of incorporation, as well as Delaware law, which preclude our Board of Directors from declaring or paying unequal per share dividends on our Class A, Class B, and Class C stock.

Specifically, Delaware law provides that amendments to our certificate of incorporation which would have the effect of adversely altering the rights, powers, or preferences of a given class of stock must be approved by the class of stock adversely affected by the proposed amendment. In addition, our certificate of incorporation provides that before any such amendment may be put to a stockholder vote, it must be approved by the unanimous consent of our Board of Directors. As a result, the undistributed earnings for each year are allocated based on the contractual participation rights of the Class A, Class B, and Class C stock as if the earnings for the year had been distributed. As the liquidation and dividend rights are identical, the undistributed earnings are allocated on a proportionate basis.

In the years ended December 31, 2021, 2022, and 2023, the net income per share amounts are the same for Class A, Class B, and Class C stock because the holders of each class are entitled to equal per share dividends or distributions in liquidation in accordance with the Amended and Restated Certificate of Incorporation of Alphabet Inc.

The following table sets forth the computation of basic and diluted net income per share of Class A, Class B, and Class C stock (in millions, except per share amounts):

	Year Ended December 31,		
	2021		
	Class A	Class B	Class C
Basic net income per share:			
Numerator			
Allocation of undistributed earnings	$ 34,200	$ 5,174	$ 36,659
Denominator			
Number of shares used in per share computation	6,006	909	6,438
Basic net income per share	$ 5.69	$ 5.69	$ 5.69
Diluted net income per share:			
Numerator			
Allocation of undistributed earnings for basic computation	$ 34,200	$ 5,174	$ 36,659
Reallocation of undistributed earnings as a result of conversion of Class B to Class A shares	5,174	0	0
Reallocation of undistributed earnings	(581)	(77)	581
Allocation of undistributed earnings	$ 38,793	$ 5,097	$ 37,240
Denominator			
Number of shares used in basic computation	6,006	909	6,438
Weighted-average effect of dilutive securities			
Add:			
Conversion of Class B to Class A shares outstanding	909	0	0
Restricted stock units and other contingently issuable shares	0	0	200
Number of shares used in per share computation	6,915	909	6,638
Diluted net income per share	$ 5.61	$ 5.61	$ 5.61

	Year Ended December 31,		
	2022		
	Class A	Class B	Class C
Basic net income per share:			
Numerator			
Allocation of undistributed earnings	$ 27,518	$ 4,072	$ 28,382
Denominator			
Number of shares used in per share computation	5,994	887	6,182
Basic net income per share	$ 4.59	$ 4.59	$ 4.59
Diluted net income per share:			
Numerator			
Allocation of undistributed earnings for basic computation	$ 27,518	$ 4,072	$ 28,382
Reallocation of undistributed earnings as a result of conversion of Class B to Class A shares	4,072	0	0
Reallocation of undistributed earnings	(230)	(30)	230
Allocation of undistributed earnings	$ 31,360	$ 4,042	$ 28,612
Denominator			
Number of shares used in basic computation	5,994	887	6,182
Weighted-average effect of dilutive securities			
Add:			
Conversion of Class B to Class A shares outstanding	887	0	0
Restricted stock units and other contingently issuable shares	0	0	96
Number of shares used in per share computation	6,881	887	6,278
Diluted net income per share	$ 4.56	$ 4.56	$ 4.56

	Year Ended December 31, 2023		
	Class A	Class B	Class C
Basic net income per share:			
Numerator			
Allocation of undistributed earnings	$ 34,601	$ 5,124	$ 34,070
Denominator			
Number of shares used in per share computation	5,922	877	5,831
Basic net income per share	$ 5.84	$ 5.84	$ 5.84
Diluted net income per share:			
Numerator			
Allocation of undistributed earnings for basic computation	$ 34,601	$ 5,124	$ 34,070
Reallocation of undistributed earnings as a result of conversion of Class B to Class A shares	5,124	0	0
Reallocation of undistributed earnings	(287)	(37)	287
Allocation of undistributed earnings	$ 39,438	$ 5,087	$ 34,357
Denominator			
Number of shares used in basic computation	5,922	877	5,831
Weighted-average effect of dilutive securities			
Add:			
Conversion of Class B to Class A shares outstanding	877	0	0
Restricted stock units and other contingently issuable shares	0	0	92
Number of shares used in per share computation	6,799	877	5,923
Diluted net income per share	$ 5.80	$ 5.80	$ 5.80

Note 13. Compensation Plans

Stock Plans

Our stock plans include the Alphabet Amended and Restated 2021 Stock Plan ("Alphabet 2021 Stock Plan") and Other Bets stock-based plans. Under our stock plans, RSUs and other types of awards may be granted. Under the Alphabet 2021 Stock Plan, an RSU award is an agreement to issue shares of our Class C stock at the time the award vests. RSUs generally vest over four years contingent upon employment on the vesting date.

As of December 31, 2023, there were 723 million shares of Class C stock reserved for future issuance under the Alphabet 2021 Stock Plan.

Stock-Based Compensation

For the years ended December 31, 2021, 2022, and 2023, total SBC expense was $15.7 billion, $19.5 billion, and $22.1 billion, including amounts associated with awards we expect to settle in Alphabet stock of $15.0 billion, $18.8 billion, and $21.7 billion, respectively.

During the year ended December 31, 2023, total SBC expense includes $432 million associated with workforce reduction costs. See Note 8 for further details.

For the years ended December 31, 2021, 2022, and 2023, we recognized tax benefits on total SBC expense, which are reflected in the provision for income taxes in the Consolidated Statements of Income, of $3.1 billion, $3.9 billion, and $4.5 billion, respectively.

For the years ended December 31, 2021, 2022, and 2023, tax benefit realized related to awards vested or exercised during the period was $5.9 billion, $4.7 billion, and $5.6 billion, respectively. These amounts do not include the indirect effects of stock-based awards, which primarily relate to the R&D tax credit.

Stock-Based Award Activities

The following table summarizes the activities for unvested Alphabet RSUs for the year ended December 31, 2023 (in millions, except per share amounts):

	Unvested Restricted Stock Units	
	Number of Shares	Weighted-Average Grant-Date Fair Value
Unvested as of December 31, 2022	324	$ 107.98
Granted	263	$ 97.59
Vested	(217)	$ 100.36
Forfeited/canceled	(32)	$ 106.56
Unvested as of December 31, 2023	338	$ 104.93

The weighted-average grant-date fair value of RSUs granted during the years ended December 31, 2021 and 2022 was $97.46 and $127.22, respectively. Total fair value of RSUs, as of their respective vesting dates, during the years ended December 31, 2021, 2022, and 2023, were $28.8 billion, $23.9 billion, and $26.6 billion, respectively.

As of December 31, 2023, there was $33.5 billion of unrecognized compensation cost related to unvested RSUs. This amount is expected to be recognized over a weighted-average period of 2.5 years.

Note 14. Income Taxes

Income from continuing operations before income taxes consisted of the following (in millions):

	Year Ended December 31,		
	2021	2022	2023
Domestic operations	$ 77,016	$ 61,307	$ 73,600
Foreign operations	13,718	10,021	12,117
Total	$ 90,734	$ 71,328	$ 85,717

Provision for income taxes consisted of the following (in millions):

	Year Ended December 31,		
	2021	2022	2023
Current:			
Federal and state	$ 10,126	$ 17,120	$ 17,125
Foreign	2,692	2,434	2,526
Total	12,818	19,554	19,651
Deferred:			
Federal and state	2,018	(8,052)	(7,482)
Foreign	(135)	(146)	(247)
Total	1,883	(8,198)	(7,729)
Provision for income taxes	$ 14,701	$ 11,356	$ 11,922

The reconciliation of federal statutory income tax rate to our effective income tax rate was as follows:

	Year Ended December 31,		
	2021	2022	2023
U.S. federal statutory tax rate	21.0 %	21.0 %	21.0 %
Foreign income taxed at different rates	0.2	3.0	0.3
Foreign-derived intangible income deduction	(2.5)	(5.4)	(4.6)
Stock-based compensation expense	(2.5)	(1.2)	(0.8)
Federal research credit	(1.6)	(2.2)	(1.8)
Deferred tax asset valuation allowance	0.6	0.9	0.6
State and local income taxes	1.0	0.8	1.0
Effect of tax law change	0.0	0.0	(1.4)
Other	0.0	(1.0)	(0.4)
Effective tax rate	16.2 %	15.9 %	13.9 %

In 2022, there was an increase in the U.S. Foreign Derived Intangible Income tax deduction from the effects of capitalization and amortization of R&D expenses starting in 2022 as required by the 2017 Tax Cuts and Jobs Act.

In 2023, the IRS issued a rule change allowing taxpayers to temporarily apply the regulations in effect prior to 2022 related to U.S. federal foreign tax credits as well as a separate rule change with interim guidance on the capitalization and amortization of R&D expenses. A cumulative one-time adjustment applicable to the prior period for these tax rule changes was recorded in 2023.

Deferred Income Taxes

Deferred income taxes reflect the net effects of temporary differences between the carrying amounts of assets and liabilities for financial reporting purposes and the amounts used for income tax purposes. Significant components of our deferred tax assets and liabilities were as follows (in millions):

	As of December 31,	
	2022	2023
Deferred tax assets:		
Accrued employee benefits	$ 955	$ 1,855
Accruals and reserves not currently deductible	1,956	2,481
Tax credits	6,002	6,609
Net operating losses	2,557	2,965
Operating leases	2,711	3,526
Capitalized research and development[1]	10,381	17,757
Other	2,289	1,951
Total deferred tax assets	26,851	37,144
Valuation allowance	(9,553)	(10,999)
Total deferred tax assets net of valuation allowance	17,298	26,145
Deferred tax liabilities:		
Property and equipment, net	(6,607)	(8,189)
Net investment gains	(2,361)	(2,405)
Operating leases	(2,491)	(2,965)
Other	(1,092)	(902)
Total deferred tax liabilities	(12,551)	(14,461)
Net deferred tax assets (liabilities)	$ 4,747	$ 11,684

(1) As required by the 2017 Tax Cuts and Jobs Act, effective January 1, 2022, our research and development expenditures were capitalized and amortized which resulted in substantially higher cash taxes starting in 2022 with an equal amount of deferred tax benefit.

As of December 31, 2023, our federal, state, and foreign net operating loss carryforwards for income tax purposes were approximately $7.1 billion, $18.6 billion, and $1.8 billion respectively. If not utilized, the federal net

operating loss carryforwards will begin to expire in 2024, foreign net operating loss carryforwards will begin to expire in 2025 and the state net operating loss carryforwards will begin to expire in 2029. It is more likely than not that the majority of the net operating loss carryforwards will not be realized; therefore, we have recorded a valuation allowance against them. The net operating loss carryforwards are subject to various annual limitations under the tax laws of the different jurisdictions.

As of December 31, 2023, our Federal and California research and development credit carryforwards for income tax purposes were approximately $600 million and $6.3 billion, respectively. If not utilized, the Federal R&D credit will begin to expire in 2037 and the California R&D credit can be carried over indefinitely. We believe the majority of the federal tax credit and state tax credit is not likely to be realized.

As of December 31, 2023, our investment tax credit carryforwards for state income tax purposes were approximately $1.0 billion and will begin to expire in 2029. We use the flow-through method of accounting for investment tax credits. We believe this tax credit is not likely to be realized.

As of December 31, 2023, we maintained a valuation allowance with respect to California deferred tax assets, certain federal net operating losses, certain state net operating losses and tax credits, net deferred tax assets relating to Other Bet companies, and certain foreign net operating losses that we believe are not likely to be realized. We continue to reassess the remaining valuation allowance quarterly, and if future evidence allows for a partial or full release of the valuation allowance, a tax benefit will be recorded accordingly.

Uncertain Tax Positions

The following table summarizes the activity related to our gross unrecognized tax benefits (in millions):

	Year Ended December 31,		
	2021	2022	2023
Beginning gross unrecognized tax benefits	$ 3,837	$ 5,158	$ 7,055
Increases related to prior year tax positions	529	253	740
Decreases related to prior year tax positions	(263)	(437)	(682)
Decreases related to settlement with tax authorities	(329)	(140)	(21)
Increases related to current year tax positions	1,384	2,221	2,346
Ending gross unrecognized tax benefits	$ 5,158	$ 7,055	$ 9,438

We are subject to income taxes in the U.S. and foreign jurisdictions. Significant judgment is required in evaluating our uncertain tax positions and determining our provision for income taxes. The total amount of gross unrecognized tax benefits was $5.2 billion, $7.1 billion, and $9.4 billion as of December 31, 2021, 2022, and 2023, respectively, of which $3.7 billion, $5.3 billion, and $7.4 billion, if recognized, would affect our effective tax rate, respectively.

As of December 31, 2022 and 2023, we accrued $346 million and $622 million in interest and penalties in provision for income taxes, respectively.

We file income tax returns in the U.S. federal jurisdiction and in many state and foreign jurisdictions. Our two major tax jurisdictions are the U.S. federal and Ireland. We are subject to the continuous examination of our income tax returns by the IRS and other tax authorities. The IRS is currently examining our 2016 through 2021 tax returns. We have also received tax assessments in multiple foreign jurisdictions asserting transfer pricing adjustments or permanent establishment. We continue to defend such claims as presented.

The tax years 2016 through 2022 remain subject to examination by the appropriate governmental agencies for Irish tax purposes. There are other ongoing audits in various other jurisdictions that are not material to our financial statements.

We regularly assess the likelihood of adverse outcomes resulting from these examinations to determine the adequacy of our provision for income taxes. We continue to monitor the progress of ongoing discussions with tax authorities and the effect, if any, of the expected expiration of the statute of limitations in various taxing jurisdictions.

We believe that an adequate provision has been made for any adjustments that may result from tax examinations. However, the outcome of tax audits cannot be predicted with certainty. If any issues addressed in our tax audits are resolved in a manner not consistent with management's expectations, we could be required to adjust our provision for income taxes in the period such resolutions occur. Although the timing of resolution, settlement, and closure of audits is not certain, it is reasonably possible that our unrecognized tax benefits from certain U.S. federal, state, and non U.S. tax positions could decrease by approximately $700 million in the next 12 months. Positions that may be resolved include various U.S. and non-U.S. matters.

Note 15. Information about Segments and Geographic Areas

We report our segment results as Google Services, Google Cloud, and Other Bets:

- Google Services includes products and services such as ads, Android, Chrome, devices, Google Maps, Google Play, Search, and YouTube. Google Services generates revenues primarily from advertising; fees received for consumer subscription-based products such as YouTube TV, YouTube Music and Premium, and NFL Sunday Ticket; the sale of apps and in-app purchases and devices.
- Google Cloud includes infrastructure and platform services, collaboration tools, and other services for enterprise customers. Google Cloud generates revenues primarily from consumption-based fees and subscriptions received for Google Cloud Platform services, Google Workspace communication and collaboration tools, and other enterprise services.
- Other Bets is a combination of multiple operating segments that are not individually material. Revenues from Other Bets are generated primarily from the sale of healthcare-related services and internet services.

Revenues, certain costs, such as costs associated with content and traffic acquisition, certain engineering activities, and devices, as well as certain operating expenses are directly attributable to our segments. Due to the integrated nature of Alphabet, other costs and expenses, such as technical infrastructure and office facilities, are managed centrally at a consolidated level. These costs, including the associated depreciation and impairment, are allocated to operating segments as a service cost generally based on usage, headcount, or revenue.

Reflecting DeepMind's increasing collaboration with Google Services, Google Cloud, and Other Bets, beginning in the first quarter of 2023 DeepMind is reported as part of Alphabet-level activities instead of within Other Bets. Additionally, beginning in the first quarter of 2023, we updated and simplified our cost allocation methodologies to provide our business leaders with increased transparency for decision-making. Prior periods have been recast to conform to the current presentation.

As announced on April 20, 2023, we brought together part of Google Research (the Brain team) and DeepMind to significantly accelerate our progress in artificial intelligence (AI). The group, called Google DeepMind, is reported within Alphabet-level activities prospectively beginning in the second quarter of 2023. Previously, the Brain team was included within Google Services.

Certain costs are not allocated to our segments because they represent Alphabet-level activities. These costs primarily include AI-focused shared R&D activities, including development costs of our general AI models; corporate initiatives such as our philanthropic activities; corporate shared costs such as certain finance, human resource, and legal costs, including certain fines and settlements. Charges associated with reductions in our workforce and office space during 2023 were not allocated to our segments. Additionally, hedging gains (losses) related to revenue are not allocated to our segments.

Our operating segments are not evaluated using asset information.

The following table presents information about our segments (in millions):

	Year Ended December 31,		
	2021	2022	2023
Revenues:			
Google Services	$ 237,529	$ 253,528	$ 272,543
Google Cloud	19,206	26,280	33,088
Other Bets	753	1,068	1,527
Hedging gains (losses)	149	1,960	236
Total revenues	$ 257,637	$ 282,836	$ 307,394
Operating income (loss):			
Google Services	$ 88,132	$ 82,699	$ 95,858
Google Cloud	(2,282)	(1,922)	1,716
Other Bets	(4,051)	(4,636)	(4,095)
Alphabet-level activities	(3,085)	(1,299)	(9,186)
Total income from operations	$ 78,714	$ 74,842	$ 84,293

See Note 2 for information relating to revenues by geography.

The following table presents long-lived assets by geographic area, which includes property and equipment, net and operating lease assets (in millions):

	As of December 31,	
	2022	2023
Long-lived assets:		
United States	$ 93,565	$ 110,053
International	33,484	38,383
Total long-lived assets	$ 127,049	$ 148,436

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

Our management, with the participation of our chief executive officer and chief financial officer, evaluated the effectiveness of our disclosure controls and procedures pursuant to Rule 13a-15 under the Exchange Act, as of the end of the period covered by this Annual Report on Form 10-K.

Based on this evaluation, our chief executive officer and chief financial officer concluded that, as of December 31, 2023, our disclosure controls and procedures are designed at a reasonable assurance level and are effective to provide reasonable assurance that information we are required to disclose in reports that we file or submit under the Exchange Act is recorded, processed, summarized, and reported within the time periods specified in the SEC's rules and forms, and that such information is accumulated and communicated to our management, including our chief executive officer and chief financial officer, as appropriate, to allow timely decisions regarding required disclosure.

Changes in Internal Control over Financial Reporting

There have been no changes in our internal control over financial reporting that occurred during the quarter ended December 31, 2023 that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Management's Report on Internal Control over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting, as defined in Rule 13a-15(f) of the Exchange Act. Our management conducted an evaluation of the effectiveness of our internal control over financial reporting based on the framework in *Internal Control—Integrated Framework* issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework). Based on this evaluation, management concluded that our internal control over financial reporting was effective as of December 31, 2023. Management reviewed the results of its assessment with our Audit and Compliance Committee. The effectiveness of our internal control over financial reporting as of December 31, 2023 has been audited by Ernst & Young LLP, an independent registered public accounting firm, as stated in its report which is included in Item 8 of this Annual Report on Form 10-K.

Limitations on Effectiveness of Controls and Procedures

In designing and evaluating the disclosure controls and procedures, management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving the desired control objectives. In addition, the design of disclosure controls and procedures must reflect the fact that there are resource constraints and that management is required to apply its judgment in evaluating the benefits of possible controls and procedures relative to their costs.

ITEM 9B. OTHER INFORMATION

10b5-1 Trading Plans

During the fiscal quarter ended December 31, 2023, the following Section 16 officer and directors adopted, modified or terminated a "Rule 10b5-1 trading arrangement" (as defined in Item 408 of Regulation S-K of the Exchange Act):

- John Hennessy, Chair of the Board of Directors, through the John L. Hennessy & Andrea J. Hennessy Revocable Trust, adopted a new trading plan on November 1, 2023 (with the first trade under the new plan scheduled for February 12, 2024). The trading plan will be effective until March 12, 2025 to sell 6,664 shares of Class C Capital Stock and 11,336 shares of Class A Common Stock.
- Ann Mather, former member of the Board of Directors, terminated her trading plan on October 30, 2023, effective with her resignation from the Board of Directors. The trading plan previously permitted the sale of 12,580 shares of Class C Capital Stock and would have been effective until June 2, 2024.
- Ruth M. Porat, President and Chief Investment Officer; Chief Financial Officer, adopted a new trading plan on November 30, 2023 (with the first trade under the new plan scheduled for March 8, 2024). The trading plan will be effective until March 8, 2025 to sell all of the (net) shares of up to 82,900 (gross) Class C Capital Stock issued upon the vesting of her Alphabet 2021 Performance Stock Units, as adjusted based on performance (net shares are net of tax withholding).

There were no "non-Rule 10b5-1 trading arrangements" (as defined in Item 408 of Regulation S-K of the Exchange Act) adopted, modified or terminated during the fiscal quarter ended December 31, 2023 by our directors and Section 16 officers. Each of the Rule 10b5-1 trading arrangements are in accordance with our Policy Against Insider Trading and actual sale transactions made pursuant to such trading arrangements will be disclosed publicly in Section 16 filings with the SEC in accordance with applicable securities laws, rules and regulations.

Required Disclosure Pursuant to Section 13(r) of the Exchange Act

As previously disclosed, Google LLC, a subsidiary of Alphabet, filed notifications with the Russian Federal Security Service pursuant to Russian encryption control requirements, which must be complied with prior to the import of covered items. The information provided pursuant to Section 13(r) of the Exchange Act in Part II, Item 5 of our Quarterly Report on Form 10-Q for the quarter ended September 30, 2023 is incorporated herein by reference.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS, AND CORPORATE GOVERNANCE

The information required by this item will be included under the caption “Directors, Executive Officers, and Corporate Governance” in our Proxy Statement for the 2024 Annual Meeting of Stockholders to be filed with the SEC within 120 days of the fiscal year ended December 31, 2023 (2024 Proxy Statement) and is incorporated herein by reference. The information required by this item regarding delinquent filers pursuant to Item 405 of Regulation S-K will be included under the caption “Delinquent Section 16(a) Reports” in the 2024 Proxy Statement and is incorporated herein by reference.

ITEM 11. EXECUTIVE COMPENSATION

The information required by this item will be included under the captions “Director Compensation,” “Executive Compensation” and “Directors, Executive Officers, and Corporate Governance—Corporate Governance and Board Matters—Compensation Committee Interlocks and Insider Participation” in the 2024 Proxy Statement and is incorporated herein by reference, except as to information disclosed therein pursuant to Item 402(v) of Regulation S-K relating to pay versus performance.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information required by this item will be included under the captions “Common Stock Ownership of Certain Beneficial Owners and Management” and “Equity Compensation Plan Information” in the 2024 Proxy Statement and is incorporated herein by reference.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

The information required by this item will be included under the captions “Certain Relationships and Related Transactions” and “Directors, Executive Officers, and Corporate Governance—Corporate Governance and Board Matters—Director Independence” in the 2024 Proxy Statement and is incorporated herein by reference.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

The information required by this item will be included under the caption “Independent Registered Public Accounting Firm” in the 2024 Proxy Statement and is incorporated herein by reference.

PART IV

ITEM 15. EXHIBITS, FINANCIAL STATEMENT SCHEDULES

We have filed the following documents as part of this Annual Report on Form 10-K:

1. Consolidated Financial Statements

Reports of Independent Registered Public Accounting Firm	48
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Consolidated Balance Sheets	51
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2. Financial Statement Schedules

Schedule II: Valuation and Qualifying Accounts

The table below details the activity of the allowance for credit losses and sales credits for the years ended December 31, 2021, 2022, and 2023 (in millions):

	Balance at Beginning of Year	Additions	Usage	Balance at End of Year
Year ended December 31, 2021	$ 1,344	$ 2,092	$ (2,047)	$ 1,389
Year ended December 31, 2022	$ 1,389	$ 2,125	$ (2,301)	$ 1,213
Year ended December 31, 2023	$ 1,213	$ 3,115	$ (2,737)	$ 1,591

Note: Additions to the allowance for credit losses are charged to expense. Additions to the allowance for sales credits are charged against revenues.

All other schedules have been omitted because they are not required, not applicable, or the required information is otherwise included.

3. Exhibits

Exhibit Number	Description	Incorporated by reference herein	
		Form	Date
2.01	Agreement and Plan of Merger, dated October 2, 2015, by and among Google Inc., the Registrant and Maple Technologies Inc.	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
3.01	Amended and Restated Certificate of Incorporation of the Registrant	Current Report on Form 8-K (File No. 001-37580)	June 3, 2022
3.02	Amended and Restated Bylaws of the Registrant, dated October 19, 2022	Current Report on Form 8-K (File No. 001-37580)	October 25, 2022
4.01	Specimen Class A Common Stock certificate	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.02	Specimen Class C Capital Stock certificate	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.03	Alphabet Inc. Deferred Compensation Plan	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.04	Transfer Restriction Agreement, dated October 2, 2015, between the Registrant and Larry Page and certain of his affiliates	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.05	Transfer Restriction Agreement, dated October 2, 2015, between the Registrant and Sergey Brin and certain of his affiliates	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.06	Joinder Agreement, dated December 31, 2021, among the Registrant, Sergey Brin and certain of his affiliates	Annual Report on Form 10-K (File No. 001-37580)	February 2, 2022

Exhibit Number			Description	Incorporated by reference herein: Form	Incorporated by reference herein: Date
4.07			Transfer Restriction Agreement, dated October 2, 2015, between the Registrant and Eric E. Schmidt and certain of its affiliates	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.08			Class C Undertaking, dated October 2, 2015, executed by the Registrant	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.09			Indenture, dated February 12, 2016, between the Registrant and The Bank of New York Mellon Trust Company, N.A., as Trustee	Registration Statement on Form S-3 (File No. 333-209510)	February 12, 2016
4.10			Registrant Registration Rights Agreement dated December 14, 2015	Registration Statement on Form S-3 (File No. 333-209518)	February 12, 2016
4.11			First Supplemental Indenture, dated April 27, 2016, between the Registrant and The Bank of New York Mellon Trust Company, N.A., as trustee	Current Report on Form 8-K (File No. 001-37580)	April 27, 2016
4.12			Form of the Registrant's 3.375% Notes due 2024 (included in Exhibit 4.11)		
4.13			Form of the Registrant's 1.998% Note due 2026	Current Report on Form 8-K (File No. 001-37580)	August 9, 2016
4.14			Form of Global Note representing the Registrant's 0.450% notes due 2025	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.15			Form of Global Note representing the Registrant's 0.800% notes due 2027	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.16			Form of Global Note representing the Registrant's 1.100% notes due 2030	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.17			Form of Global Note representing the Registrant's 1.900% notes due 2040	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.18			Form of Global Note representing the Registrant's 2.050% notes due 2050	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.19			Form of Global Note representing the Registrant's 2.250% notes due 2060	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.20			Description of Registrant's Securities	Annual Report on Form 10-K (File No. 001-37580)	February 3, 2023
10.01	◆		Form of Indemnification Agreement entered into between the Registrant, its affiliates and its directors and officers	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
10.02	◆	*	Form of Offer Letter for Directors		
10.03	◆		Offer Letter, dated March 20, 2015, between Ruth Porat and Google Inc., as assumed by the Registrant on October 2, 2015	Current Report on Form 8-K (File No. 001-36380)	March 26, 2015
10.04	◆		Compensation Plan Agreement, dated October 2, 2015, between Google Inc. and the Registrant	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
10.05	◆		Director Arrangements Agreement, dated October 2, 2015, between Google Inc. and the Registrant	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
10.06	◆		Alphabet Inc. Deferred Compensation Plan	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015

Exhibit Number		Description	Incorporated by reference herein: Form	Incorporated by reference herein: Date
10.08	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan	Current Report on Form 8-K (file No. 001-37580)	June 3, 2022
10.08.1	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet Restricted Stock Unit Agreement	Quarterly Report on Form 10-Q (file No. 001-37580)	July 28, 2021
10.08.2	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet Restricted Stock Unit Agreement	Quarterly Report on Form 10-Q (File No. 001-37580)	July 26, 2023
10.08.3	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet 2022 CEO Performance Stock Unit Agreement	Annual Report on Form 10-K (File No. 001-37580)	February 3, 2023
10.08.4	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet 2022 Non-CEO Performance Stock Unit Agreement	Annual Report on Form 10-K (File No. 001-37580)	February 4, 2020
10.08.5	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet 2023 Non-CEO Performance Stock Unit Agreement	Quarterly Report on Form 10-Q (File No. 001-37580)	July 26, 2023
10.09	◆	Alphabet Inc. Company Bonus Plan, as amended	Annual Report on Form 10-K (File No. 001-37350)	February 2, 2023
21.01	*	Subsidiaries of the Registrant		
23.01	*	Consent of Independent Registered Public Accounting Firm		
24.01	*	Power of Attorney (incorporated by reference to the signature page of this Annual Report on Form 10-K)		
31.01	*	Certification of Chief Executive Officer pursuant to Exchange Act Rules 13a-14(a) and 15d-14(a), as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002		
31.02	*	Certification of Chief Financial Officer pursuant to Exchange Act Rules 13a-14(a) and 15d-14(a), as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002		
32.01	‡	Certifications of Chief Executive Officer and Chief Financial Officer pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002		
97.01	*	Clawback Policy		
101.INS	*	Inline XBRL Instance Document - the instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document		
101.SCH	*	Inline XBRL Taxonomy Extension		

Exhibit Number		Description	Incorporated by reference herein	
			Form	Date
101.PRE	*	Inline XBRL Taxonomy Extension Presentation Linkbase Document		
104		Cover Page Interactive Data File (embedded within the Inline XBRL document and contained in Exhibit 101)		

◆ Indicates management compensatory plan, contract, or arrangement.

* Filed herewith.

‡ Furnished herewith.

ITEM 16. FORM 10-K SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this Annual Report on Form 10-K to be signed on its behalf by the undersigned, thereunto duly authorized.

Date: January 30, 2024

ALPHABET INC.

By: /S/ SUNDAR PICHAI

Sundar Pichai

Chief Executive Officer

(Principal Executive Officer of the Registrant)

POWER OF ATTORNEY

KNOW ALL PERSONS BY THESE PRESENTS, that each person whose signature appears below constitutes and appoints Sundar Pichai and Ruth M. Porat, jointly and severally, his or her attorney-in-fact, with the power of substitution, for him or her in any and all capacities, to sign any amendments to this Annual Report on Form 10-K and to file the same, with exhibits thereto and other documents in connection therewith, with the Securities and Exchange Commission, hereby ratifying and confirming all that each of said attorneys-in-fact, or his or her substitute or substitutes, may do or cause to be done by virtue hereof.

Pursuant to the requirements of the Securities Exchange Act of 1934, this Annual Report on Form 10-K has been signed below by the following persons on behalf of the Registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Sundar Pichai **Sundar Pichai**	Chief Executive Officer and Director (Principal Executive Officer)	January 30, 2024
/s/ Ruth M. Porat **Ruth M. Porat**	President and Chief Investment Officer; Chief Financial Officer (Principal Financial Officer)	January 30, 2024
/s/ Amie Thuener O'Toole **Amie Thuener O'Toole**	Vice President, Corporate Controller and Principal Accounting Officer	January 30, 2024
/s/ Frances H. Arnold **Frances H. Arnold**	Director	January 30, 2024
/s/ Sergey Brin **Sergey Brin**	Co-Founder and Director	January 30, 2024
/s/ R. Martin Chavez **R. Martin Chávez**	Director	January 30, 2024
/s/ L. John Doerr **L. John Doerr**	Director	January 30, 2024
/s/ Roger W. Ferguson Jr. **Roger W. Ferguson Jr.**	Director	January 30, 2024
/s/ John L. Hennessy **John L. Hennessy**	Director, Chair	January 30, 2024
/s/ Larry Page **Larry Page**	Co-Founder and Director	January 30, 2024
/s/ K. Ram Shriram **K. Ram Shriram**	Director	January 30, 2024
/s/ Robin L. Washington **Robin L. Washington**	Director	January 30, 2024